

Captured weight and boundary leakage bound the error of sample-based spectral functions

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Sample-based quantum diagonalization returns a subspace of measured configurations; Rayleigh–Ritz on it yields a spectral function, and we ask what bounds its error. The captured Born weight w alone does not: Rayleigh–Ritz displaces the poles it keeps, an analytic counterexample forces any bound indexed on it to its trivial value, and on the probe’s support, where $w = 1$, the error is still 0.35–0.43 at $\eta = 0.18t$ against a ceiling of 2. What bounds it, with the missed weight, is the Hamiltonian coupling across the subspace boundary: for Rayleigh–Ritz on any orthogonal projector, given the exact ground state, we prove that the relative L_1 error lies between $1 - w$ and $\min\{1 + w, (1 + \sqrt{w})(\sqrt{1 - w} + \hat{\Lambda}(\eta)/\eta)\}$, with $\hat{\Lambda}$ the boundary coupling at resolution η , from the Ritz pairs and the retained probe. For the 1D Hubbard $A(\omega)$, built from classical Born draws, or their infinite-shot ranking, of a real-time evolved probe (one device run, an execution record, aside), the bound first beats the trivial one at a true error of $1.2\text{--}7.3 \times 10^{-3}$ in all sixteen (L, η) cells over $L = 6\text{--}12$; a two-constant rule in $1 - w$, fitted here without guarantee, is tighter on 11 of 14 splits; the Born-ranked $A(k, \omega)$ at 85% of its sector is bounded at 350–900 times its true error. Yet at all five sizes up to dimension 10 306 296 the bound is vacuous at every operating fraction of the resource scan, by 1.66–8.61, and informative only well above them. In the determinant basis the probe’s Krylov space has, at $L = 6\text{--}14$, coordinate support on every symmetry-allowed determinant, so zero leakage at full weight needs all of them; a momentum probe measured in plane-wave orbitals stays in one block, $1/L$ of the sector, a gain of $1/L$ against an exponential. On pairing-model fibers every orbital-invariant leakage bound is vacuous. At full weight a second-order certificate follows; below it, where every operating fraction lies, one is open.

I. THE OPERATIONAL QUESTION

A sample-based quantum calculation returns a list of bitstrings. Diagonalizing the Hamiltonian inside the span of the configurations they name yields an energy, and—as this work and several others show—a frequency-resolved response as well. What it does not yield is a statement of how wrong that response is. This paper addresses the question a user is left with once the shots are spent: *given the subspace those shots populated, how wrong is the spectral function built on it?* The natural diagnostic, the Born weight the subspace captures, does not answer it: Rayleigh–Ritz displaces the poles it keeps instead of deleting the ones it misses, so a subspace holding the probe’s entire Born support still carries an error of order one. What enters a bound on it, together with the missed weight, is the Hamiltonian coupling across the boundary of the subspace. Given the exact ground state, that bound is provable on any subspace, and we measure where it is informative—not at the sector fractions at which the reconstructions here operate—and why, and price the method in determinants and shots.

Three statements hold on any subspace: two results and a classical property whose exact reach we state. First, the captured weight alone cannot bound the error: an analytic counterexample forces the constant of any bound indexed on it to its trivial value, and on the probe’s own Born support, where the captured weight is exactly one, the relative error is still 0.43 at $L = 6$ and 0.35 at $L = 8$ at $\eta = 0.18t$ on the periodic rings (0.28 on the open chain of Sec. IV, a different lattice). Second, a two-sided bound (Theorem 1) whose boundary-leakage term is a posteriori: for Rayleigh–Ritz reconstruction on any orthogonal projector, the L_1 error of the broadened spectral function lies between the Born weight of the probe that the subspace failed to capture and that weight plus the Hamiltonian coupling that leaks across the boundary of the subspace at the resolution η , a quantity with a closed form in the Ritz pairs the reconstruction already produces and in the retained part of the probe; both terms take the probe and the energy reference from the exact ground state, which makes the bound an instrument for benchmarking a sample-based pipeline where exact diagonalization is available; and by the identity behind it, the error that survives full captured weight is generated by the coupling across the boundary alone (Sec. III D), whatever the bound’s own value there. Third, what the subspace structure does guarantee is moment exactness, the classical moment-matching property of Krylov

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projections (Sec. IV): a subspace containing the order- K Krylov space reproduces the spectral moments through order $2K + 1$ and no further, a reach we state exactly and measure on coordinate subspaces (Theorem 2), and a non-circular count gives the shots that capture every high-probability configuration (Proposition 1).

Evaluated on the subspaces of the resource scan, at five sizes up to a sector of dimension 10 306 296, the leakage branch of the bound is worse than the trivial bound at every operating fraction of the resource scan, by factors of 1.66 to 8.61, and at $L = 12$ that verdict is converged in Krylov depth; it becomes informative only well above those fractions, and at this paper’s 5% accuracy criterion only from 0.98 of the sector at $L = 8$. The Born-ranked (infinite-shot) reconstruction of Fig. 3, at 85% of its sector, is non-vacuous in all sixteen channels, at a relative bound of 0.77–1.75 against the trivial 2, 350–900 times its true error (Sec. V). An independent implementation recovers the acceptance fractions of the scan to under 2%. The crossing of the trivial bound occurs at a true relative error between 1.2×10^{-3} and 7.3×10^{-3} over sixteen (L, η) cells whose true errors span five decades, a spread of 5.9, so the certificate is calibrated: once it improves on the trivial bound, the true error is already at that level. A two-constant rule in the missed weight, fitted out of sample on these systems and carrying no guarantee, localizes the same crossing more tightly on eleven of fourteen splits; what distinguishes the certificate is that it is an inequality with a proof, given the exact probe. And in the determinant basis in which the device measures, the Krylov space of the probe has coordinate support on every determinant that the reflection about the seed site allows, measured from $L = 6$ to $L = 14$, so at full captured weight only a subspace holding all of them has zero leakage; if the device instead samples in the plane-wave orbital basis, at a cost of about $L(L - 1)$ further two-qubit gates whose effect on the sampled distribution is unmeasured, a momentum-resolved probe confines its Krylov support to one momentum block, $1/L$ of the sector, a gain of $1/L$ against an exponential (Sec. VE).

No orbital-rotation invariant can bound the boundary leakage usefully on every state—on fibers of a pairing model every such bound is vacuous—so none replaces the certificate by an a priori predictor computed from the state before any circuit runs (Sec. VI). The natural candidates are the efficiently measurable monotones of fermionic non-Gaussianity, and on number-conserving states the one-body member collapses to the single invariant $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$ (Eq. (17)). An invariant of the occupation spectrum cannot see a boundary: along a geminal orbit every order of it is fixed while the orbit contains a point of classical cost $O(K)$ in the fixed computational basis (the pairing basis: 2^K determinants, but bond dimension 2; Theorem 3) and points at which Born sampling is plausibly hard (Sec. S5), a hardness we do not establish, and on fibers where every invariant and the captured weight are frozen the subspace-aware certificate still separates a non-vacuous value from the trivial bound (Proposition 2). Within the six (L, N) strata of the Hubbard interaction sweep the rank correlation of $\mathcal{F}_1$ with the determinant support is exactly -1 (exact permutation $p = 3.35 \times 10^{-13}$ against independent orderings, a null that six strata sharing one mechanism do not satisfy), a deterministic monotone relation whose sign reverses between families, as an orbital invariant compared with a basis-dependent count must (Sec. VI). Sections VII and VIII then price the method in support, resolution and shots, and check the full-sector references against the opposite finite-size scaling of the charge and spin gaps of the Hubbard chain.

Four bodies of work bound what is new here. (i) Patel, Rangi and Tam [1] construct Green’s functions from sampled Krylov subspaces for dynamical mean-field theory [2], independently and concurrently, and cite this work as “a closely related sample-based approach to dynamical spectral functions, constructed directly from bitstring-sampled subspaces”. (ii) Dynamical DMRG computed all three momentum-resolved channels reported here for the half-filled extended Hubbard chain at $L = 60$ –120 in 2007 [3], and correction-vector DMRG treats the Hubbard chain at $L = 48$, the Heisenberg chain at $L = 64$ (the $k = \pi$ cut to $L = 128$) and the t - J ladder at 48×2 sites [4]; the classical toolbox for such spectra is broader still [5–10]. (iii) Frequency-resolved responses of lattice models, all but one of them correlated, have been measured on devices far larger than the 12-qubit spectral run of this work: a retarded spin Green’s function of the Fermi–Hubbard chain at up to 30 qubits [11], an angle-resolved-photoemission-like $A(k, \omega)$ of a free-fermion 27-site chain at 54 qubits [12], $S(q, \omega)$ on 50 qubits against neutron scattering in KCuF_3 [13], and quench spectroscopy of $L = 101$ spins [14]; at a scale comparable to ours, $S(q, \omega)$ of a 20-site Heisenberg chain was computed on a trapped-ion device and compared with neutron data for copper sulfate [15]. The two device runs of this work are execution records only: the spectral one retained its entire sector, so its agreement with exact diagonalization is exact by coverage rather than a test of the method (Sec. IX A). (iv) Computable Gaussian-invariant quantifiers of fermionic non-Gaussianity include the antifatness of Ref. [16] used here. Tarabunga *et al.* [17] separate covariance-matrix occupation entropies, two members of which are strong monotones that lower-bound the non-Gaussian gate count, from full-state natural-orbital participation entropies, which upper-bound the classical simulation cost in an orthonormal Gaussian basis; the quadratic member, which coincides with the $k = 1$ antifatness, is one of the two, and Leone and Bittel [18] independently prove it an efficiently measurable strong pure-state Gaussian monotone, both in July 2026; the priority for those measures is theirs, and what this work adds is their contrast with the basis-dependent determinant support, a constructive instance of it, and its extension from the determinant support to the leakage. Every system studied here is classically tractable, Hubbard rings at $L \leq 14$ and nineteen active spaces with a largest minimal bond dimension of 42, and no quantum advantage is claimed: the one quantitative statement this work makes

about its own cost—that the basis-dependent bond dimension tracks the determinant support—argues, wherever that bond dimension is small, for a tensor network rather than for a sampler (Sec. IX).

The paper is organized as follows. Section II fixes the sampling primitive and states which reconstructions carry a shot budget and which are full-sector references. Section III proves the bound and shows that no bound on the captured weight alone can replace it; Sec. IV states what the subspace structure certifies; Sec. V measures where the certificate is informative; Sec. VI addresses a priori predictors; Secs. VII and VIII price the method and validate the references; and Sec. IX reports the device runs, the molecular suite and the relation to classical methods. The systems are the half-filled one-dimensional Hubbard model at $U/t = 8$, rings and open chains at $L = 4$ to 14, and nineteen molecular active spaces. Three different counts share the symbol $\mathcal{S}$ and are kept apart: the static ground-state support of Sec. VI, the dynamical support of Sec. VII, and the set of configurations the sampler returned, the $\mathcal{S}$ of Theorem 1. The proof of Theorem 1 is in Appendix A; the other proofs, full tables and extended discussions are in the Supplemental Material [19], cited as Sec. Sn.

II. ONE SAMPLING PRIMITIVE, FOUR CHANNELS: WHICH ARE RECONSTRUCTED AND WHICH ARE REFERENCES

The quantum processor is asked for one thing: computational-basis bitstrings of a time-evolved response seed. The subspace those bitstrings name, its diagonalization and every spectral function built on it are computed classically, and Table I states, channel by channel, which objects here were reconstructed from measurements and which are full-sector references.

A. Construction

For a Hamiltonian $\hat{H}$ with N -electron ground state $|0\rangle$ of energy E_0 , the single-particle spectral function has an electron-addition and an electron-removal branch,

$$A_p(\omega) = \sum_n |\langle n^{N+1} | \hat{c}_p^\dagger | 0 \rangle|^2 \delta(\omega - (E_n^{N+1} - E_0)) + \sum_m |\langle m^{N-1} | \hat{c}_p | 0 \rangle|^2 \delta(\omega - (E_0 - E_m^{N-1})), \quad (1)$$

energies measured from E_0 ; the chemical potential $\mu = \frac{1}{2}(E_0^{N+1} - E_0^{N-1})$ places the Fermi level mid-gap for the particle-hole-symmetric Hubbard model at half filling. The subspace is built in three steps. (1) Prepare the response seeds $|\phi^\pm\rangle = \hat{c}_p^\dagger |0\rangle, \hat{c}_p |0\rangle$ in the $(N\pm 1)$ sectors (every simulation here takes $|0\rangle$, the seeds and $e^{-i\hat{H}t_k}$ exact; how the non-unitary seed is prepared on a device is not specified by this work). (2) Time-evolve each, $|\phi^\pm(t_k)\rangle = e^{-i\hat{H}t_k} |\phi^\pm\rangle$, and measure computational-basis bitstrings at each of the $K+1$ snapshots; the union of the *distinct* bitstrings returned is the determinant subspace $\mathcal{S}^\pm$. (3) Form $\hat{H}_{\mathcal{S}^\pm} = P_{\mathcal{S}^\pm} \hat{H} P_{\mathcal{S}^\pm}$, diagonalize it as $\sum_m \tilde{E}_m^\pm |\tilde{m}^\pm\rangle \langle \tilde{m}^\pm|$, and evaluate the Lehmann sum inside $\mathcal{S}^\pm$. The retarded Green's function is then assembled entirely on the classical processor,

$$G_{pq}(\omega) = \sum_m \frac{\langle 0 | \hat{c}_p | \tilde{m}^+ \rangle \langle \tilde{m}^+ | \hat{c}_q^\dagger | 0 \rangle}{\omega - (\tilde{E}_m^+ - E_0) + i\eta} + \sum_m \frac{\langle 0 | \hat{c}_q^\dagger | \tilde{m}^- \rangle \langle \tilde{m}^- | \hat{c}_p | 0 \rangle}{\omega - (E_0 - \tilde{E}_m^-) + i\eta}, \quad (2)$$

whose imaginary part is the Lorentzian-broadened $A_{pq}(\omega)$ at resolution η . The seed selects the channel: charged seeds $\hat{c}_q^\dagger |0\rangle$ in the $(N\pm 1)$ sectors give $A(\omega)$ and, by Fourier transform of G_{pq} , $A(k, \omega)$; number-conserving density and spin seeds $\hat{n}_q |0\rangle, \hat{S}_q^z |0\rangle$ in the neutral N sector give $S(q, \omega)$ and $S^{zz}(q, \omega)$. Figure 1 schematizes the primitive—what one circuit and one readout can address; Table I states separately what was actually run for each channel here.

B. Relation to quantum subspace and Krylov methods

For equally spaced snapshots the span of the $K+1$ time-evolved seeds is the order- K Krylov space of the propagator $e^{-i\hat{H}\Delta t}$ [23], a real-time counterpart of the polynomial Krylov space $\text{span}\{\phi, \hat{H}\phi, \dots, \hat{H}^K \phi\}$, which block encoding builds exactly without time evolution [24]; diagonalizing $\hat{H}$ inside a subspace containing such a space is the route of quantum subspace expansion [25], filter diagonalization [26] and real-time quantum Krylov algorithms [27], whose error

TABLE I. **Provenance of every dynamical object in this paper: the subspace it was built on, and the measurements that built it.** “Full sector” means no subspace selection of any kind—the continued fraction runs on the whole sector at the stated recursion depth n_ℓ , so the row is a reference against which reconstructions are scored, not a reconstruction. “Observed bitstrings” means the subspace is the union of the distinct bitstrings actually returned by T draws split evenly over the $K+1$ snapshots, which is the protocol of Sec. II; one *stream* is one independent pseudorandom realization of that draw. “Born ranking” means the subspace is the top fraction of the sector under the accumulated Born weight of the evolved state, which is the $T \rightarrow \infty$ limit of the same protocol and carries no shot budget. Sector dimensions are exact; budgets are per channel and per stream.

Channel and system	Shown in	Determinant subspace	Shots T
$A(\omega)$; open Hubbard chain $L=6$, $U/t=8$, $\eta=0.15t$; dim = 300	Fig. 2	observed bitstrings, plus the largest-amplitude determinant of the probe, inserted and not drawn; $ S = 245$ of 300 in panel (a), 242.5 on average in panel (b)	5.2×10^4 per stream (4×10^3 at each of 13 snapshots, $K=12$), 8 streams
$A(\omega)$, orbital projected; 19 molecules at dissociation; dim = 2–300	Table S12	observed bitstrings, plus the largest-amplitude determinant of the probe, inserted and not drawn; the fraction $ S /D$ is the last column of that table	2.16×10^5 per molecule (8×10^3 at each of 27 snapshots, $K=26$), one stream
$A(\omega)$; N_2 along dissociation, CAS(6,6)	Fig. S6(a)	full CAS sector	none
$A(\omega)$; hardware, <code>ibm_fez</code> , $L=6$, $U/t=4$, $\eta=0.15t$; dim = 300	Fig. 9(a)	post-selected from device bitstrings read in reversed bit order, so every retained shot is a device error (Sec. IX A); $ S = 300$ of 300, the entire sector, so the reconstruction equals the exact $A(\omega)$ by coverage	3.5×10^5 : 5×10^4 on each of 7 circuits, $k=0$ (the $t=0$ reference) to $k=6$
$A(k, \omega)$; $L=12$, $\eta=0.18t$; dim = 731 808	Fig. 4	full sector, $n_\ell = 150$ per momentum	none
$A(k, \omega)$; $L=8$, $\eta=0.18t$; dim = 3920	Fig. 3(a,b)	Born ranking, 3332 of 3920 per channel, over 17 snapshots ($K=16$)	none: (a)–(b) are the $T \rightarrow \infty$ limit; at finite shots 5.2×10^6 per channel suffice and 2.6×10^6 do not (see text)
$A(\omega)$; $L=8$, $\eta=0.15t$; dim = 3920	Fig. 3(c)	observed bitstrings; sector fraction 0.835 ± 0.0025 at the largest budget	up to 2.60×10^6 over 19 snapshots ($K=18$), 8 streams
$S(q, \omega)$; $L=12$, $\eta=0.18t$; dim = 853 776	Fig. 7	full N -particle sector, $n_\ell = 200$	none
$S^{zz}(q, \omega)$; $L=12$, $\eta=0.10t$; dim = 853 776	Fig. 8	full N -particle sector, $n_\ell = 200$	none
Required sector fraction versus L ; $L=4$ –14, $\eta=0.15t$	Fig. 5(b,c)	both selections, plotted together: Born ranking (open squares) and observed bitstrings (filled circles)	measured: 2.4×10^4 , 2.6×10^5 , 3.1×10^6 , 1.9×10^7 at $L = 6, 8, 10, 12$ (Table IV, 4–6 seeds per size)

theory is now characterized [28]. What distinguishes the present method is that these states are used not to *estimate matrix elements* [29–34]—the step requiring Hadamard tests, compute–uncompute overlaps, Hamiltonian-moment or quantum equation-of-motion measurements on the device—but as a *sampling distribution* selecting a determinant subspace [35–41], extended since to embedded fragments and to auxiliary-field post-processing [42, 43], and here lifted to the $(N \pm 1)$ manifold and to a frequency-resolved observable. Apart from the concurrent Green’s-function extension of Ref. [1] (Sec. I), that lineage stops at energies, static correlators and, at most, discrete levels: the closest published results are a QSCI quasiparticle band structure returning discrete poles [44], a molecular spectrum built from a neutral-sector autocorrelation with classically projected long-time dynamics [45], and, from the measured-overlap side rather than the sampled one, a dynamical structure factor of a Kitaev spin liquid obtained by quantum subspace expansion [46], or a hybrid quantum–classical framework for many-fermion response and structure [47]. Closest on lattice models, Nogaki *et al.* [48] apply symmetry-adapted SQD to ground- and excited-state *energies* and a *static*

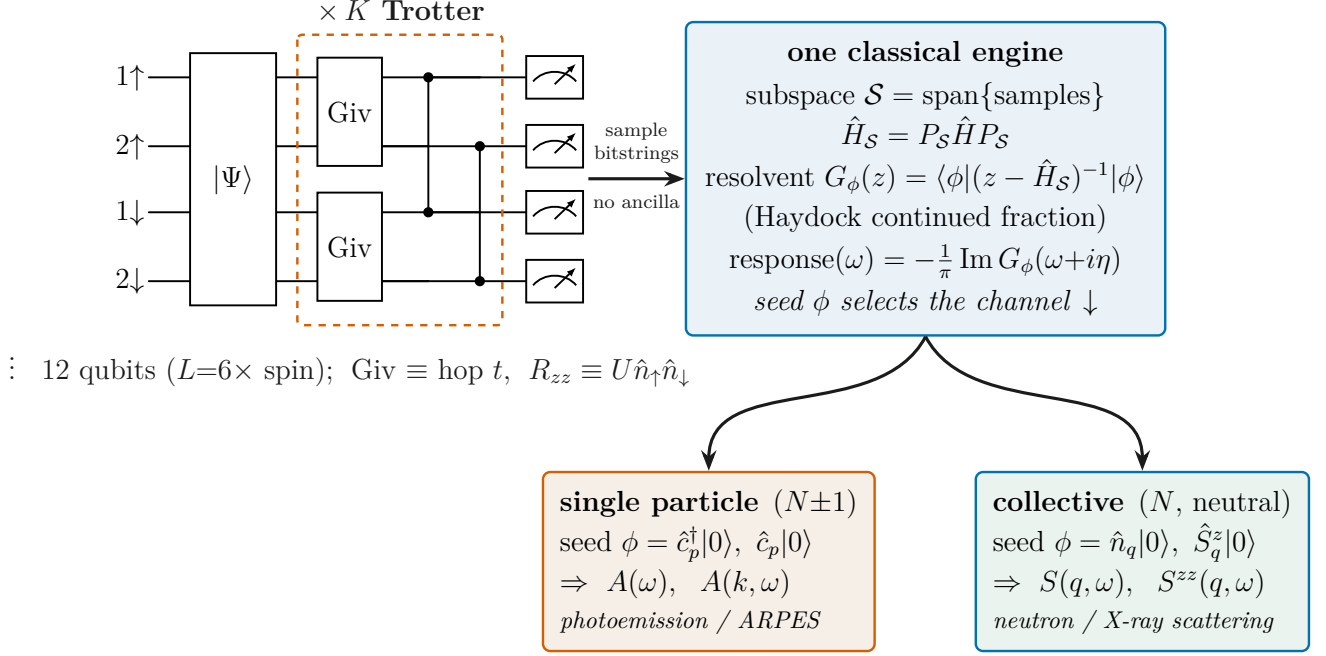


FIG. 1. **One sampling primitive, four dynamical responses.** An ancilla-free circuit—state preparation and a Trotter layer repeated K times (Givens rotations for the hopping t , native R_{zz} for the on-site $U \hat{n}_{\uparrow} \hat{n}_{\downarrow}$)—is measured in the computational basis; no Hadamard test or controlled unitary is run on the device. The sampled bitstrings define a sector-specific subspace $\mathcal{S}$ and feed a single classical engine—a Haydock continued-fraction resolvent [20] $G_{\phi}(z) = \langle \phi | (z - \hat{H}_{\mathcal{S}})^{-1} | \phi \rangle$, whose single-particle case ($\phi = \hat{c}_p^{\dagger} |0\rangle$) is the Green’s function of Eq. (2). The seed ϕ then selects the channel: particle addition/removal seeds $\hat{c}_p^{\dagger} |0\rangle, \hat{c}_p |0\rangle$ in the $(N \pm 1)$ sectors give the single-particle spectral functions $A(\omega)$ and $A(k, \omega)$ (photoemission/ARPES), while number-conserving density and spin seeds $\hat{n}_q |0\rangle, \hat{S}_q^z |0\rangle$ in the N sector give the dynamical structure factors $S(q, \omega)$ and $S^{zz}(q, \omega)$ (neutron [21] and resonant inelastic X-ray [22] scattering)—all from *one* sampling primitive, the seed selecting the sector and hence the subspace. Here $A(\omega)$ is reconstructed from sampled bitstrings (Fig. 2) and $A(k, \omega)$ on a Born-ranked subspace, the infinite-shot limit of the protocol (Fig. 3); the $A(k, \omega)$ of Fig. 4, $S(q, \omega)$ and $S^{zz}(q, \omega)$ are full-sector references, not reconstructed here (Table I).

superconducting pair correlation of the two-leg Hubbard ladder, and Wray *et al.* [49] measure how SQD converges on a variable-length cuprate chain. The readout is that of SQD, the circuits are the Trotterized Krylov evolutions of Krylov quantum diagonalization [50], sampled as in sample-based Krylov diagonalization [37, 51], and all dynamical information is assembled classically through Eq. (2) by one continued-fraction resolvent [20, 52].

C. What the subspace reconstruction is: Rayleigh–Ritz, not restriction

Step (3) is the definition the rest of this paper turns on. With $P_{\mathcal{S}}$ the orthogonal projector onto the span of the determinants in $\mathcal{S}$ and ϕ the seed,

$$A_{\mathcal{S}}(\omega) = \frac{\eta}{\pi} \left\| (\omega + E_0 + i\eta - H_{\mathcal{S}})^{-1} P_{\mathcal{S}} \phi \right\|^2, \quad H_{\mathcal{S}} = P_{\mathcal{S}} \hat{H} P_{\mathcal{S}}, \quad (3)$$

the spectral function of the *Rayleigh–Ritz operator* $P_{\mathcal{S}} \hat{H} P_{\mathcal{S}}$, whose poles are the Ritz values $\tilde{E}_m$ of Eq. (2)—not eigenvalues of $\hat{H}$ —and not the restriction of A to a subset of its own eigenpairs. The reproduction repository computes exactly this, at the small sizes by forming and diagonalizing the submatrix $\tilde{H}[\mathcal{S}, \mathcal{S}]$ and at the large ones by a matrix-free restricted matrix–vector product inside the same continued fraction; the consequence is the mechanism of this paper: *spectral weight the subspace retains is not kept in place, it is moved in ω* . Truncation does not delete Lehmann poles, it displaces every surviving one, so the weight a subspace captures cannot by itself bound how far the reconstruction has moved—which is why no weight-only bound survives contact with Eq. (3) (Sec. IIID), and why Theorem 1 carries a second term indexed on the coupling $P_{\mathcal{S}} \hat{H} Q_{\mathcal{S}}$ across the boundary of $\mathcal{S}$, with $Q_{\mathcal{S}}$ the complementary projector—exactly

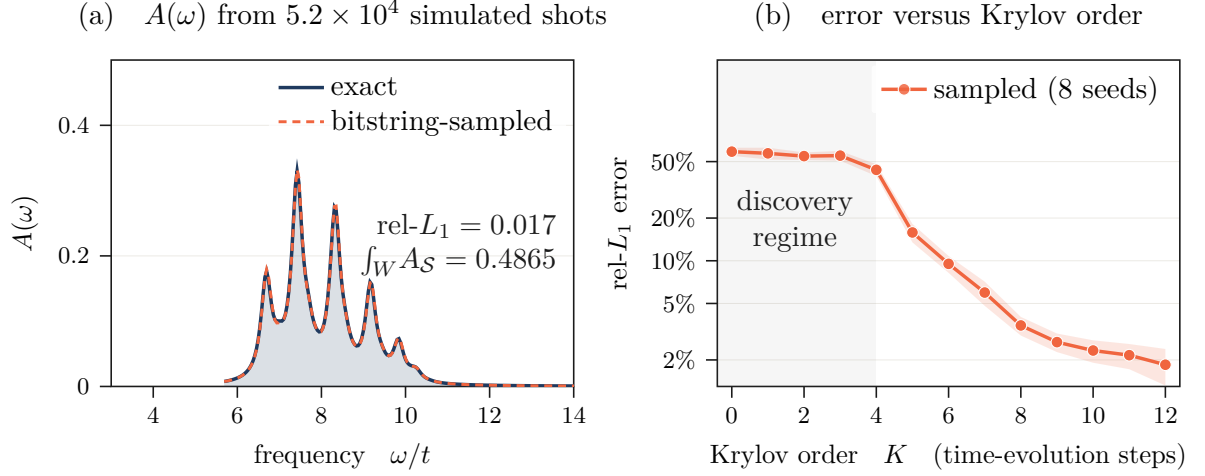


FIG. 2. **The method validated at $L = 6$, and its error against Krylov order.** (a) The single-particle addition branch $A^+(\omega)$, reconstructed from classically simulated shots (independent draws from the exact Born distribution of the evolved probe, with the probe’s largest-amplitude determinant inserted into each subspace rather than drawn), reproduces the exact one (open Hubbard chain, $L = 6$, $U/t = 8$; broadening $\eta = 0.15t$) to relative- L_1 0.017—the exact (filled) and reconstructed (dashed) curves are indistinguishable. The analytic addition-branch sum rule $\int A^+ d\omega = 1 - \langle n_p \rangle = 0.5$ is an identity of the ground state and not a check on the reconstruction; integrated over the plotted window the exact and reconstructed curves give 0.486588 and 0.486471, agreeing with each other to 2.4×10^{-4} relative and both falling 2.7% short of 0.5 because Lorentzian tails leave the window (Sec. S7). (b) Convergence of the relative- L_1 error with the Krylov order K (number of time-evolution steps sampled), seed-averaged over eight independent runs (band = standard deviation), $\eta = 0.15t$, so the budget is $T = 4000(K+1)$ and reaches 5.2×10^4 at $K = 12$. The error falls by a factor 32 across the thirteen points, from 0.59 at $K = 0$ to 0.0185 at $K = 12$, with one non-monotone step at $K = 3$ inside the discovery phase, while the mean subspace returned grows from 75 to 242 determinants of the 300-dimensional sector. No geometric rate is fitted to the window $4 \leq K \leq 9$ here (Sec. IV B): one is not implied by moment exactness, and inside that window the decay in K is confounded with the growth of $|S|$.

what Eq. (3) discards. Two consequences used later, that determinants outside $\text{supp}(\phi)$ still move the Ritz spectrum and that the error does not vanish on $\text{supp}(\phi)$ itself, are stated in Sec. S1 and measured in Sec. III D.

D. Provenance: what was reconstructed from shots, channel by channel

Table I gives, for every dynamical object in this paper, the subspace it was built on and the measurements that built it. The neutral channels $S(q, \omega)$ and $S^{zz}(q, \omega)$, the $A(k, \omega)$ of Fig. 4 and the N_2 series of Fig. S6 are full-sector references computed by continued fraction, not reconstructions. The Born ranking of Fig. 3(a,b) and of the published curve of Fig. 5 is the infinite-shot limit of the declared protocol, and its finite-shot replacement is measured with its budget: at $L = 8$, 2.60×10^6 shots over 19 snapshots reach sector fraction 0.835 ± 0.0025 at relative L_1 error $(4.81 \pm 0.27) \times 10^{-3}$ (Fig. 3(c)), and the literal configuration of Fig. 3(a,b) reaches the accuracy of panel (b) at 5.2×10^6 shots per channel and not at half that budget. The full declarations, including the branch sum rules and what is guaranteed in advance, are in Sec. S1.

III. TWO BUDGETS: THE MISSED WEIGHT AND THE BOUNDARY

The L_1 error of a spectral function reconstructed on a sampled subspace lies between the Born weight the subspace missed and that weight plus a boundary-leakage term, $\Lambda_P(\eta)/\eta$, computed from the Ritz pairs the reconstruction has already produced and one matrix–vector product per Ritz vector (Theorem 1). The leakage term does not need the exact A ; it does need the exact ground state (H4). The bound is two-sided, it holds for *any* orthogonal projector and therefore for any set of configurations a sampler might return, and its upper branch has two terms: the first is fixed by the captured weight and vanishes once the probe’s support is held; the second is not controlled by the weight, although it falls as the subspace grows. Section V measures where the resulting certificate is informative, Sec. VI shows that no invariant of the state can replace it, and Sec. VII prices both terms in shots.

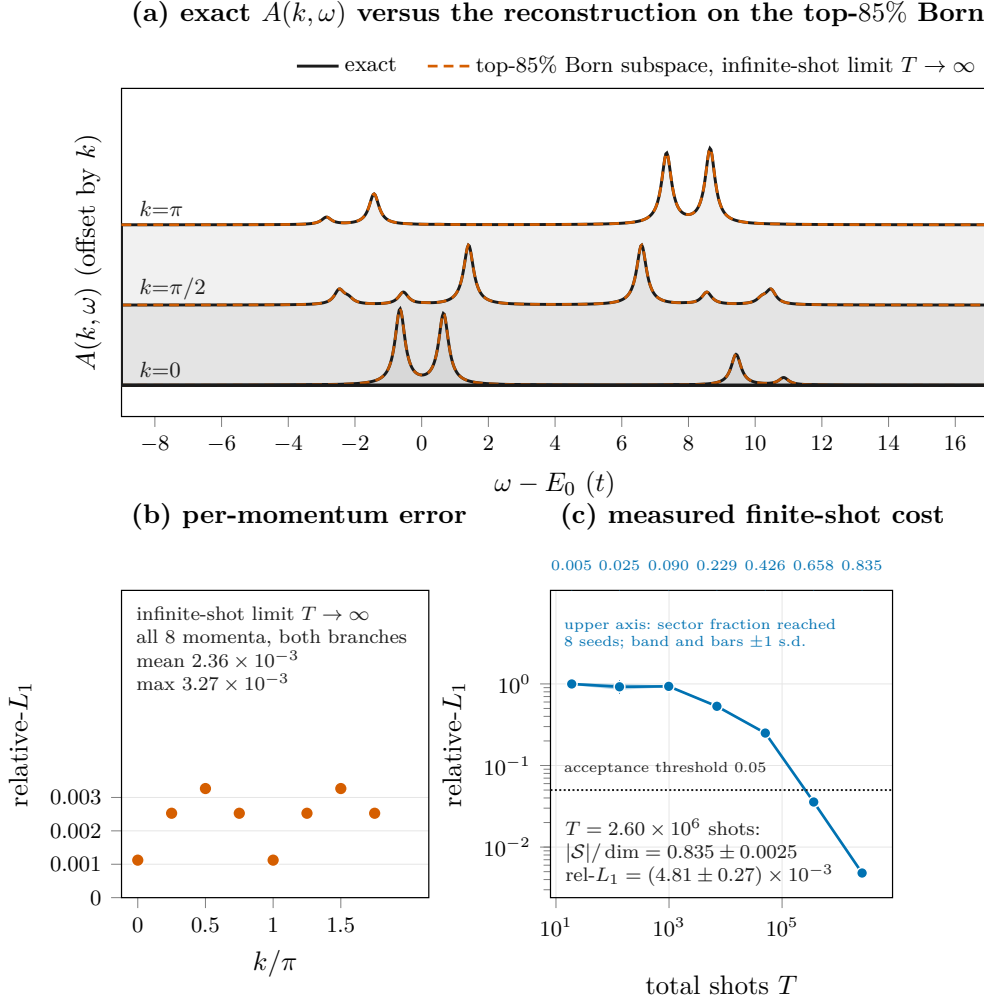


FIG. 3. **The momentum-resolved spectral function on a Born-ranked subspace, and what that subspace costs in shots.** (a) Exact $A(k, \omega)$ (solid, shaded) and the reconstruction obtained by diagonalizing the top-85% of the $(N \pm 1)$ sector under the time-averaged Born weight of the evolved state (dashed), at three momenta ($k = 0, \pi/2, \pi$; Hubbard ring, $L = 8$, $U/t = 8$, $\eta = 0.18t$, $K = 16$ time slices, $\delta t = 0.5$; curves offset vertically by k). Frequencies are measured from E_0 with no chemical-potential shift, so the removal branch lies below $\omega - E_0 = 1.41t$ and the addition branch above $6.59t$; the plotted and scored window, $\omega - E_0 \in [-9t, 17t]$, is symmetric about $\mu = U/2$ and holds both branches. The subspace is 3332 of the 3920 determinants of each channel. The ranking is taken on the *exact* Born distribution, so panels (a) and (b) are the **infinite-shot limit** $T \rightarrow \infty$ of the protocol, not a finite-shot experiment; no shot budget is attached to them. (b) Relative- L_1 error of that reconstruction at all 8 physical momenta, addition and removal branch summed over that window: mean 2.36×10^{-3} , maximum 3.27×10^{-3} . (c) What a finite shot budget buys, measured. Same ring and the same sector dimension ($L = 8$, $\dim(N+1) = 3920$) but a separate run: local seed $c_{0\uparrow}^\dagger |\Psi_0\rangle$, $\eta = 0.15t$, $K = 18$ time slices, T simulated shots split evenly over the 19 snapshots, subspace = the support observed in those T shots, 8 independent seeds. Markers are the seed mean; the shaded band and the error bars are ± 1 s.d. over the 8 seeds (at most 8.1% of the mean at any budget, thinner than the line). Upper axis: the sector fraction $|S|/\dim$ reached at that budget; the dotted line is the acceptance threshold $\text{rel-}L_1 = 0.05$ used by the resource scan. At the largest budget measured, $T = 2.60 \times 10^6$ shots, the observed subspace covers 0.835 ± 0.0025 of the sector and the reconstruction reaches $\text{rel-}L_1 = (4.81 \pm 0.27) \times 10^{-3}$. **The shot budget of (a)–(b), measured.** Panel (c) is at another resolution and seed, so the configuration of (a)–(b) was run at finite shots separately: same $\eta = 0.18t$ and $K = 16$, all 8 momenta, both branches, the subspace being the determinants observed in T multinomial draws per channel, 4 seeds, 32 (momentum, seed) pairs, both branches summed in each, scored on the window of (a)–(b); $\pm$ is the s.d. over the pairs. At $T = 5.2 \times 10^6$ shots per channel— 8.3×10^7 over the 16 channels—the reconstruction reaches $\text{rel-}L_1 = (2.16 \pm 0.23) \times 10^{-3}$, maximum 2.46×10^{-3} : panel (b)’s accuracy. It reaches it on a *larger* subspace, covering 0.882 of the addition sector against the 0.850 of (a)–(b), so the budget buys that accuracy together with three extra points of fraction, and not the accuracy at the fraction this figure announces. At half the budget, $T = 2.6 \times 10^6$, the observed subspace covers 0.839—just under the 0.850 of (a)–(b)—and the error is $(6.16 \pm 0.42) \times 10^{-3}$, a factor 2.61 above the mean of (b). That factor, and not the 2.05 transferred from Table S6 ($L = 8$, $\eta = 0.15t$, $|S| = 2180$), is the finite-shot penalty at this configuration; the transferred value understated it by 21% ($1 - 2.05/2.61$).

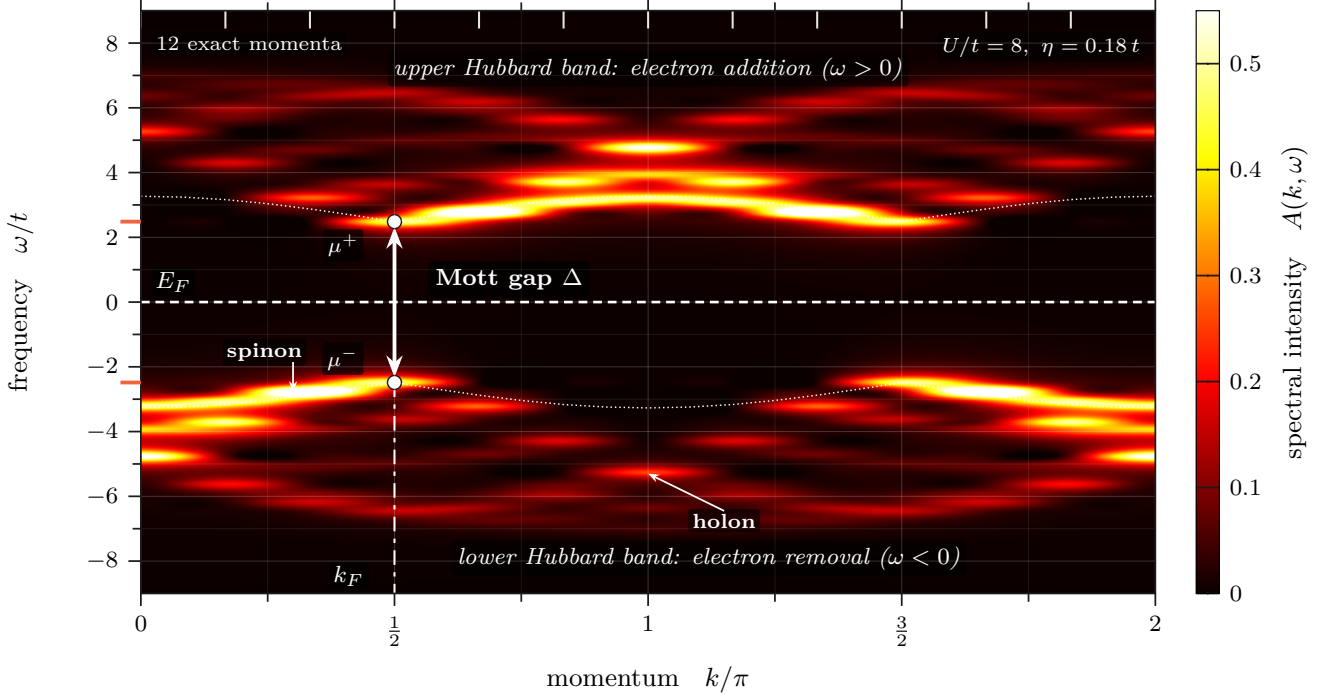


FIG. 4. Momentum-resolved spectral function: the dynamical target. The single-particle spectral function $A(k, \omega)$ of the half-filled periodic Hubbard ring ($U/t = 8$, $L = 12$, Lorentzian broadening $\eta = 0.18t$), computed from the resolvent $\langle \phi | (z - \hat{H})^{-1} | \phi \rangle$ by a Haydock continued fraction [20] of depth $n_\ell = 150$ over the full $(N \pm 1)$ sectors—a finite-depth continued fraction, that is a 150-node Gauss quadrature of the spectral measure [52] and not the Lehmann sum (Sec. S7)—the frequency- and momentum-resolved observable that sample-based diagonalization targets but its energy-only output cannot itself deliver. The two dispersive Hubbard bands—electron removal (lower, $\omega < 0$; photoemission) and electron addition (upper, $\omega > 0$; inverse photoemission)—are separated by the correlation-driven *Mott gap* $\Delta = \mu^+ - \mu^- = 4.97t$ between the highest occupied state (μ^- , lower edge) and the lowest unoccupied state (μ^+ , upper edge), obtained from the exact $(N \pm 1)$ ground-state energies (ticks at $\pm 2.484t$); the broadened spectral peaks of the plotted $A(k_F, \omega)$ sit slightly wider, at $\pm 2.49t$ (a $4.99t$ splitting), the $\sim 0.02t$ excess being the $\eta = 0.18t$ Lorentzian broadening; the thermodynamic Lieb–Wu value [53] is $4.68t$, the $L = 12$ excess being finite-size. The gap is *direct* at the Fermi momentum $k_F = \pi/2$. Particle–hole symmetry places the Fermi level at mid-gap ($\omega = 0$; here ω is measured from the chemical potential $\mu = U/2$, whereas Figs. 2, 3 and 9 measure it from E_0), verified to double-precision round-off, and each momentum obeys the analytic spectral sum rule $\int A(k, \omega) d\omega = 1$ (the plotted broadened grid recovers it to $\sim 2\%$ within the $\pm 9t$ window). In the thermodynamic limit the lower Hubbard band is spin–charge separated; at $L = 12$ the plot is compatible with a narrow spinon-like branch (bandwidth $\pi J/2 = 0.79t$ [54], $J = 4t^2/U$; dotted, drawn with its mirror image at $\omega > 0$, validated against the exact peak dispersion to $< 0.1t$) that rides a broad holon (charge, $\sim 4t$) band, as expected from the Bethe-ansatz solution [53], and is not offered as evidence of it. At $L = 12$ that structure is three to seven poles per momentum (Table S7), and the in-panel labels “spinon” and “holon” name those branches, not thermodynamic-limit excitations. The momentum axis is a periodic cubic-spline interpolation of the 12 physical momenta (ticks, top; the in-panel “12 exact momenta” counts these allowed momenta, not an exact spectrum). Faithfully *sampling* this full momentum-resolved observable—as opposed to the ground-state energy—requires a large fraction of the $(N+1)$ sector (Fig. 5), the cost quantified in Sec. VII.

A. The objects

Let ϕ be the probe— $\hat{c}_p^\dagger |\Psi_0\rangle$ or $\hat{c}_p |\Psi_0\rangle$ for the charged channels, $\hat{n}_q |\Psi_0\rangle$ or $\hat{S}_q^z |\Psi_0\rangle$ for the neutral ones— H the Hamiltonian restricted to the sector of ϕ , $(N \pm 1)$ or N , and E_0 the ground-state energy of the N -electron sector. Write $z = \omega + E_0 + i\eta$ and $R(z) = (z - H)^{-1}$. The η -broadened spectral function of Eq. (2) is

$$A(\omega) = \frac{\eta}{\pi} \|R(z)\phi\|^2, \quad \text{so that} \quad \int_{\mathbb{R}} A(\omega) d\omega = \|\phi\|^2 \quad (4)$$

for every $\eta > 0$, the sum rule holding because the Poisson kernel integrates to one on each Lehmann pole. Let P be an orthogonal projector on the sector, $Q = I - P$, and

$$\phi_P = P\phi, \quad \phi_Q = Q\phi, \quad w_P = \frac{\|\phi_P\|^2}{\|\phi\|^2}, \quad H_P = PHP, \quad R_P(z) = (z - H_P)^{-1}|_{\text{ran } P}, \quad (5)$$

with Ritz pairs $H_P|\tilde{m}\rangle = \tilde{E}_m|\tilde{m}\rangle$ and $c_m = \langle \tilde{m}|\phi\rangle$. The reconstruction of Eq. (3) is $A_P(\omega) = (\eta/\pi)\|R_P(z)\phi_P\|^2$, and it obeys the same sum rule with $\|\phi_P\|^2$ on the right. When $P = P_S$ is diagonal in the determinant basis—the only case a device produces—we write S for the configuration set and w_S, Λ_S for the quantities below.

The magnitude the theorem is indexed on is the Hamiltonian coupling that crosses the boundary of the subspace, averaged over the resolution window rather than over the spectrum. With $g_m = QH|\tilde{m}\rangle$,

$$\Lambda_P(\eta)^2 := \frac{\eta}{\pi} \int_{\mathbb{R}} \|QH R_P(z)\phi_P\|^2 d\omega = \sum_{m,m'} c_m \bar{c}_{m'} \langle g_{m'}, g_m \rangle \frac{2i\eta}{(\tilde{E}_{m'} - \tilde{E}_m) + 2i\eta}, \quad (6)$$

the second equality being exact and proved in Appendix A; the sum is real, the (m, m') and (m', m) terms being complex conjugates. Two features of Eq. (6) carry the practical content of this paper. It is assembled from objects the reconstruction has already produced—the Ritz vectors and their coefficients—plus one matrix–vector product $H|\tilde{m}\rangle$ per Ritz vector, so evaluating it costs a small multiple of the diagonalization that produced A_P and *does not require* A . And it vanishes exactly when $\text{span}(S)$ contains $\mathcal{K}(H, \phi_P)$, the Krylov space of the retained part of the probe: $QH R_P(z)\phi_P$ vanishes for every z exactly when $\mathcal{K}(H_P, \phi_P)$ is invariant under H , and it then coincides with $\mathcal{K}(H, \phi_P)$. For a subspace that holds the whole probe, $w_P = 1$, the condition is that $\text{span}(S)$ contain $\mathcal{K}(H, \phi)$ itself, which for the local probes here, in the site basis, was measured at $L = 6$ –14 to mean every reflection-allowed determinant of the sector (Sec. V E).

B. The bound

Theorem 1 (a posteriori leakage certificate). *Let $H = H^\dagger$ act on the sector of ϕ , let $\eta > 0$, let P be any orthogonal projector on that sector, and let A, A_P, w_P and $\Lambda_P(\eta)$ be as in Eqs. (4)–(6). Then*

$$\|\phi\|^2(1 - w_P) \leq \|A - A_P\|_1 \leq \min \left\{ \|\phi\|^2(1 + w_P), (\|\phi\| + \|\phi_P\|) \left(\underbrace{\|\phi_Q\|}_{\text{missed weight}} + \underbrace{\Lambda_P(\eta)/\eta}_{\text{boundary leakage}} \right) \right\}, \quad (7)$$

equivalently, dividing by $\|\phi\|^2$ and writing $\hat{\Lambda}_P = \Lambda_P/\|\phi\|$,

$$1 - w_P \leq \frac{\|A - A_P\|_1}{\|\phi\|^2} \leq \min \left\{ 1 + w_P, (1 + \sqrt{w_P})(\sqrt{1 - w_P} + \hat{\Lambda}_P(\eta)/\eta) \right\}.$$

Four hypotheses carry it, each a property of the *pipeline* that produced A_P rather than of the inequality: (H1) A and A_P are referred to the same E_0 ; (H2) the L_1 norm is taken over the whole line and normalized by $\|\phi\|^2$; (H3) A is the exact spectral function of the full sector; (H4) the probe ϕ is built from the exact N -electron ground state. Each is load-bearing and each is checked against this paper’s own pipeline in Sec. S11. A shot-based pipeline breaks (H1), and in general (H4) with it (Sec. III C). If only E_0 is estimated on a sampled subspace, the probe being kept exact, and A_P is referred to $E_0 + \delta$, the bound is guaranteed only once an additive $\|\phi_P\|^2(4/\pi) \arctan(|\delta|/2\eta)$ is added to it, a term the certificate does not control; at $L = 8$, with E_0 from 60% of the N sector, that term moves the bound by +0.20% while the measured error grows by a factor 16 (Sec. S11). The proof, from the Feshbach–Löwdin split of the resolvent (Eq. (A1)) and Cauchy–Schwarz in ω , is in Appendix A.

C. Reading the two branches

What the weight does not control. The first summand of the leakage branch, $\|\phi_Q\| = \|\phi\|\sqrt{1 - w_P}$, is the Born weight of the probe the subspace failed to hold; Table I prices it channel by channel. The second, $\Lambda_P(\eta)/\eta$, reads the off-diagonal block PHQ , and the captured weight does not control it: along the sampler’s ranking it falls as S grows (Sec. V), but a subspace can hold the whole probe and still leak, and Sec. III D measures how much.

A certificate, not an error estimate. The gap between the bound and the measured error is large, and it is not the accident of a constant. At *fixed* P the looseness is stable—within $\pm 17\%$ ($L = 6$) and $\pm 25\%$ ($L = 8$) over a decade in η —but across subspaces it varies by a factor 51, and at the threshold where the leakage branch first improves on the trivial one it is 3×10^2 to 2.4×10^3 (Sec. S4). Equation (7) is therefore an exclusion, not an error estimate; Sec. V reports what it excludes, and at what fraction of the sector.

Neither term is free of the ground state. Given the probe, $\Lambda_P(\eta)$ is computable from H , $\text{ran } P$, ϕ_P and the Ritz vectors alone, without A ; for $P = P_S$, ϕ_P consists of the amplitudes of Ψ_0 on the determinants that the probe operator maps into S , which the reconstruction A_P uses as well. The weight term, $\|\phi_Q\| = \|\phi\|\sqrt{1 - w_P}$, needs the full norm of the probe as well as its retained part ($\|\phi\|^2 = 1 - \langle \hat{n}_p \rangle$ for electron addition). Under (H4) both terms are therefore evaluable without A , but neither without Ψ_0 . If Ψ_0 is itself a sampled-subspace estimate, (H4) fails: the probe is another vector, and the theorem bounds how far the reconstruction lies from that vector’s own spectral function, not from A ; w_P estimated against it is biased in the optimistic direction, and Λ_P is evaluated for the wrong probe, with an error of unknown sign. Nothing the pipeline computes then bounds the distance from its reconstruction to A , and no certificate in this work has been evaluated with such a probe. The complete bound is therefore *not* evaluable by a user who does not already have the ground state. Its regime is the one in which Sec. V evaluates it: exact-diagonalization sizes, where the exact ground state is available and the bound is applied, subspace by subspace, to the configuration sets a sampled pipeline returns; there Sec. VD gives the conservative substitution for w_P that makes the decision itself a posteriori, given the exact probe.

What is new, and the prior art. What this section adds is the conjunction of three things: a closed form for $\Lambda_P(\eta)$ from the Ritz pairs on a subspace of *coordinates*, with no Krylov structure, invariance or gap assumption; a bound two-sided on the same object; and a measured calibration of where it crosses the trivial bound (Sec. VD). Each step of the proof is standard. The split is Löwdin partitioning [55], the algebra of the Feshbach projection [56]; Rayleigh–Ritz and Krylov error analysis, for resolvents and matrix functions [52, 57] and for eigenpairs [28, 58–60], with Paige’s loss-of-orthogonality analysis and the restart convergence theory of Beattie, Embree and Sorensen [61, 62], contains every rung of its steps, and the integration step is the reduced-basis bound on a Galerkin projection with inf-sup constant η [63, 64]. A posteriori estimates for Krylov approximations to matrix functions are due to Frommer and Simoncini, Jia and Lv, and Frommer and Schweitzer [65–67], with Lanczos and stochastic-Lanczos-quadrature versions [68–70]; on quantum Krylov, Oliveira, Ziemas and Glaser assess reliability without knowledge of the true spectrum [71]; and a response-targeted selection criterion for classical selected configuration interaction is Reinholdt, Kjellgren and Kongsted’s [72]. The priority for the a posteriori framing is theirs. At the fractions sampling reaches here the certificate is vacuous (Sec. V, which also measures where it is informative); the full comparison with that literature is in Sec. S2.

D. The captured weight alone does not bound the error

The certificate one would want for a sampled reconstruction indexes the error on the captured weight alone,

$$\|A - A_S\|_1 \leq 2\|\phi\|^2(1 - w_S) + C\rho^{-K_S}, \quad (8)$$

with $w_S = \|P_S\phi\|^2/\|\phi\|^2$ and K_S the largest Krylov order contained in $\text{span}(S)$ (we set $K_S = -1$ when $\phi \notin \text{span}(S)$). It is false, for every constant C and every ρ , and it fails hardest in exactly the regime a sampling experiment aims for.

A two-level counterexample forces the trivial constant. Take $H = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}$, $\phi = e_0$, $S = \{0\}$, $E_0 = 0$. Then $P_S\phi = \phi$, so $w_S = 1$ exactly and the weight term vanishes identically; and $\mathcal{K}_0 = \text{span}\{\phi\} \subseteq \text{span}(S)$ while $\mathcal{K}_1 \not\subseteq \text{span}(S)$, so $K_S = 0$ exactly and the second term is the bare constant C . The exact A is a pair of Lorentzians of weight $\frac{1}{2}$ at $\pm g$; the reconstruction, built on $H_S = P_S H P_S = 0$, is one Lorentzian of weight 1 at the origin, so $\|A - A_S\|_1 \rightarrow 2\|\phi\|^2$ as $g/\eta \rightarrow \infty$. Since g is free, Eq. (8) requires $C \geq 2\|\phi\|^2$: false if C is a constant, vacuous if it may depend on the instance. The same holds at every Krylov order: on a chain of $K + 2$ sites with $\phi = e_0$ and S its first $K + 1$ sites, $w_S = 1$ and $K_S = K$, and the exact and reconstructed spectra have strictly interlacing poles, so the same limit requires $C\rho^{-K} \geq 2\|\phi\|^2$ for every K , which no fixed C satisfies when $\rho > 1$. With $w_S < 1$ strictly the weight term alone fails without bound: in a three-level family its violation diverges as ε^{-2} in the amplitude ε of the discarded component (Sec. S9). By contrast $\|A - A_S\|_1 \leq \|\phi\|^2(1 + w_S)$ holds for every S under no hypothesis at all, A and A_S being nonnegative with known integrals: that trivial bound is the only surviving member of the family of Eq. (8), and Theorem 1 keeps it explicitly, as the first branch of its minimum.

The violation is systematic, and worst where the reconstruction is best. In a pooled sweep of 606 subspaces from five Hamiltonian families the weight term of Eq. (8) alone—its $C = 0$ member, $\|A - A_S\|_1 \leq 2\|\phi\|^2(1 - w_S)$, the constant having been disposed of analytically above—fails on 469, and on all 363 with $w_S \geq 0.99$, the mildest by a factor 2.00 (Fig. S5, Sec. S9). In the configuration of Fig. 3 its median violation factor rises from 18.5 to 29.9 between

sector fractions 0.50 and 0.95 while the measured error falls by three orders of magnitude. The sweep and its declared limits are in Sec. S2; the conclusion does not depend on it, the counterexample being analytic.

Why it fails. The reconstruction is Rayleigh–Ritz on $H_S = P_S H P_S$, not a restriction of A to a subset of its eigenpairs: weight the subspace keeps is *displaced* in ω , not discarded, and the two-level example is the extreme case. What the integrals do give is the lower bound $\|\phi\|^2(1 - w_S) \leq \|A - A_S\|_1$, with 0 violations across the 606 subspaces, the left-hand half of Theorem 1.

The two terms are mutually exclusive, and one is off at every operating point here. S is a set of determinants, so P_S is diagonal in the working basis and $P_S \phi = \phi \iff \text{supp}(\phi) \subseteq S \iff w_S = 1$, while $K_S \geq 0$ means precisely $\phi = H^0 \phi \in \text{span}(S)$. Hence the exclusion lemma

$$K_S \geq 0 \iff w_S = 1, \quad (9)$$

verified with 0 discrepancies across the 378 deposited subspaces, of which 357 have $K_S = -1$ (Sec. S9). The two terms of Eq. (8) are therefore never both informative: either $w_S = 1$ and the whole bound is the second term—the two-level counterexample—or $K_S = -1$ and $C\rho^{-K_S} = C\rho$, an undefined constant with no decay in it. Counting settles which branch applies here. The seed is $\phi = \hat{c}_{0\uparrow}^\dagger |\Psi_0\rangle$ at half filling, so $\text{supp}(\phi)$ lies in the set of $(N+1)$ determinants with orbital $0\uparrow$ occupied, a fraction $n_\uparrow/L = \frac{1}{2} + \frac{1}{L}$ of the sector, and fills it except where symmetry zeroes an amplitude of Ψ_0 . By Eq. (9), $K_S \geq 0$ demands $|S| \geq |\text{supp}(\phi)|$, a floor of 0.571 at $L = 8$ (2237 of 3920), 0.600 at $L = 10$, and above 0.54 at $L = 12$ and 14 at any amplitude threshold from 10^{-12} to 10^{-10} , against the sector fractions 0.5561, 0.3625, 0.1700, 0.0800 at which the resource scan of Sec. VII operates (Fig. 5; the coarser replication grid of Table S5 reads 0.180 at $L = 12$ for the same operating point). From $L = 8$ on, every operating point of that scan lies below its own floor, so $K_S = -1$ by counting alone, with no numerics, no ranking and no protocol, and any point below the same line inherits the verdict (Sec. S9). Theorem 1 carries no such bookkeeping: its two terms are defined and simultaneously active for every S , neither is free of Ψ_0 (Sec. III C), and $\Lambda_S(\eta)$ is evaluated, subspace by subspace, in Sec. V.

The magnitude, not the constant. Two measurements at the same, exactly unit, captured weight show that Eq. (8) was indexed not on a bad constant but on the wrong quantity. First, let $S^* = \text{supp}(\phi)$, the smallest determinant subspace of unit captured weight—fixed by the seed, with no ranking, no tie-breaking and no zero-amplitude padding—of size $(\frac{1}{2} + \frac{1}{L})D$ at $L = 6$ and $0.571D$ at $L = 8$, where 426 of the 4900 ground-state amplitudes vanish by symmetry. Its weight term is identically zero, and its relative L_1 error is 0.43 at $L = 6$ and 0.35 at $L = 8$ on the periodic rings at $\eta = 0.18t$, rising to 1.15 and 0.73 at $\eta = 0.05t$, against a hard ceiling of $1 + w_S = 2$. Second, S^* is the $K = 0$ rung of the Krylov ladder $S_K = \bigcup_{a \leq K} \text{supp}(H^a \phi)$, every rung of which contains ϕ and so carries $w_S = 1$ identically, while its moment reach grows: S_K reproduces the exact spectral moments through order $2K+1$ —to the 10^{-15} level, and in general failing at $2K+2$, as on the open chain of Table II at both rungs that do not fill the sector—whereas coordinate subspaces of the same size drawn at random miss already μ_0 , in all of 300 draws (Sec. IV). Along a family of constant, exactly unit captured weight the error therefore runs from order unity to moment-exact through order $2K+1$. The captured weight, identical on every rung, does not order them.

What does is elementary. Writing μ_j, μ_j^S for the moments of the unbroadened measures of Sec. IV, every determinant subspace with $w_S = 1$ reproduces μ_0 and μ_1 exactly, and the first moment that can fail is μ_2 , with deficit

$$\mu_2 - \mu_2^S = \|Q_S H P_S \phi\|^2, \quad Q_S = I - P_S, \quad (10)$$

the squared coupling of the Hamiltonian across the boundary of S (E_0 drops out because $Q_S \phi = 0$), zero exactly when S also holds $\text{supp}(H\phi)$ —the $K_S \geq 1$ condition. That boundary coupling is the magnitude the replacement is indexed on: it controls the leakage term through $\Lambda_S(\eta) \leq \|P_S H Q_S\| \|\phi_S\|$, and Theorem 1 averages it over the resolution window rather than over the spectrum. The captured weight says what the subspace *contains*; the error is set by what the Hamiltonian does to the seed at the subspace’s *edge*. Both halves of this paper follow from that displacement: Sec. V measures where the boundary term is small enough for the certificate to say something, and Sec. VI shows why no orbital-rotation invariant can predict it usefully on every state in advance.

IV. MOMENT EXACTNESS ON A KRYLOV-ADEQUATE SUBSPACE

A subspace that contains the order- K Krylov space of the probe reproduces its spectral moments exactly through order $2K+1$ (Theorem 2); the statement is classical, and what is measured here is its exact reach on the coordinate subspaces this work builds. That reach is sharp: on the construction of Table II the moments part company at order $2K+2$, while coordinate subspaces of the same size drawn at random fail already at the zeroth moment. The guarantee is structural: it depends on which determinants the subspace holds, not on how many, and it is decidable from the subspace and the probe, without the exact spectral function. It is not an error bound, and it does not imply convergence in K (Sec. IV B).

TABLE II. **Moment exactness and its exact reach.** Open Hubbard chain, $L = 6$, $U/t = 8$; seed $|\phi\rangle = \hat{c}_{0\uparrow}^\dagger|0\rangle$; $(N+1, S_z = +\frac{1}{2})$ sector of dimension $D = 300$; $\mathcal{S} = \bigcup_{a \leq K} \text{supp}(H^a \phi)$. The largest Krylov order inside $\text{span}(\mathcal{S})$, $K_{\mathcal{S}}$, and the captured Born weight $w_{\mathcal{S}}$ are measured, not assumed. ϵ_j is the *relative* discrepancy of the j -th moment, quoted as the largest value over six equivalent implementations of the same identity (the deposited generator, `src/moments_table.py`, regenerates the roundoff residues to the same order and every other entry to the precision printed); the absolute discrepancy at order $2K+2$ follows in parentheses. Only relative residues are quoted below that order, because μ_j spans five decades between $j = 0$ and $j = 6$ and the absolute roundoff residue is route-dependent (Sec. S10.2). At $K = 2$ the ladder has filled the sector, so $\text{span}(\mathcal{S})$ carries every eigenvector of nonzero Lehmann weight, $A_{\mathcal{S}} = A$ identically, and there is no order- $2K+2$ failure to exhibit; that row is kept to mark where the ladder terminates, and its single entry is the roundoff level of the moments.

K	$ \mathcal{S} $	$ \mathcal{S} /D$	$w_{\mathcal{S}}$	$K_{\mathcal{S}}$	$\max_{j \leq 2K+1} \epsilon_j$	$\longrightarrow$	ϵ_{2K+2} (absolute)
0	200	0.667	1	0	1.8×10^{-15}	$\longrightarrow$	9.5×10^{-4} (3.10×10^{-2})
1	280	0.933	1	1	3.2×10^{-15}	$\longrightarrow$	6.0×10^{-8} (1.34×10^{-4})
2	300	1.000	1	full sector	7.5×10^{-15}	$\longrightarrow$	no failure: $A_{\mathcal{S}} = A$

A. Statement, proof and measured reach

Write $\mathcal{K}_K = \text{span}\{\phi, H\phi, \dots, H^K\phi\}$ for the order- K Krylov space of the seed $|\phi\rangle = \hat{c}_p^\dagger|0\rangle$, and let $d\mu$ and $d\mu_{\mathcal{S}}$ be the unbroadened Stieltjes measures whose Lorentzian convolutions are the exact A and the reconstruction $A_{\mathcal{S}}$, the Lehmann sum evaluated inside $\text{span}(\mathcal{S})$ as in Sec. II (Sec. S10).

Theorem 2 (Moment exactness). *Let P be the orthogonal projector onto $\text{span}(\mathcal{S})$ and $H_{\mathcal{S}} = PHP$. If $\text{span}(\mathcal{S}) \supseteq \mathcal{K}_K$, then*

$$\int \omega^j d\mu_{\mathcal{S}}(\omega) = \int \omega^j d\mu(\omega) = \langle \phi | (H - E_0)^j | \phi \rangle, \quad 0 \leq j \leq 2K + 1. \quad (11)$$

Proof. Because $\mathcal{K}_K \subseteq \text{span}(\mathcal{S})$ we have $PH^a\phi = H^a\phi$ for $0 \leq a \leq K$, whence by induction $H_{\mathcal{S}}^b\phi = PH^b\phi$ for $0 \leq b \leq K + 1$. Then for any $a \leq K$ and $b \leq K + 1$, using only $P = P^\dagger = P^2$ and $H = H^\dagger$,

$$\langle \phi | H_{\mathcal{S}}^{a+b} | \phi \rangle = \langle H_{\mathcal{S}}^a \phi | H_{\mathcal{S}}^b \phi \rangle = \langle H^a \phi | PH^b \phi \rangle = \langle PH^a \phi | H^b \phi \rangle = \langle \phi | H^{a+b} | \phi \rangle, \quad (12)$$

and every $j \leq 2K + 1$ is such an $a + b$. Both measures are shifted by the same E_0 , so the identity survives term by term in the binomial expansion of $(H - E_0)^j$. $\square$

The proof is three lines of Rayleigh–Ritz, and the statement is the moment-matching property of Krylov projections that underlies Lanczos–Gauss quadrature [52]. The proof does not identify $A_{\mathcal{S}}$ with a Gaussian-quadrature rule, which is not the object built here: every reconstruction diagonalizes $P_{\mathcal{S}}HP_{\mathcal{S}}$ on a coordinate subspace, so $d\mu_{\mathcal{S}}$ carries $|\mathcal{S}|$ nodes, not $K + 1$ (Sec. S3).

Table II measures Eq. (11) on the construction this work uses—the union of the determinant supports of $\{\phi, H\phi, \dots, H^K\phi\}$ —for the open Hubbard chain at $L = 6$, $U/t = 8$, in the $(N+1, S_z = +\frac{1}{2})$ sector of dimension $D = 300$. Through order $2K + 1$ the moments agree to a relative 10^{-14} ; at order $2K+2$ they part company by eleven orders of magnitude at $K = 0$ and seven at $K = 1$. The order $2K + 1$ is the exact reach of the construction, not merely a sufficient claim.

A negative control fixes what the table is about. Over 300 coordinate subspaces of the *same* size $|\mathcal{S}| = 200$ —two thirds of the sector—drawn uniformly from it, every single draw fails already at the zeroth moment, against 4×10^{-16} for the Krylov-support subspace of identical size (Sec. S10.2). What Theorem 2 certifies is *which* determinants the subspace holds, not *how many*.

The ladder terminates once the subspace contains the cyclic subspace generated by ϕ , and there the reconstruction is exact: at $L = 6$ that is $K = 2$ on the open chain, and already $K = 1$ on the periodic rings of Sec. VII, where it stalls at 296 of 300 determinants with $A_{\mathcal{S}} = A$ to 10^{-14} (Sec. S10.2). These three rows are the whole validation, and they say nothing about the large- K behavior of $\|A - A_{\mathcal{K}_K}\|_1$, which is the open question of Conjecture 1.

B. What moment exactness does not buy

Moment exactness is a list of $2K + 2$ linear identities, and the L_1 distance between two Lorentzian-broadened measures is not a finite linear functional of them, so geometric convergence in the Krylov order does not follow from it.

Conjecture 1 (Geometric convergence in the Krylov order). *Fix $\eta > 0$ and let $A_{\mathcal{K}_K}$ be the Rayleigh–Ritz reconstruction on $\mathcal{K}_K$. Are there $C(\eta) < \infty$ and $\rho(\eta) > 1$, uniform over a class of Hamiltonians containing the models studied here, such that $\|A - A_{\mathcal{K}_K}\|_1 \leq C(\eta) \rho(\eta)^{-K}$ for every K ?*

If the class contains the two-level family of Sec. III D, that family already forces $C(\eta) \geq 2\|\phi\|^2$ at $K = 0$, so any content the conjecture has must be carried by $\rho(\eta)^{-K}$ alone; and the chain family of the same subsection, on which $\text{span}(S) = \mathcal{K}_K$, forces $C(\eta)\rho(\eta)^{-K} \geq 2\|\phi\|^2$ at every K as $g/\eta \rightarrow \infty$, so the class must bound the couplings relative to η . Nor is Eq. (11) an error bound: for a coordinate subspace, containing $\mathcal{K}_K$ forces $w_S = 1$, and at unit captured weight the error is $O(1)$ —0.28 on the open chain of Table II at $\eta = 0.18t$, and up to 0.43 on the periodic rings (Sec. III D). The panel of Fig. 2 read against the conjecture, and the full discussion, are in Sec. S3.

C. The shot count

Proposition 1 counts the shots that populate such a subspace.

Proposition 1 (Sampling capture). *Let $p(x) = \frac{1}{K+1} \sum_{k=0}^K p_k(x)$ with $p_k(x) = |\langle x | e^{-iHt_k} \phi \rangle|^2 / \|\phi\|^2$, draw T bitstrings independently from p , or, for T a multiple of $K+1$, $T/(K+1)$ independently from each p_k as in the protocol of Sec. II, and let $\mathcal{S}$ be the set of distinct outcomes. For $\varepsilon, \delta \in (0, 1)$, if*

$$T \geq \varepsilon^{-1} \log(1/(\varepsilon\delta)) \quad (13)$$

then with probability at least $1 - \delta$ the set $\mathcal{S}$ contains every configuration x with $p(x) \geq \varepsilon$.

The count is written in ε and not in $|\mathcal{S}|$ because the form $T = O(\varepsilon^{-1} \log(|\mathcal{S}|/\delta))$ is circular: $|\mathcal{S}|$ is the random output of the very T draws the bound is meant to fix. With $\mathcal{N}_\varepsilon = \{x : p(x) \geq \varepsilon\}$, each member contributing at least ε to a probability distribution, $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$ *deterministically*; a union bound over $\mathcal{N}_\varepsilon$ of the event “ x never drawn in T trials” gives $\Pr[\mathcal{N}_\varepsilon \not\subseteq \mathcal{S}] \leq \varepsilon^{-1} e^{-\varepsilon T}$, hence Eq. (13). For the stratified draws, $\Pr[x \notin \mathcal{S}] = \prod_k (1 - p_k(x))^{T/(K+1)} \leq e^{-Tp(x)}$, so the same count holds verbatim; and wherever the sampler returns more than ε^{-1} distinct configurations this logarithm is the smaller of the two as well (Sec. S10.4).

The parameter ε thresholds a Born probability, not a spectral accuracy: capturing the high-probability configurations is a statement about the sampler, not about the spectrum (Sec. IV B); what the resulting subspace costs is priced in Sec. VII (Sec. S3 gives the two compositions of the bound this work does not make).

V. WHERE THE LEAKAGE CERTIFICATE IS INFORMATIVE: ITS NON-VACUITY THRESHOLD, MEASURED

Whether the certificate of Theorem 1 says anything on a given subspace is decided by one comparison, computed from the Ritz pairs and the retained probe, given the exact probe (H4). Its *non-vacuity threshold* FR^* is the sector fraction above which the leakage branch of the bound first improves on the trivial bound $\|\phi\|^2(1 + w_S)$, a property of the inequality chain that proves the theorem rather than of the reconstruction. Swept at $L = 6$ –12, the certificate first improves on the trivial bound at a true relative error of 1.2×10^{-3} to 7.3×10^{-3} in every one of sixteen (L, η) cells (Sec. V D); above the threshold it gives converged bounds below the trivial one (Sec. V C); evaluated on the published subspaces at five sizes up to $L = 14$, at the operating points of the resource scan, which lie well below the threshold, it is vacuous (Sec. V B); and the displacement of the threshold is located rung by rung in the proof, with the basis structure behind it (Sec. V E).

Non-vacuity is a condition on one number. With $\hat{\Lambda}_S = \Lambda_S / \|\phi\|$, the leakage branch of Eq. (7) improves on the trivial branch exactly when

$$\frac{\hat{\Lambda}_S(\eta)}{\eta} < \tau(w_S) := \frac{1 + w_S}{1 + \sqrt{w_S}} - \sqrt{1 - w_S}, \quad \tau(1) = 1, \quad (14)$$

so a subspace holding the entire Born weight of the probe certifies something as soon as its leakage, in units of the resolution it was asked for, drops below unity.

TABLE III. **The certificate on the subspaces this paper publishes, including the size the sweep never reached.** Each row is Theorem 1 evaluated at the integer determinant count $|\mathcal{S}| = \text{round}(\text{frac} \times \text{dim})$ that Fig. 5 reports, not at the nearest point of the grid of Table S1. Columns 5–8 are the vacuity factor—the leakage branch of Eq. (7) over the trivial branch $\|\phi\|^2(1 + w_S)$ —so a value above 1 means the certificate says nothing the trivial bound does not. All twenty evaluations are the $T \rightarrow \infty$ limit of the ranking protocol; the sixteen up to $L = 12$ pass the depth test of Sec. VA, and the $L = 14$ row its indicator half only (see below); controls and costs are in Sec. S11. The last column is the measured relative L_1 error at the resolution and against the threshold that *define* the published fraction ($\eta = 0.15t$, $\text{rel-}L_1 < 0.05$), from code sharing neither reference, normalization nor recursion depth with the scan that produced the fractions. At $L = 14$ the ground state is the stored checkpoint, the count is one determinant above the deposited 824 503, and the $K = 18$ ranking, re-run at that size, reproduces the stored selection as a set (Sec. S4). The $L = 14$ row is taken at the deepest recursion run, depth 250, where Λ_{out} was not recorded, so only the indicator half of the depth test applies: the depth indicator is 0.045 and the factor at $\eta = 0.10t$ still falls by 0.17% between depths 200 and 250 ($8.627 \rightarrow 8.613$); the vacuity is not at risk.

L	$\dim(N+1)$	published subspace		vacuity factor at $\eta/t =$				rel- L_1 at $\eta = 0.15t$
		fraction	$ \mathcal{S} $	0.10	0.15	0.18	0.25	
6	300	0.8200	246	4.54	2.94	2.40	1.66	0.0477
8	3 920	0.5561	2 180	5.65	3.69	3.05	2.17	0.0478
10	52 920	0.3625	19 183	6.50	4.28	3.54	2.52	0.0484
12	731 808	0.1700	124 407	8.04	5.20	4.27	3.00	0.0485
14	10 306 296	0.0800	824 504	8.61	5.59	4.61	3.26	0.0463

A. What is evaluated

We evaluate Theorem 1 on the subspaces of Fig. 5 at $L = 6, 8, 10$ and 12 , with the ranking protocol of Sec. VII, over fractions 0.20 – 0.95 and $\eta = 0.10$ – $0.25t$: 204 evaluations, of which 187 are converged in Krylov depth by the depth test of the sweep—the internal indicator $\Lambda_{\text{in}}/\Lambda_{\text{out}}$, the truncation part of Λ over its boundary part, below 0.05 , and Λ_{out} constant to 10^{-3} relative over the last three depths of the ladder—and no row flagged not converged is quoted. To that grid we add twenty evaluations at the published subspaces themselves, at the integer counts 246, 2180, 19 183, 124 407 and 824 504 at $L = 6$ – 14 . The sixteen up to $L = 12$ pass the depth test and the $L = 14$ row its indicator half only (Table III). The subspaces are the $T \rightarrow \infty$ limit of the protocol, not finite-shot draws. What is certified is the object the code forms, the reorthogonalized Krylov reconstruction A_K at a declared depth, against a reference whose drift is declared, and no measured error within a factor 20 of that drift is quoted. The protocol, the controls and the costs are in Secs. S4 and S11.

B. The certificate at the operating fractions of the resource scan

Evaluated on the published subspaces themselves, the leakage branch of the bound exceeds the trivial bound $\|\phi\|^2(1 + w_S)$ at all five sizes and all four resolutions, by factors running from 1.66 ($L = 6$, $\eta = 0.25t$) to 8.61 ($L = 14$, $\eta = 0.10t$), and by 2.40 to 4.61 at the $\eta = 0.18t$ of the momentum-resolved channel; Table III carries the twenty values. They are monotone in both arguments—rising with L at every resolution, falling with η at every size—so the certificate is furthest from non-vacuity at the largest sector and the finest resolution, the regime the method targets.

The largest classically converged case carries the evidence. At $L = 12$ and the published subspace— $|\mathcal{S}| = 124 407$ of 731 808 determinants— $w_S = 0.998622$, so Eq. (14) asks for $\hat{\Lambda}_S/\eta < \tau(w_S) = 0.9625$; the measured value is 4.236 at $\eta = 0.18t$, converged in depth (Sec. S4). Over the 606 live subspaces of the pooled sweep the leakage branch is the smaller of the two on only 61. The reconstructions at the operating fractions of the resource scan—Fig. 5 and the operating points of Secs. VII and VIII—are therefore uncertified; the Born-ranked reconstruction of Fig. 3, at 85% of its sector, lies above the threshold (Sec. VC).

Two controls come with the table. Applying this paper’s own acceptance criterion to the same sweep with that independent code returns the reported fractions to under 2% (Table S1), and the last column of Table III evaluates the defining criterion at the published subspaces themselves, 0.0463–0.0485 against the threshold 0.05. The reconstructions and the curve are thus reproduced independently; what those fractions lack is a certificate.

C. What the certificate certifies, and at what shot cost

What the sweep does certify, at each size the strongest statement whose bracket is converged, is collected in Table S3: at $L = 12$, 60% of a 731 808-dimensional sector gives relative $L_1 \leq 0.717$, a factor 2.8 below the trivial cap, tightening to ≤ 0.218 at 70% and $\eta = 0.25t$. One further point of the map lies above the threshold: for the Born-ranked (infinite-shot) reconstruction of Fig. 3, at 85% of its sector, the certificate of the depth-500 Krylov reconstruction is non-vacuous in all sixteen channels, at a relative bound of 0.77–1.75 against the trivial 2, and loose, 350–900 times the true error of the channel integrated over the whole frequency line (`cert_akw.json`). At the acceptance criterion used throughout this paper, relative $L_1 < 0.05$, which the reconstruction at $L = 8$, $\eta = 0.18t$ reaches at $\text{FR} \approx 0.52$, the certificate requires at least 0.98 of the sector at $L = 8$, the crossing lying in $[0.977, 0.983]$, and on our grid it is not reached at all at $L = 6, 10$ or 12 : at that accuracy Theorem 1 does not certify the operating points of Fig. 5, and the regime where it would costs more shots than the method saves, since reaching the non-vacuous band multiplies the shots per snapshot needed for the marginal determinant by 7.5–10 at $L = 8$ and by 20–35 at $L = 12$ (Table S2).

D. The crossing of the trivial bound is calibrated

Across all sixteen (L, η) cells the crossing of the trivial bound occurs at a true relative L_1 error of 1.2×10^{-3} to 7.3×10^{-3} , median 2.8×10^{-3} : a spread of 5.9 over true errors spanning five decades, and in every cell at least 47 times the drift of our own reference. The spread is a drift, not scatter, and it runs the conservative way in L : the larger the system, the smaller the true error at which the certificate first says anything. The crossing of $\tilde{\Lambda}_S(\eta)/\eta$ therefore locates a true error to within that factor, and it is computed from H , S , the retained probe ϕ_S , the probe norm $\|\phi\|$ and η , without the spectral function it bounds, given the exact probe (H4), with four limits (Sec. S4). It locates one error level per cell and is not an error estimate. A user without w_S can discharge $\tau(w_S)$ conservatively: an assumed floor $w_S \geq 0.995$ gives $\tau \geq 0.9280$, moving the threshold by at most 7%. The calibration rests on these sixteen cells, $L = 6$ – 12 , and is not extrapolated to $L = 14$. A two-constant rule in $1 - w_S$, $1 - w_S < c\eta^p$ with c and p fitted on some sizes and tested on the others, localizes the crossing error more tightly than the certificate on eleven of the fourteen train/test splits (`null_ws7.py`): as a tracker of the error the certificate has no edge in sharpness over the captured weight. Its edge is provability. It is an inequality with a proof, whose decision survives replacing w_S by a floor, whereas the fitted rule is a regularity of these systems with no guarantee outside its fitted range—no inequality indexed on the captured weight alone bounds the error (Sec. III D)—and it asks for $1 - w_S$ to 10^{-5} – 10^{-4} and, stripped of its second parameter, is the looser of the two on thirteen of the fourteen splits.

E. Where the threshold comes from

Measured rung by rung along the chain of inequalities that proves Theorem 1, at $L = 8$, the displacement of the threshold is 44% the pointwise step to a Hilbert-space distance, 38% the operator-norm step $\|R(z)\| \leq 1/\eta$, 11% the Feshbach/Minkowski split and 4.6% Cauchy–Schwarz in ω , whose total margin for tightening is a factor 1.05 typically and 3.29 at worst. The subspace adds one structural fact. For a subspace holding the whole probe, $\Lambda_S = 0$ requires $\text{span}(S) \supseteq \mathcal{K}(H, \phi)$, and the reflection $j \rightarrow -j$ about the seed site commutes with H and has ϕ as an eigenvector, so every vector of $\mathcal{K}$ vanishes on the reflection-fixed determinants of the opposite parity—4, 18, 48, 200 and 600 at $L = 6$ – 14 —while a reflection-projected power iteration (`z_suppK_sym.py`) finds every other determinant in the support: 296, 3902, 52 872, 731 608 and 10 305 696 at $L = 6$ – 14 . At full captured weight, therefore, no coordinate subspace short of all of them has zero leakage. That is a statement that $\Lambda_S \neq 0$, not about its size; the vacuity at the operating fractions of the resource scan, all below full captured weight, is measured in Table III, not deduced from it. Nor is the ranking the weak link: a Krylov-diagonal oracle ranking gives a *worse* certificate than the Born ranking at $L = 6$, $\eta = 0.18t$, at every fraction checked from 0.50 on. At full captured weight the leakage branch is of first order in the boundary coupling and the error of second, and the same identity gives an a posteriori certificate of second order,

$$\|A - A_P\|_1 \leq \frac{\Lambda_P(\eta) \tilde{\Lambda}_P(\eta)}{\eta^2} \quad (\phi_Q = 0), \quad (15)$$

where $\tilde{\Lambda}_P$ is Eq. (6) with the energy difference in its kernel reversed in sign (equal to Λ_P for a real Hamiltonian and probe) (Sec. S4); whether one exists below full captured weight, where every published fraction lies, is open. The measurement basis is a choice, and for the momentum-resolved probe of Fig. 3 it is a forced one: the Krylov space lies inside one momentum block of exactly $\dim(N+1)/L$ plane-wave determinants, on which the certificate becomes non-vacuous at 48–50 of 50 determinants ($L = 6$) and 455–483 of 490 ($L = 8$) (`z_kblock.py`), a factor 5–7 fewer than

the site determinant basis requires. The block exists only after the measurement is rotated to the plane-wave orbitals, about $L(L-1)$ further two-qubit gates whose effect on the sampled distribution is unmeasured (Sec. S4); in the site basis the same probes at $L=8$ have support on 3848 to 3920 of the 3920 determinants (Sec. S2). The gain is $1/L$ against an exponential, and whether the support fills the block at $L \geq 10$ is unmeasured.

VI. NO SHORTCUT TO THE BOUNDARY: AN OBSTRUCTION AND ITS WITNESS

Section III leaves one question open: can the a posteriori certificate be replaced by an a priori predictor computed from the state before any circuit runs? The natural candidates are the monotones of fermionic non-Gaussianity, efficiently measurable from two-point functions (Sec. S5 reviews them). None can play that role, for either term of Eq. (7): not for the static determinant support $|\mathcal{S}|_{\varepsilon_w}$ of the ground state that the shots must populate, and not for the leakage $\Lambda_S(\eta)/\eta$ of the coordinate subspace the sampler returned. What replaces the predictor is the measured map of Sec. V. Three results establish this. On number-conserving states the one-body antiflatness is a single invariant of the one-particle density matrix (Eq. (17)); a geminal family carries antiflatness of every order at bond dimension 2 (Theorem 3); and on fibers on which every invariant is frozen the subspace-aware certificate still separates, so the leakage bound that invariants do supply is vacuous there (Proposition 2).

A. The collapse: one-body magic is a single 1-RDM invariant

We work with the fermionic AntiFlatness (FAF) [16],

$$\mathcal{F}_k = 2n_o - \frac{1}{2} \text{tr}[(\Gamma^\top \Gamma)^k] = 2n_o - \sum_{j=1}^{2n_o} \lambda_j^{2k}, \quad \Gamma_{mn} = -\frac{i}{2} \langle [\gamma_m, \gamma_n] \rangle, \quad (16)$$

with Γ the $4n_o \times 4n_o$ Majorana covariance matrix, $\{\lambda_j\} \in [0, 1]$ the moduli of its eigenvalue pairs $\pm i\lambda_j$, and $2n_o$ the number of spin-orbital *modes* (twice the spatial-orbital count), distinct from the electron number of the $(N \pm 1)$ sectors. Each $\mathcal{F}_k$ vanishes if and only if the state is Gaussian and is invariant under Gaussian (free-fermion) unitaries by construction [16]; the index k is the replica (moment) order of $\text{tr}[(\Gamma^\top \Gamma)^k]$, and every $\mathcal{F}_k$ is efficiently measurable from two-point functions [16, 17, 73].

On the number-conserving states of electronic structure the anomalous correlators $\langle c_i c_j \rangle$ vanish, so Γ is fixed by the one-particle reduced density matrix γ and its singular values are $\lambda_i = |2g_i - 1|$, with $g_i \in [0, 1]$ the natural spin-orbital occupations. The $k=1$ member then reduces to a single 1-RDM invariant,

$$\boxed{\mathcal{F}_1 = \sum_i [1 - (2g_i - 1)^2] = 4 \sum_i g_i(1 - g_i) = 4 \text{tr}[\gamma(1 - \gamma)]}, \quad (17)$$

exact on any number-conserving state and verified state by state to 10^{-13} on the 38 molecular ground states of this work. For spin-restricted natural orbitals it equals $2N_u$, the effectively-unpaired-electron index [74, 75], and $\text{tr}[\gamma(1 - \gamma)]$ is one quarter of the one-body quadratic (linear) entropy [76], a trace-form one-body entanglement entropy [77]. The algebra is an immediate number-conserving specialization of the antiflatness of Ref. [16], noted independently for fixed-particle-number states in a study of interacting integrable models [78]. Being a function of the occupation spectrum alone, $\mathcal{F}_1$ is invariant under orbital rotations (the spin-polarized cases and the dictionary are in Sec. S5).

B. The geminal witness, and what an invariant cannot certify

What obstructs the operative quantities—the thresholded support $|\mathcal{S}|_{\varepsilon_w}$ and the leakage—is a symmetry, exhibited by a family whose amplitudes are $\Theta(1)$ and therefore threshold-robust: an antisymmetrized product of K strongly orthogonal singlet geminals (perfect-pairing / generalized-valence-bond),

$$|\Psi_K\rangle = \bigotimes_{i=1}^K \left(\cos \theta \hat{b}_{i\uparrow}^\dagger \hat{b}_{i\downarrow}^\dagger + \sin \theta \hat{a}_{i\uparrow}^\dagger \hat{a}_{i\downarrow}^\dagger \right) |0\rangle, \quad (18)$$

on $2K$ spatial orbitals ($4K$ modes, $N = 2K$ electrons), each geminal occupying a disjoint bonding/antibonding pair (b_i, a_i) .

Theorem 3 (Geminal orbit witness). *At maximal pairing $\theta = \pi/4$ the family (18) satisfies, exactly,*

$$\mathcal{F}_1 = 2N_u = 4K, \quad \mathcal{F}_2 = 4K, \quad |\mathcal{S}|_{\text{pair}} = 2^K, \quad \chi = 2, \quad (19)$$

with $\mathcal{F}_{1,2}$ the order-1 and order-2 antifatness of Eq. (16), $|\mathcal{S}|_{\text{pair}}$ the determinant support in the pairing basis, and χ the minimal matrix-product-state bond dimension in the pairing ordering. In that localized (pairing) basis the state is a bond-dimension-2 tensor network, representable, contractible and Born-samplable in $O(K)$.

One-body magic, higher-order magic and determinant support therefore all diverge with K : magic of every order is unbounded at bond dimension 2, where the simulation cost grows only linearly, as $O(K)$. The proof, four block-diagonal computations, is given in the Supplemental Material (Sec. S12). The values of $\mathcal{F}_1$, $\mathcal{F}_2$ and $|\mathcal{S}|_{\text{pair}}$ are reproduced by exact diagonalization for $K = 1, \dots, 4$, and so is $\chi = 2$, the largest Schmidt rank over all cuts of the pairing ordering (Fig. S1 and Sec. S12). The orbit is the argument: $\mathcal{F}_k$ is invariant under number-conserving orbital rotations [16], so the whole orbit $\{U|\Psi_K\rangle\}$ carries the same $(\mathcal{F}_1, \mathcal{F}_2) = (4K, 4K)$ while the cost in the fixed computational basis is $O(K)$ at the pairing point and, at generic rotated points, not established here (Born sampling there is plausibly hard, Sec. S5). A Gaussian-invariant fermionic magic therefore does not lower-bound the bond dimension, nor the cost beyond linear order in K , and it determines the cost only if the plausibly hard rotated points are in fact easy. The same orbit reaches the second budget of Eq. (7):

Proposition 2 (Fibers on which every orbital-rotation invariant gives a vacuous leakage certificate). *Let $\mathcal{U}_U$ be the Gaussian unitary implementing $U \in \mathcal{U}(n_o)$, and let the same physical problem written in the rotated orbital basis be $(H_U, \Psi_U, \phi_U) = (\mathcal{U}_U^\dagger H \mathcal{U}_U, \mathcal{U}_U^\dagger \Psi, \mathcal{U}_U^\dagger \phi)$, the determinant basis—in which the device measures, and of which S is a coordinate subspace—held fixed. Along this orbit the spectrum of H , E_0 , $\|\phi\|^2$, every Lehmann weight and hence $A(\omega)$ pointwise are constant, as is every orbital-rotation invariant of the state, in particular $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$ and all $\mathcal{F}_k$. $\Lambda_S(\eta)/\eta$ is not. An upper bound on $\Lambda_S(\eta)/\eta$ is available from invariants alone—combining $\Lambda_S \leq \|P_S H Q_S\| \|\phi_S\|$ with $\|Q_S H P_S\| = \|Q_S(H - c)P_S\| \leq \min_c \|H - c\|$ gives $\Lambda_S(\eta)/\eta \leq (W/2)\|\phi\|\sqrt{w_S}/\eta$, W the spectral width of the $(N \pm 1)$ sector—but it is vacuous by a factor of at least $W/2\eta$, which grows extensively with system size. That no invariant bound can be useful on these fibers is seen at fixed invariants: any $B(\text{invariants}, |\mathcal{S}|, w_S)$ valid for $\Lambda_S(\eta)/\eta$ is constant on a fiber of fixed arguments and must therefore exceed the fiber maximum. In the geminal family APSG($\pi/4$) on $2K$ orbitals with $H = \varepsilon \hat{N} - g \sum_i A_i^\dagger A_i + \delta h$ we exhibit fibers ($K = 2, 3, 4$; $\delta = 0.01$; $\eta = 0.10$; $|\mathcal{S}| = 2^{K-1}$; spectrum, Lehmann weights and all $\mathcal{F}_k$ matched to 10^{-14} , w_S to 2.2×10^{-16}) whose endpoints carry $\hat{\Lambda}_S(\eta)/\eta = 0.237$ and 2.326 , 0.323 and 2.336 , 0.391 and 2.346 . Since the larger value exceeds $\tau(w_S) = 0.795$ of Eq. (14), above which the leakage branch of Theorem 1 no longer improves on the trivial bound $\|\phi\|^2(1 + w_S)$, every valid invariant bound is vacuous on these fibers—while the subspace-aware certificate is non-vacuous at the aligned endpoint (0.856 against a trivial 1.962). The statement is one of non-determination and non-vacuity, not of non-existence: invariants do not fail to bound the leakage, they fail to bound it usefully.*

The fibers (Table S4), an exhaustive search over the 276 subspaces of that size at $K = 2$, and four limits of the exhibit—a matched pair at one angle, a margin over the trivial bound that shrinks from 2.29 to 1.69 between $K = 2$ and $K = 4$, a small-leakage endpoint realized only by the stabilizer of the state, and sector dimensions of at most 3920—are in Sec. S5.

The raw determinant support—the unthresholded Slater rank—is obstructed separately, by continuity: no continuous functional of a fixed set of reduced density matrices can upper-bound the count more tightly than the sector dimension near a single determinant, states arbitrarily close to one in every such functional carrying any support up to the whole sector, rank being coefficient-insensitive and not upper-semicontinuous (though, unlike the CP tensor rank, whose sets of rank at most r are not closed [79], it is lower-semicontinuous); and at half filling the invariant $\mathcal{F}_1$ cannot lower-bound the count beyond 2, the two-determinant cat having maximal $\mathcal{F}_1$. That continuity argument (Sec. S13) does *not* transfer to the operative, coefficient-sensitive $|\mathcal{S}|_{\varepsilon_w}$, which collapses to 1 along the same path as the functionals; the symmetry above does.

The cost side: a bilateral obstruction whose sign is not stable. Unlike $\mathcal{F}_1$, the determinant support is basis dependent, and both directions of the decoupling are realized (Fig. S2): a free-fermion ground state has $\mathcal{F}_1 = 0$ and a site-basis support $|\mathcal{S}|_{\varepsilon_w}$ (open chain, $\varepsilon_w = 2.5 \times 10^{-3}$) of 35, 336, 3496 and 37361 at $L = 4$ –10; along a Hubbard chain raising U raises $\mathcal{F}_1$ while lowering $|\mathcal{S}|_{\varepsilon_w}$; across the nineteen molecules the two rise together. Within each of the six (L, N) strata of the Hubbard sweep the rank correlation of $\mathcal{F}_1$ with $|\mathcal{S}|_{\varepsilon_w}$ is exactly -1 (stratified Kendall $\tau = -1.000$, exact one-sided permutation $p = 3.35 \times 10^{-13}$ against independent orderings within each stratum, a null that six strata sharing one mechanism do not satisfy), while it is $+1.000$ in the chain and ladder families and $+0.86$ across the molecules: an excellent rank predictor whose sign reverses between families, as an orbital invariant compared with a basis-dependent count must. The bond dimension χ , a rigorous floor on the raw support, tracks the thresholded one

better (mean within-stratum $\rho = 0.948$), but it is itself basis dependent, measured rather than predicted, and blind to the leakage. Computable Gaussian-invariant measures of non-Gaussianity include the antiflatness used here [16] and the two families of Tarabunga *et al.* [17], whose natural-orbital participation entropies, themselves Gaussian invariant, also upper-bound the classical simulation cost in an orthonormal Gaussian basis; their quadratic occupation entropy, which equals $\mathcal{F}_1$, is a strong pure-state monotone, proved by Tarabunga *et al.* and, independently, by Leone and Bittel [18]. The contrast between an orbital invariant and a basis-dependent count is our observation, and what the construction adds is a family on which every order of the invariant is maximal at $O(K)$ cost, and fibers on which the invariants are frozen while the certificate of Theorem 1 still separates. The stratified statistics, the molecular set, which does not discriminate, and the two families where $\mathcal{F}_1$ wins are in Secs. S5 and S14.

VII. THREE PRICES: SUPPORT, RESOLUTION, AND SHOTS

On half-filled periodic Hubbard rings at $U/t = 8$ ($L = 4$ –14, 8 to 28 qubits) the method is priced on three axes, each measured here: the determinant support the sampler must populate grows as $e^{1.05L}$, more slowly than the sector's $e^{1.30L}$; the sector fraction, unlike the support, is a property of the question put to the model rather than of the model, its exponent moving by a factor 2.2 over a decade of the resolution η ; and the shot budget T that populates the support, measured at $L = 4$ –12, lies two orders of magnitude above it and grows as $e^{1.12L}$ over $L = 6$ –12, its $L = 14$ value being a floor of 1.3×10^8 shots for a single seed and branch.

The dynamical support. $|\mathcal{S}|_\eta$ here is the number of determinants needed for the reconstruction of $A(\omega)$ to reach relative $L_1 < \text{thr}$ at resolution η and recursion depth n_ℓ , so it depends on all three (`scaling_lanczos_mf.py`, `stage_frac`: `thr = 0.05`, $\eta = 0.15t$, $n_\ell = 150$). It is distinct from the static ground-state support $|\mathcal{S}|_{\varepsilon_w}$ of Sec. VI and from the returned set S of Theorem 1: the three coincide only by accident, and no statement here transfers to the other two (Sec. I).

A. The support exponent and its resolution dependence

Across $L = 4$ –14 (8 to 28 qubits, half-filled *periodic* rings at $U/t = 8$; the three largest sizes reached by matrix-free Lanczos/Haydock and validated against dense diagonalization at $L \leq 8$), the determinant support needed to reconstruct $A(\omega)$ to relative $L_1 < 0.05$ at $\eta = 0.15t$ and $n_\ell = 150$ grows exponentially,

$$|\mathcal{S}|_{\eta=0.15} = 22, 246, 2180, 19183, 124407, 824503 \implies |\mathcal{S}|_\eta \sim e^{1.05L} \quad (L = 4\text{--}14, R^2 = 0.998), \quad (20)$$

more slowly than the $(N+1)$ sector itself, $\dim = 24, 300, 3920, 52920, 731808, 10306296 \sim e^{1.30L}$. Restricted to $L = 4$ –12, the five sizes of the replication grid, the fit gives 1.082 at $R^2 = 0.998$. The fraction is the support divided by the sector dimension, exactly, so the fraction and support exponents differ by the constant $b_{\dim} = 1.291$ and move together, by 0.148 over the decade $\eta = 0.05 \rightarrow 0.50t$ (Table S5): a seventh of the support exponent, three quarters of the fraction exponent. We therefore report the support, the absolute resource a device has to populate, and the fraction as the derived, resolution-dependent quantity that it is.

B. The sector fraction is a property of the question, not of the model

At $\eta = 0.15t$ the fraction needed for a fixed spectral accuracy falls from 0.92 at $L = 4$ to 0.08 at $L = 14$, but its exponent is set by η : over the decade of η it moves by a factor 2.2, from -0.1266 to -0.2743 (Table S5), and even at fixed η the local logarithmic slopes between consecutive published points steepen by a factor 6.8 across five intervals, so the fraction is not a single exponential either. The reference is itself a depth-150 Lanczos computation whose drift between depths 100 and 200 at $L = 12$ is 0.93% at $\eta = 0.15t$ and 3.0% at $\eta = 0.10t$, so the fraction is reported as a declared window, $\eta \geq 0.15t$ at $n_\ell = 150$, with the full grid and the finite-depth caveats in Sec. S6.

C. The third price: shots

Populating a subspace by measurement is a coupon-collector problem on the Born distribution of the time-evolved seed, so the budget exceeds the support it buys. Table IV reports that budget *measured*, by multinomial sampling of

TABLE IV. **The third price, measured at every size but the largest.** T is the total shot budget, split evenly over the 19 snapshots of the protocol, for a single orbital seed and a single branch; it is the budget at which the finite-shot reconstruction itself first reaches relative $L_1 < 0.05$, and $\pm$ is the sample standard deviation across seeds, not the standard error of the mean. Run A is `sampled_honest.json` (8 seeds, $L \leq 8$); run B is `honest_sampling.json` (4 seeds, 6 at $L = 10$), and the two are separate runs with different seeds and different budget grids, listed separately at $L = 6$ and $L = 8$. At $L = 4$ run B is censored by its own cap of 10^8 shots per snapshot, the 24-state sector being degenerate for this purpose, and run A is quoted there. The last column divides T by the *published* support; dividing instead by the support each run actually populated gives 95, 107, 141 and 124 at $L = 6$ –12. Only $L = 14$ is a floor: the overhead held at its $L = 12$ value, which is a lower bound and not an estimate.

L	$\dim(N+1)$	$ \mathcal{S} _\eta$	T	$T/ \mathcal{S} _\eta$	status
4	24	22	$(7.8 \pm 2.0) \times 10^2$	35	measured, run A, 8 seeds
6	300	246	$(2.43 \pm 0.34) \times 10^4$	99	measured, run A, 8 seeds
6	300	246	$(2.44 \pm 0.23) \times 10^4$	99	measured, run B, 4 seeds
8	3920	2180	$(2.70 \pm 0.13) \times 10^5$	124	measured, run A, 8 seeds
8	3920	2180	$(2.61 \pm 0.08) \times 10^5$	120	measured, run B, 4 seeds
10	52920	19183	$(3.09 \pm 0.04) \times 10^6$	161	measured, run B, 6 seeds
12	731808	124407	$(1.91 \pm 0.01) \times 10^7$	154	measured, run B, 4 seeds
14	10306296	824503	$\gtrsim 1.3 \times 10^8$	154	floor, not measured

the declared protocol, at $L = 4$ –12, in two independent runs that agree where they overlap to 0.2% and 3.2%; every budget is for *one* orbital seed and *one* branch.

The budget is two orders of magnitude larger than the support it buys, already at $L = 6$, and it grows exponentially, 7.8×10^2 to 1.9×10^7 over $L = 4$ –12, a log-linear fit over $L = 6$ –12 giving $T \sim e^{1.12L}$ at $R^2 = 0.996$. At $L = 8$ the sixteen-channel momentum-resolved configuration of Fig. 3 reached its own resolution and accuracy with 8.3×10^7 , and not with half as many; the $16 \times 2.7 \times 10^5 \approx 4 \times 10^6$ shots that sixteen channels at the local-seed budget would cost is a transfer and not a measurement, because the momentum seeds have larger support. The $L = 14$ floor of 1.3×10^8 shots for a *single* seed and branch (Table IV) exceeds the largest shot budget drawn anywhere in this work, the 8.3×10^7 shots of the sixteen-channel configuration of Fig. 3. The overhead per determinant is a level and not a growth law—35, 99, 124, 161, 154 against the published support across $L = 4$ –12, rising by a factor 3.5 over the first three sizes (35 to 124), then to 161 at $L = 10$, and not rising over the last step, 161 to 154. Its exponent is not resolved by these data. Fitted on $L = 6$ –12 it is 0.08 per site against the published support (95% interval $[-0.03, 0.19]$) and 0.054 ($[-0.066, 0.174]$) against the support each run populated; both intervals contain zero, so *these data do not establish that the shot budget grows with a larger exponent than the support*, only that it is larger in absolute terms at every size measured.

What the published curve is. The published curve is the $T \rightarrow \infty$ limit of the declared protocol: the ranking that produced the published fractions is `argsort` applied to the Born distribution of the time-evolved state accumulated over the 19 snapshots of the protocol—the distribution the device would itself measure with infinitely many shots—and not a clairvoyant ordering by exact eigenvector amplitudes. With finite shots the fraction needed rises, from 0.8200 to 0.8571 ± 0.0135 at $L = 6$ and from 0.5561 to 0.6275 ± 0.0031 at $L = 8$ (Table S6), so the published curve is a strict lower bound on the finite-shot cost. At fixed size the penalty is budgetary, not structural: at $L = 8$ raising the budget by a factor 12.7 takes the ratio of the finite-shot error to the infinite-shot-ordering error from 1.96 ± 0.12 to 1.06 ± 0.01 . Finite-shot fractions are therefore reported only with their budget and seed count attached.

VIII. PHYSICAL VALIDATION: THE TWO GAPS SCALE IN OPPOSITE DIRECTIONS

Across $L = 6$ –12 the exact full-sector references against which the sampled reconstructions are scored reproduce the opposite finite-size scaling of the two collective scales of the half-filled Hubbard ring. On the same lattice, the charge gap decreases monotonically to within 6.2% of its Lieb–Wu thermodynamic value, while the spin gap closes as $\approx 2/L$, with an independent effective-Heisenberg control (Table V). The answer is known exactly and independently of this work, so the scaling, reported across four sizes rather than as a picture at a single size, is a check of the references and not a result (Sec. VIII C).

A. Charge converges to a finite value, spin closes as $1/L$

Charge. $\Delta(L)$ decreases monotonically and stays above the Lieb–Wu value $4.67952t$ at every size, the residual falling from $0.679t$ at $L = 6$ to $0.289t$ at $L = 12$, 6.2% of the thermodynamic value; $\Delta(L = 12) = 4.96876t$ is the

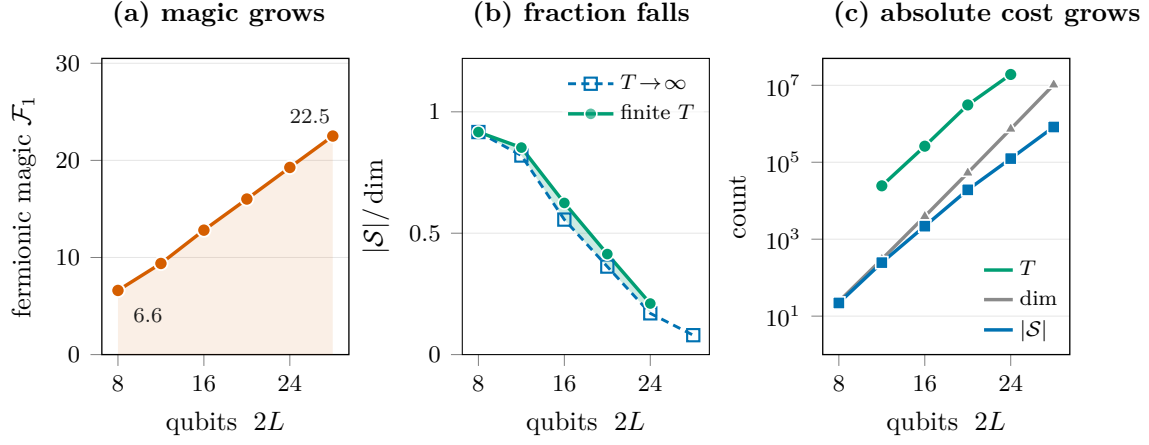


FIG. 5. The resource scan: one-body magic, the required sector fraction and the absolute counts. One-dimensional Hubbard ring, $U/t = 8$, half filling, periodic boundary conditions; the subspace is built in the $(N+1, S_z = +\frac{1}{2})$ sector seeded by $c_{0\uparrow}^\dagger |\Psi_0\rangle$ from the $K+1 = 19$ time snapshots of the declared protocol ($K = 18$, $\delta t = 0.5$), and a subspace is accepted when its Haydock reconstruction reaches relative- $L_1 < 0.05$ against the exact spectrum at broadening $\eta = 0.15t$. All fractions below are at that single η . **(a)** One-body fermionic magic of the ground state, 6.6074 at $L = 4$ to 22.5050 at $L = 14$ (periodic ring; the open-chain families elsewhere in this paper carry different values and must not be substituted here). It is not a cost: its rank correlation with the support reverses sign between families (Sec. VI). **(b)** The required fraction of the sector, twice. Open squares, dashed: the infinite-shot limit of the same protocol — the subspace obtained by ranking the exact Born weights of the snapshots — 0.9167, 0.8200, 0.5561, 0.3625, 0.1700, 0.0800 for $L = 4$ –14, the published fractions quoted throughout this paper. Filled circles, solid: the same acceptance criterion reached by actually drawing shots (classically simulated multinomial draws from the exact time-evolved state), 0.9167 ± 0.0000 , 0.8527 ± 0.0096 , 0.6246 ± 0.0033 , 0.4141 ± 0.0012 , 0.2104 ± 0.0004 (mean $\pm$ s.d. over 4 independent shot seeds; 6 seeds at $L = 10$); the error bars are drawn and are smaller than the markers. Finite sampling needs +4.0%, +12.3%, +14.2% and +23.7% more of the sector than the infinite-shot limit at $L = 6, 8, 10, 12$: the penalty does not cancel, it grows with size, and the shaded strip is that difference. An independent 8-seed run of a separately written sampler reproduces 0.8571 ± 0.0135 ($L = 6$) and 0.6275 ± 0.0031 ($L = 8$), agreeing with the curve to within one standard deviation. No finite-shot point is shown at $L = 14$: its Born weights were not computed, and they are not extrapolated. **(c)** The absolute counts that panel (b) normalizes away, plus the third one. Sector dimension $\dim$ (grey) and the support $|\mathcal{S}|$ of the reported subspace (blue) both grow exponentially; the measured number of shots T that reaches the finite-shot operating point of panel (b) (green) is 2.4×10^4 , 2.6×10^5 , 3.1×10^6 , 1.9×10^7 at $L = 6, 8, 10, 12$ (s.d. over seeds 2.3×10^3 , 8.1×10^3 , 3.8×10^4 , 1.2×10^5). $L = 4$ carries no shot count: its 24-state sector is degenerate for this purpose and its budget is censored by the 10^8 -per-snapshot cap. Log-linear fits over the six sizes drawn in (c) (four for T , which has no $L = 4$ and no $L = 14$ point) give $\dim \sim e^{1.30L}$ ($R^2 = 0.9999$), $|\mathcal{S}| \sim e^{1.05L}$ ($R^2 = 0.998$) and $T \sim e^{1.12L}$ ($R^2 = 0.996$, four sizes). Those exponents are fits to the points this panel plots, not to the five-size fits stored in `honest_sampling.json` (1.108 and 1.082). Over $L = 6$ –12 the exponent of T exceeds that of the populated support by 0.054 per site ($[-0.066, 0.174]$) and that of the plotted support by 0.08 ($[-0.03, 0.19]$): these data do *not* establish that the shot budget grows with a larger exponent than the support, only that it is larger in absolute terms at every size measured (Table IV). **Resolution dependence.** Over $\eta/t \in [0.05, 0.50]$ the exponent of $|\mathcal{S}|$ moves from 1.165 to 1.017 and that of the fraction in panel (b) by a factor 2.2, from -0.1266 to -0.2743 (Table S5, $L = 4$ –12). In absolute value both move by the same 0.148: they are one exponent, and neither is stable.

TABLE V. The two collective scales, across four system sizes. Charge: the Mott gap $\Delta = \mu^+ - \mu^-$ from exact ($N \pm 1$) ground-state energies (half-filled Hubbard ring, $U/t = 8$; particle-hole symmetry $\mu^+ + \mu^- = U$ verified to 10^{-8} at every size, ARPACK checked against dense `eigh` on every sector of dimension ≤ 5000). $\Delta_\infty = 4.67952t$ is the Lieb–Wu thermodynamic value [53], evaluated here from its exact integral representation to an absolute quadrature error of 2×10^{-9} and cross-checked against a second exact representation, the two agreeing to 10^{-13} . Spin: the lowest pole of $S^{zz}(q = \pi, \omega)$, a pole position of the tridiagonal recursion and therefore independent of the broadening used to display it. Right-hand block: the same quantity in the pure Heisenberg ring at the effective exchange $J = 4t^2/U = 0.5t$, a model with no charge degree of freedom at all. The residual column is quoted to four decimals so that it equals the Δ/t column minus $\Delta_\infty/t = 4.67952$; at five it would not be.

L	charge		spin (Hubbard)		spin (Heisenberg, $J = 0.5t$)	
	Δ/t	$(\Delta - \Delta_\infty)/t$	Δ_s/t	$L \Delta_s/t$	Δ_{ST}/t	$L \Delta_{ST}/t$
6	5.35813	0.6786	0.34857	2.091	0.34237	2.054
8	5.18584	0.5063	0.23958	1.917	0.26134	2.091
10	5.02224	0.3427	0.19956	1.996	0.21162	2.116
12	4.96876	0.2892	0.16635	1.996	0.17792	2.135

Hubbard-band separation of Fig. 4, and every pole of $S(q, \omega)$ at $L = 12$ carrying at least 1% of the largest weight at its momentum—97.5% or more of the weight at each q —lies above it (Fig. 7); weaker poles are not resolved and are not bounded by this statement. No extrapolation exponent is fitted: a straight line in $1/L$ undershoots Lieb–Wu by $0.12t$, which is evidence that the form is not $1/L$, not evidence about the limit.

Spin. The lowest pole of $S^{zz}(q = \pi, \omega)$ gives $L\Delta_s = 2.091, 1.917, 1.996, 1.996$: a $1/L$ closing with a coefficient of about 2 (mean 2.000, range 1.917–2.091), and a fit $L\Delta_s = d_0 + d_1L$ has a slope consistent with zero ($d_1 = -0.010 \pm 0.018$). A linear-in- $1/L$ intercept of Δ_s is -0.020 ± 0.020 , against 4.561 ± 0.044 for the charge gap; but the pure Heisenberg ring, whose gap is known to vanish, gives $0.0142 \pm 0.0014t$ at these sizes, so a vanishing intercept is not by itself evidence that a gap closes, and the flatness of $L\Delta_s$ is what carries the statement. That Heisenberg ring at $J = 4t^2/U = 0.5t$ —a different Hamiltonian, Hilbert space and code path—reproduces the $1/L$ law, $L\Delta_{\text{ST}} = 2.054\text{--}2.135$, differing from the Hubbard value by 6.5% at $L = 12$ with a residual that changes sign: it confirms the scaling and the coefficient to within 10% and does not pin the coefficient (Sec. S7).

The two channels therefore behave *qualitatively oppositely* under a change of system size—one converging to a finite thermodynamic value, the other extrapolating to zero as $1/L$ —with a factor 29.9 between them at $L = 12$, and with $1/L$ intercepts that separate. That opposition, measured at four sizes with an independent control on the smaller scale, is the physical check this section makes. It is a statement about pole positions, not about lineshapes, so it does not depend on the broadening at which either response is plotted.

B. Resolution: why the statement rests on pole positions

A spin–charge statement cannot be read off a lineshape at a single size, for a reason of resolution. At $L = 12$ the spin scale is $0.16635t$ against the $\eta = 0.10t$ used for Fig. 8: the gap is 1.66η , i.e. 0.83 of the Lorentzian full width at half maximum. At that resolution a spin response with a $0.17t$ gap and one with none are not distinguishable by eye in a single-size plot. Terms that presuppose a resolved dense spectrum—“continuum”, “gapless” and “edge of the continuum”—would need that distinction, and none is used for any channel of this work at this size.

What establishes the physics instead is the four-size scaling of Table V, plotted in Fig. 6, and a pole census (Table S7): between $L = 6$ and $L = 12$ the sector grows by a factor 2440 while the largest per-momentum count of poles above 1% of the channel maximum grows from 3 to 7 in the addition branch, from 4 to 9 in the charge channel and from 2 to 3 in the spin channel, a discrete spectrum whose lines are broadened together rather than a dense one forming. The census, the sum rule as measured, and the size convention of the exact-diagonalization literature together with the dynamical-DMRG resolution criterion [80] this work does not meet at $L = 12$, are in Sec. S7.

C. Relation to the known spin–charge physics

That the spin and charge responses of the one-dimensional Hubbard model live on separate energy scales has been known exactly since Lieb and Wu [53], is textbook [81], was seen in angle-resolved photoemission thirty years ago [82, 83], is already legible in the spectral weight of twelve-site clusters embedded by cluster perturbation theory [84], and has been computed by dynamical DMRG since 2007 [3] in all three of the momentum-resolved channels reported here—in that reference’s notation the charge factor $N(q, \omega)$ at $L = 60$ (at $q \approx \pi$ up to $L = 120$), the spin factor $S(q, \omega)$ at $L = 100$ and the one-particle function $A(q, \omega)$ at $L = 90$, which are respectively our $S(q, \omega)$, our $S^{zz}(q, \omega)$ and our $A(k, \omega)$, for the extended Hubbard model of which ours is the $V = 0$ line; the same literature treats the Hubbard chain at $L = 48$ with $\eta = 0.1t$ by correction-vector DMRG, and computes the full Heisenberg-chain $S(k, \omega)$ at $L = 64$, with the $k = \pi$ cut to $L = 128$ [4]. Here it serves as a check, against a known answer, of the exact full-sector references against which the sampled reconstructions are scored, with the des Cloizeaux–Pearson lower boundary [54] and the Müller upper bound [85] recovered as thermodynamic-limit references rather than as findings. At $L = 12$ the spin peaks follow that lower boundary for $q \leq \pi/2$ and lie above it toward $q = \pi$ (Fig. 8).

IX. HARDWARE RUNS, CLASSICAL REACH, AND SCOPE

A. The two hardware runs, and what “exact by coverage” means

Two device runs were made, and they are the only quantum hardware used in this work. The first, on the IBM Heron processor `ibm_fez`, sampled the $L = 6$ periodic Hubbard ring at $U/t = 4$ with 12 qubits and 3.5×10^5 shots, and post-selection on particle number and S_z , with no configuration recovery, retained all 300 determinants of the $(N+1)$ sector, so the reconstructed A_{hw} equals the exact A *by coverage*: Rayleigh–Ritz on the whole sector is its diagonalization,

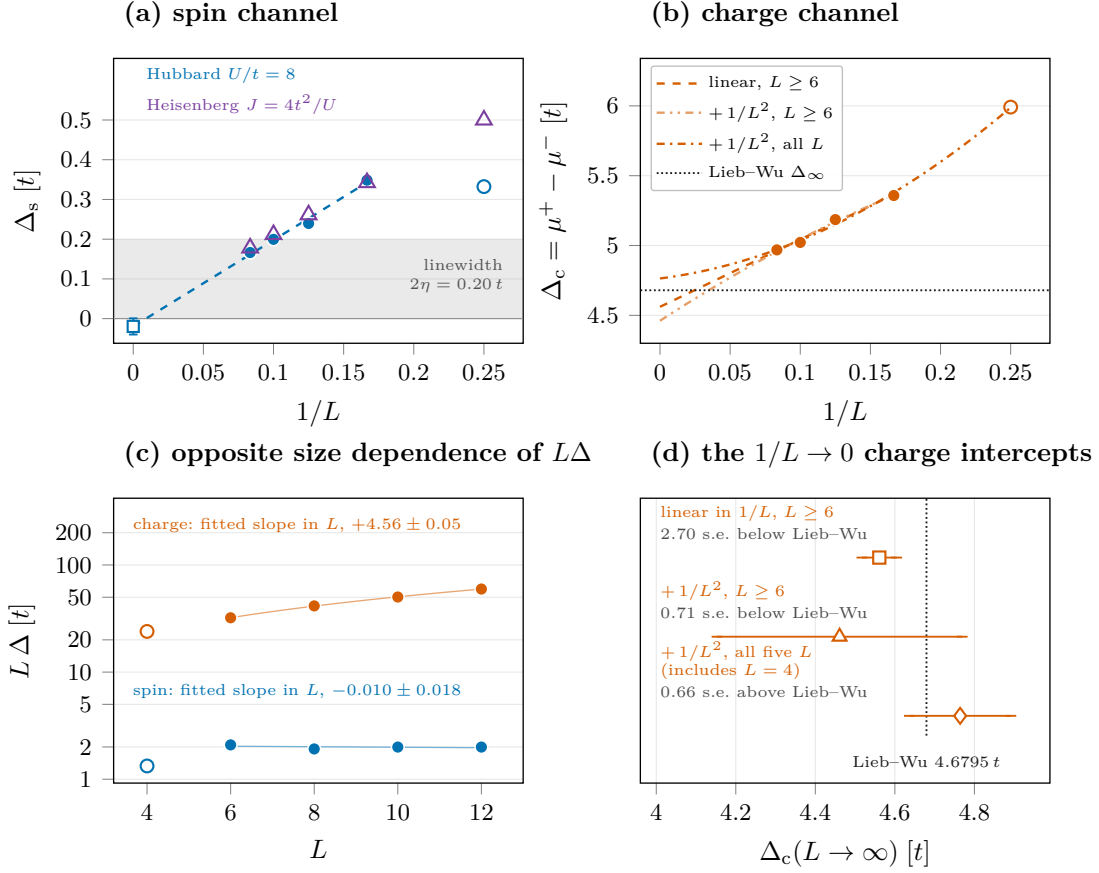


FIG. 6. The two gaps of the half-filled Hubbard ring scale in opposite directions. Periodic ring, $U/t = 8$, $L = 4, 6, 8, 10, 12$; in (a) and (b) the abscissa is $1/L$, so the sizes run $L = 12, 10, 8, 6, 4$ right to left, and open circles are $L = 4$, which the $L \geq 6$ fits exclude (the other open markers are not $L = 4$: the Heisenberg triangles of (a) are drawn open at every size, and the open square of (a) and the markers of (d) are fit intercepts). Every gap is an exact Lanczos pole, full reorthogonalization, *no* broadening. **(a)** Spin gap, lowest pole of $S^{zz}(q = \pi, \omega)$: $0.34857, 0.23958, 0.19956, 0.16635 t$ at $L = 6, 8, 10, 12$ ($0.33232 t$ at $L = 4$). Dashed: linear fit in $1/L$, $L \geq 6$ ($R^2 = 0.988$); the open square is its intercept $-0.0196 \pm 0.0204 t$, 0.96 s.e. from zero. Triangles: the pure Heisenberg ring $J = 4t^2/U = 0.50 t$, an independent control on a different code path, whose lowest $S^{zz}(q = \pi)$ pole reproduces its singlet-triplet gap to 8×10^{-15} . For $L \geq 6$ the two channels agree to within 8.3% (ratios $1.018, 0.917, 0.943, 0.935$); at $L = 4$ they do not (0.665), which is why that size is excluded. Grey band: the linewidth actually used for the published spin figure, FWHM $= 2\eta = 0.20 t$ at $\eta = 0.10 t$. The $L = 12$ gap lies inside it ($0.16635 t = 0.83$ FWHM), and broadened by this η it still carries 26.5% of its peak height at $\omega = 0$. One size alone therefore cannot separate a closing gap from no gap. **(b)** Charge gap $\Delta_c = \mu^+ - \mu^-$ from the $N \pm 1$ sectors: $5.3581, 5.1858, 5.0222, 4.9688 t$ at $L = 6, 8, 10, 12$ ($5.9914 t$ at $L = 4$). At $L = 12$ that is $27.6\eta = 13.8$ FWHM for the $\eta = 0.18 t$ of the one-particle channel, 16.6 times further from $\omega = 0$ in units of its own linewidth than the spin gap. Three extrapolations are drawn and they disagree. The ordinate omits zero because the question here is which finite value the gap tends to, not whether it vanishes. **(c)** The same data with no extrapolation model, logarithmic ordinate: $L\Delta_s$ is flat over $L \geq 6$ (mean $2.0 t$, the four values spanning 8% and not monotone, so no third figure is quoted; fitted slope in L -0.010 ± 0.018 , 0.57 s.e. from zero) while $L\Delta_c$ grows linearly ($+4.56 \pm 0.05$, 101 s.e.). The thin lines are those two linear fits, curved here only because the ordinate is logarithmic. **(d)** Those three intercepts on a scale where their standard errors are legible; on the scale of (b) the $\pm 0.0439 t$ bar is thinner than its marker. Linear, $L \geq 6$: $4.5609 \pm 0.0439 t$, *undershooting* the exact Lieb-Wu gap $4.6795 t$ by $0.1186 t$, 2.70 s.e. Same protocol plus a $1/L^2$ term: $4.4611 \pm 0.3084 t$, undershooting by $0.2184 t$ (0.71 s.e., standard error 2.4 times larger). Only the $1/L^2$ fit over all five sizes brackets Lieb-Wu ($4.7641 \pm 0.1278 t$, 0.66 s.e. above), and it is the one fit here that includes $L = 4$, the size (a) and (b) mark anomalous. These sizes therefore fix the $L \rightarrow \infty$ charge gap only to about ± 0.1 – $0.3 t$: compatible with Lieb-Wu once a $1/L^2$ term is allowed, not evidence for it. What all three exclude is zero, the weakest by 14.5 s.e. and the linear one by 104 s.e. *Caveat, because it limits (a)*. A vanishing intercept is not by itself the evidence: the Heisenberg control, whose gap is known to vanish in the thermodynamic limit, has a linear-in- $1/L$ intercept $0.0142 \pm 0.0014 t$, 10.0 s.e. above zero, and its own $L\Delta$ drifts $+6.8\%$ from $L = 4$ to $L = 12$, both from its logarithmic corrections. The model-free statement is (c).

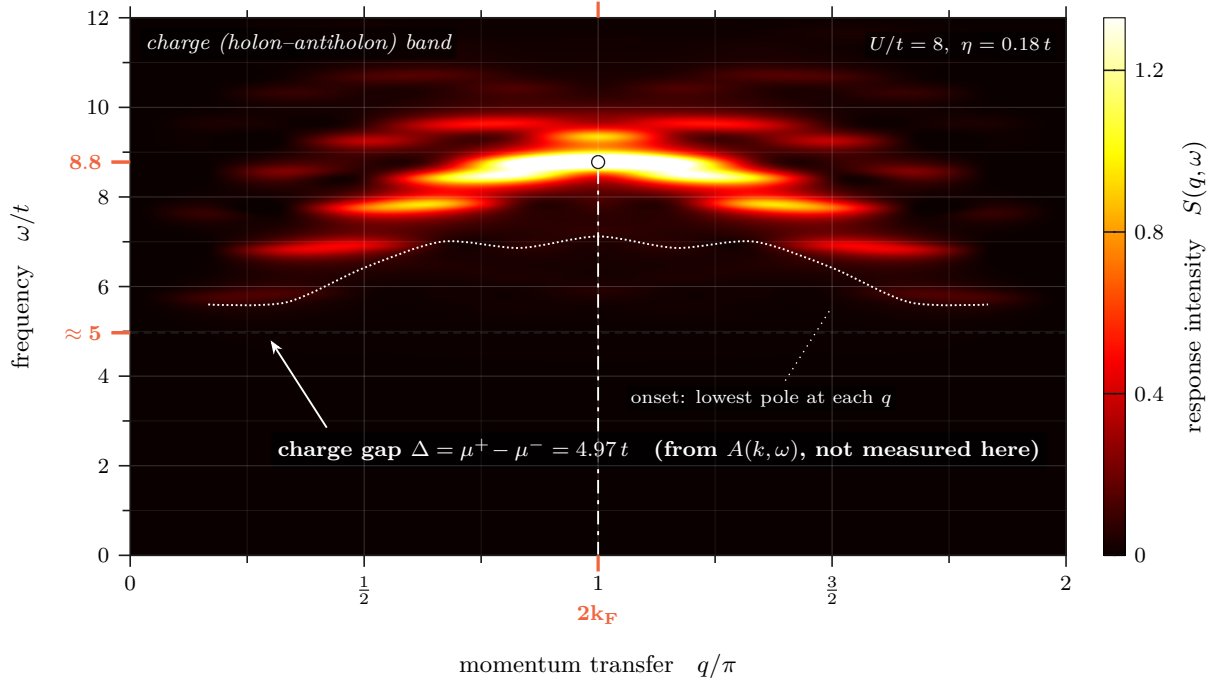


FIG. 7. **Dynamical structure factor: the density response.** $S(q, \omega)$ of the half-filled periodic Hubbard ring ($U/t = 8$, $L = 12$, Lorentzian broadening $\eta = 0.18t$)—the density–density response, a number-conserving (neutral) excitation, computed by a Haydock continued fraction of depth $n_\ell = 200$ in the N -particle sector—a finite-depth quadrature of the spectral measure, not the Lehmann sum (Sec. S7). Color is the *response intensity* $S(q, \omega)$ (units $1/t$): bright means the system responds strongly at that (q, ω) , dark means it does not. The weight forms the two-particle charge (holon–antiholon) band, whose dotted curve is the onset, the lowest pole resolved at each q , and sets the low-frequency scale of the band and traces its lower-boundary dispersion; the diffuse upper boundary is left unmarked. At $L = 12$ this band is six to nine discrete poles per momentum merged by the broadening, not a dense spectrum (Table S7). Two features are decisive. (i) The band is gapped on the scale of the charge gap: every resolved pole lies above $\Delta = \mu^+ - \mu^- = 4.97t$ (exact $(N \pm 1)$ ground-state energies), the lowest onset reached here being $5.60t$ at $q/\pi = 1/6$. The limit $q \rightarrow 0$ is inaccessible to this observable, $S(q=0, \omega>0)$ vanishing identically by density conservation, so Δ is drawn as a horizontal reference level imported from the single-particle channel and nothing is claimed about that limit. It is the *same* Mott gap that separates the Hubbard bands in the single-particle channel (Fig. 4): two independent dynamical observables, one correlation gap. (ii) The band peaks at $q = 2k_F = \pi$ at $\omega = 8.78t$ (coral tick, at the measured peak), within 10% of the bare Hubbard scale $U = 8t$ of a doubly-occupied site; the tick marks the peak, not U . The reflection $S(q) = S(2\pi - q)$ holds to $\sim 10^{-4}$ (Lanczos precision), and the analytic f-sum rule $\int \omega S(q, \omega) d\omega \propto (1 - \cos q)$ is recovered by the plotted (finite-window) grid to $\sim 0.3\%$. This is the collective observable the same bitstring-sampled primitive targets; as for $A(k, \omega)$, faithfully sampling the *full* response carries the sector-fraction cost of Sec. VII. This channel carries no shot budget: it is a full-sector reference (Table I).

and the vanishing error is an identity rather than a measurement of device fidelity (Fig. 9(a)). The circuits prepared a computational-basis $(N+1)$ determinant rather than $\hat{c}_p^\dagger |\Psi_0\rangle$ and applied $R_{xx}R_{yy}$ hopping without Jordan–Wigner strings, so they are not the primitive of Fig. 1. The post-selection, moreover, read each bitstring in reversed qubit order, which reflects the ring and exchanges the two spin species: under that reading none of the 3.5×10^5 shots of a noiseless simulation of the same seven circuits is retained, against all of them under the correct reading, so every one of the 300 retained determinants arose from a device error (`src/hw_bitorder_check.py`). The run is therefore an execution record: the steps of the pipeline execute end to end on a processor, but the run does not show that the device sampled the intended distribution, and it establishes nothing about accuracy at complete or incomplete coverage. The second, stretched N_2 at 24 qubits on `ibm_marrakesh`, reaches 0.59 ± 0.14 mHa of the exact active-space energy after recovery, below the 29.5 mHa of the noiseless simulation of the same circuit (Fig. 9(b)). The record does not store the recovered subspace sizes on which that comparison turns, so it cannot say whether device noise helps, although a study under a simple noise model reports the mechanism [86]. That study also finds that recovery from randomly generated configurations alone builds accurate CI spaces, and no such control was run here. What is controlled is recovery against no recovery on the same corrupted bitstrings, and there the gain grows with the noise (Figs. S7 and S8). Neither run is a spectral record in the sense of Refs. [11, 12, 15], whose spectra are determined by device data; the full account, the

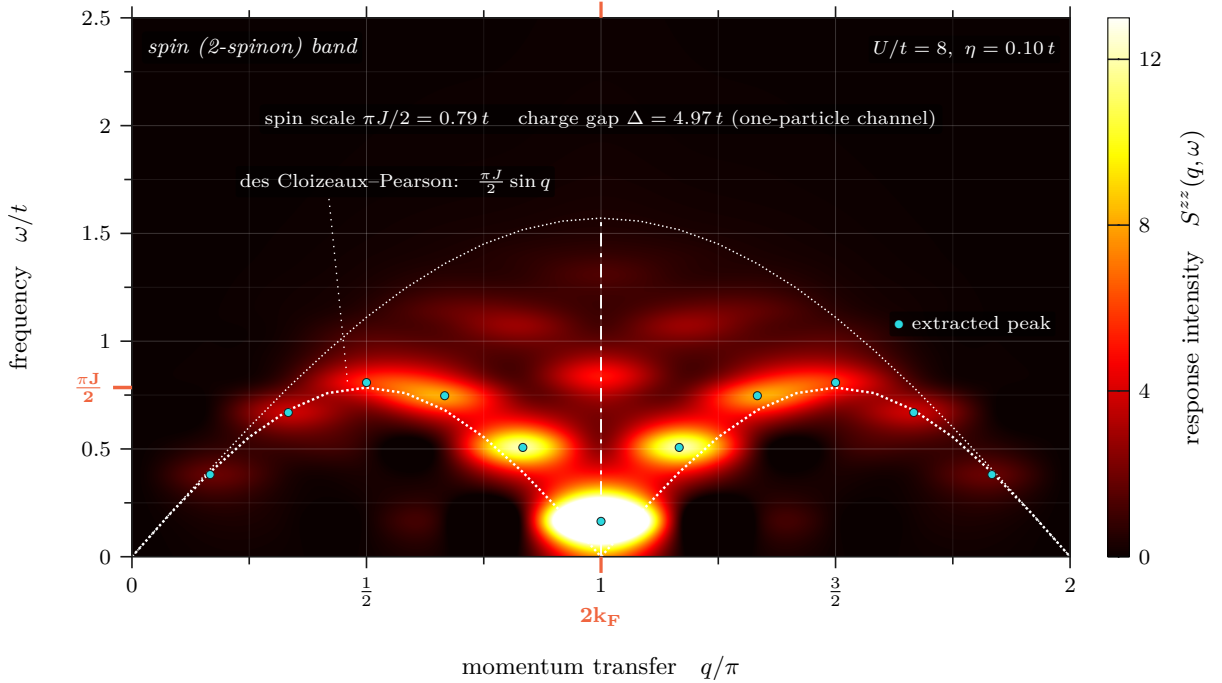


FIG. 8. **Spin structure factor: the low-frequency companion of the charge response.** The dynamical spin structure factor $S^{zz}(q, \omega)$ of the same half-filled periodic Hubbard ring ($U/t = 8$, $L = 12$; here $\eta = 0.10 t$, finer because the spin scale is small), computed by a Haydock continued fraction of depth $n_\ell = 200$ from a spin seed $S_q^z|\psi_0\rangle$ in the N -particle sector; at this η a reference of that kind is not converged in general (the single-particle one moves by 3.0% between $n_\ell = 100$ and 200, Sec. S6), so the lineshape is quoted with its depth and not as exact (Sec. S7). *Contrast Fig. 7:* the charge scale stays at the Mott gap $\Delta \approx 5 t$ at every size measured, while the spin scale is $0.16635 t$ at $L = 12$ and closes as $\approx 2/L$ (Table V). At this size the spin response is one dominant pole plus at most two satellites per momentum (Table S7), not a dense band, and what the figure shows is a set of broadened lines; whether the band touches $\omega = 0$ in the thermodynamic limit is not established here and is not claimed. This is spin-charge separation in the collective (two-particle) channel: charge and spin live in energetically separate worlds, the spin set by the superexchange $J = 4t^2/U = 0.5 t$, some $16\times$ smaller than U . The spectral weight piles up on the des Cloizeaux-Pearson [54] lower boundary $\frac{\pi J}{2}|\sin q|$ (dotted), bounded above by $\pi J|\sin \frac{q}{2}|$ [85]; the numerically-extracted peak dispersion (cyan markers: the maximum of each broadened line on the $0.006 t$ frequency grid) tracks this lower boundary to within about 3% for $q \leq \pi/2$ (and $q \geq 3\pi/2$); toward $q = \pi$ it lies above it, by 10% at $q = 2\pi/3$ and 29% at $q = 5\pi/6$, and at $q = \pi$, where the boundary vanishes, it sits at the finite-size spin gap; the antiferromagnetic soft mode at $q = \pi$ carries the dominant weight. The reflection $S^{zz}(q) = S^{zz}(2\pi - q)$ holds to double-precision round-off and $S^{zz}(q \rightarrow 0) \rightarrow 0$ (total- S^z conservation). The same bitstring-sampled primitive that targets $A(k, \omega)$ and $S(q, \omega)$ targets this collective response as well. This channel carries no shot budget: it is a full-sector reference (Table I).

job identifiers and the shot arithmetic are in Sec. S8.

The molecular suite. On the seven Hamiltonians of the nineteen-molecule suite (Sec. S15.1, Table S11) whose sectors have dimension 20 to 300, the same primitive reproduces the orbital-projected $A(\omega)$ to a relative L_1 between 2.2×10^{-6} and 1.0×10^{-4} from 21% to 50% of the sector, a consistency check of the primitive outside the Hubbard family and not a scaling statement. The other twelve reconstructions are exact by construction, because their sampled subspaces, of at most twelve determinants, already contain the probe's Krylov space, so the suite cannot show that the reconstruction is efficient across chemistry.

B. What the cost tracker implies for classical simulation

The cost tracker of Sec. VI, the bond dimension χ , is by definition the cost parameter of a matrix-product representation of the same state: wherever it says the determinant support is small, a matrix-product state is small too. A sampler can therefore be needed only for a target whose minimal bond dimension is large in every ordering while the sampled support stays compact, a property of geometry and orbital connectivity that two-dimensional lattices and molecular orbital graphs would test and that nothing here measures (Sec. S8). Every target measured

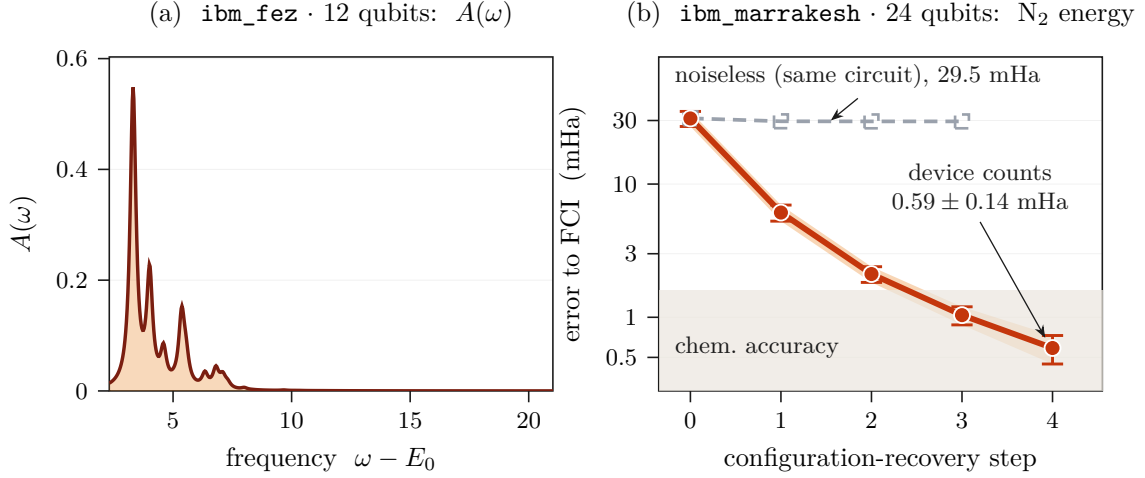


FIG. 9. **The complete quantum-hardware record of this work: two runs on IBM Heron processors.** (a) Electron-addition branch of the local spectral function, $A_{0\uparrow}^+(\omega)$, of the $L = 6$ periodic Hubbard ring at $U/t = 4$, $\eta = 0.15t$ (12 qubits; the classical validation of Fig. 2 is a different instance, $U/t = 8$), computed on the subspace post-selected from 350,000 computational-basis bitstrings—50,000 from each of seven circuits—measured on **ibm_fez** (measurement and Pauli twirling and dynamical decoupling; no readout-error mitigation and no configuration recovery). The retained shots covered the complete $(N+1)$ sector ($|\mathcal{S}| = 300/300$), so the reconstruction is exact by coverage; they were post-selected in reversed bit order, so every retained shot is a device error (Sec. IX A). The panel is an execution record, and neither a fidelity measurement nor a claim of advantage. (b) Ground-state energy error of stretched N_2 ($R = 2.0$ Å, CAS(10e,12o)/cc-pVDZ, 24 qubits) versus self-consistent configuration-recovery step, on **ibm_marrakesh** (10,000 shots, CCSD-seeded local unitary cluster-Jastrow (LUCJ) circuit). Recovery on the device counts (solid) reaches 0.59 ± 0.14 mHa of the exact FCI value—well below chemical accuracy—whereas the noiseless statevector simulation of the *same* circuit (dashed) is stranded at 29.5 mHa (its recorded recovery history ends at step 3, unchanged since step 2). That contrast is not read here as device noise *helping*: the mechanism such a reading needs is decided by the two recovered subspace sizes, and the record stores neither (Sec. IX A). The solid curve is the mean over 8 recovery seeds on the cached device counts; the shaded envelope and the capped whiskers at every step both show the ± 1 population s.d. (normalized by 8, not 7) across those 8 seeds (widest at step 0, 31.1 ± 4.0 mHa, where it starts together with the noiseless curve, and tapering as recovery converges to 0.59 ± 0.14 mHa; best seed 0.40 mHa). No random-bitstring control was run, so the panel cannot separate what the device counts contributed from what recovery alone supplies; the controlled recovery-versus-no-recovery comparison, under a simulated bit-flip model, is Fig. S8. These two runs are the entirety of the quantum hardware used here; every other result in this work is an exact classical simulation.

here—Hubbard rings at $L \leq 14$, nineteen active spaces, and a largest minimal bond dimension of 42 over the 74 deposited exact ground states—lies on the other side: there is no computation performed here that a classical method cannot perform, and we make no advantage claim at any size reached. The literature shows the same pattern. The 7260 trajectories of a 120-qubit Fermi–Hubbard dynamics experiment [87] were reproduced by tensor contraction at bond dimension 32 [88], the pattern in which tensor-network and Pauli-propagation methods have overtaken earlier hardware demonstrations [89–91], with Pauli propagation since extended to real-time correlators [92]; the energy of any single-layer unitary cluster-Jastrow circuit, the largest (77-qubit) experiment of Ref. [35] included, is now computed classically in polynomial time [93], and an extended matchgate simulator evaluates, to additive approximation, the Born-rule probabilities of hardware-sampled cluster-Jastrow bitstrings at a cost exponential only in the magnitudes of the controlled-phase angles [94]; on the ground-state energy the strong classical selectors already win on cost [95, 96], and even optimally ordered SQD subspaces need a number of configurations that grows exponentially with lattice size for Heisenberg and Hubbard ground states [97].

A dequantization test. On an emulated, under-sampled head for N_2 ($R = 2.5$ Å; 1000 shots drawn from the exact FCI marginal and corrupted by a synthetic Heron-class noise model, ten paired seeds; no device data), a classical Epstein–Nesbet prior [98] applied to the same histogram cuts the energy error by 36%, from 38.22 to 24.38 mHa (paired $t_9 = 12.37$, all ten seeds), while a generative flow network [99] fused onto that prior moves it a further 1.80 mHa, a difference ten paired seeds do not resolve ($t_9 = 1.19$) and whose sign reverses between two five-seed halves; at 500 shots the fused generator is worse than the classical control (Sec. S16). On this instance the dominant determinants are supplied by the sampled histogram (emulated, so the test does not measure whether a device state would supply them) and completed by a classical prior rather than manufactured by a learned generator, and the largest improvement available to an under-sampled head costs no quantum resources at all.

C. Scope

What the work establishes is four things, each with its measurement attached: one sampling primitive that targets four frequency-resolved channels, of which the single-particle ones are reconstructed here from ranked or sampled subspaces and the two neutral ones are full-sector references (Table I); a two-sided bound on the error of the reconstruction, whose leakage term is a posteriori, with its non-vacuity threshold measured at five sizes up to a sector of dimension 10 306 296 and its crossing calibrated to a true relative error of $\approx 3 \times 10^{-3}$ within a factor of three (Sec. V); an obstruction showing that no orbital-rotation invariant supplies an a priori predictor of that bound useful on every state (Sec. VI); and three prices—support, resolution and shots—each measured here (Sec. VII).

The momentum-resolved spectra are one-dimensional Hubbard rings at $L \leq 12$ with $\eta = 0.10\text{--}0.18t$ (the support, certificate and shot measurements reach $L = 14$), which is the exact-diagonalization convention for this observable—Xu, Lu, Tohyama and Shao compute it at $L = 12$ and $\eta = 0.2t$ at half filling, and at $L = 16$ at quarter filling, in the extended Peierls–Hubbard chain [100]—but the resolution criterion the dynamical-DMRG literature applies at those sizes is not met here, so Sec. VIII rests its spin–charge statement on pole positions at four sizes rather than enlarge η .

D. Data availability statement

The data that underlie the figures and the quoted numbers of this work are openly available in the reproduction repository [101] at <https://github.com/nicolasbonilla/dynamical-spectral-functions-sqd>, archived on Zenodo under DOI 10.5281/zenodo.22967220, the code under the MIT license and the manuscript, the deposited datasets and the reproduction documentation under CC BY 4.0. Every figure’s plotted values are deposited—as a JSON table in `data/` or as a plain-text table in `paper/figs/`—and `docs/FIGURE_PROVENANCE.md` names the file that is authoritative for each; the one exception is Fig. 1, a circuit schematic that plots no data. The classical data—the exact-diagonalization spectral functions of the one-dimensional Hubbard model, the molecular suite and the resource statistics—are additionally *regenerable*: the computation scripts that produced them are deposited in `src/`, and `docs/REPRODUCE.md` gives the reproduction commands together with a measured table of what a clean clone does and does not reach. An automated guardian (`python src/verify.py`) recomputes the underlying physics by exact diagonalization and checks the deposited artifacts against it—1280 checks over 14 563 numeric assertions, exiting non-zero on any discrepancy. It covers the deposited data files and the numbers printed in the figure sources, in the tables and in the abstract; it is not a proof that every sentence of this manuscript is verified, and four items are printed by name on every run, the known open defects; the ratios plotted in Fig. S5 are re-derived by it from the deposited certificate rows.

The quantum-hardware results are deposited as the arrays that enter the figures, together with the backend name, the shot count and the IBM Quantum job identifier for each of the two runs (`data/heron_spectral.json`, `data/hw_lucj_n2_result.json`). *The raw measured bitstrings are not part of the deposit*: the counts can be obtained only from the IBM Quantum service, by retrieving the recorded job identifier (`service.job(...)`) in place of the submission cell of the notebooks in `notebooks/`, which requires an account and a job whose results are still retained, so the classical post-processing of the raw counts cannot be re-run from the deposited files alone. The hardware runs themselves cannot be regenerated in any case—the jobs are closed, and a device executed today has a different calibration, so new counts would be new data rather than a reproduction. The dequantization control of Sec. S16 is regenerated seed by seed by `src/gflow_dequant.py`, which needs `pyscf` and `torch` and writes `data/gflow.json`. The $L = 14$ ground-state checkpoint (94 MB), on which the largest vacuity factor and the $L = 14$ support count rest, is deposited in `ckpt/` with its SHA-256 checksums. Known gaps between the deposited code and the deposited artifacts are enumerated in `docs/KNOWN_DISCREPANCIES.md`; those still open are the rasters embedded in Figs. 4, 7 and 8, which are deposited as rendered and cannot be regenerated from the deposit; the third panel of Fig. 5 and the captions of Figs. 3 and 6, edited by hand after generation; two computation scripts without deposited output; and the parts of the pipeline that require `pyscf` or an IBM Quantum account.

X. CONCLUSION

For a spectral function reconstructed by Rayleigh–Ritz on a sampled subspace, the captured Born weight does not bound the error; the weight the subspace missed, together with the Hamiltonian coupling across its boundary, does. Theorem 1, for any orthogonal projector and given the exact ground state, bounds the L_1 error from below by the Born weight the subspace failed to capture and from above by that weight plus the coupling that leaks across the subspace boundary at the resolution asked for, a quantity computed from the Ritz pairs and the retained part of the probe, without the spectral function it bounds. The captured weight alone cannot replace it: an analytic counterexample and

a counting argument rule out any bound indexed on it, and on the probe's own Born support, where the captured weight is exactly one, the relative error is still 0.43 at $L = 6$ and 0.35 at $L = 8$ at $\eta = 0.18t$, because the weight a subspace keeps is displaced in ω , not discarded (Sec. III D). What a Krylov-adequate subspace does guarantee is the classical moment exactness of Krylov projections, through order $2K + 1$ and no further, a reach stated exactly and measured on coordinate subspaces (Theorem 2). And no orbital-rotation invariant bounds the boundary *usefully* on every state in advance, on pairing-model fibers every such bound being vacuous: one-body fermionic magic is constant along an orbit that contains a point of classical cost $O(K)$, at bond dimension 2 and 2^K determinants in the pairing basis, and points at which sampling is plausibly hard, a hardness not established here, and on fibers on which the subspace-aware certificate still separates (Theorem 3, Proposition 2), consistent with the separation drawn by Tarabunga *et al.* [17] between covariance-matrix quantities, two of which lower-bound the non-Gaussian gate count, and full-state natural-orbital participation entropies, which upper-bound the classical simulation cost in an orthonormal Gaussian basis.

Evaluated on the subspaces of the resource scan at five system sizes up to a sector of dimension 10 306 296, the certificate is vacuous at every operating fraction of the resource scan, by factors of 1.66 to 8.61, converged in Krylov depth at $L = 12$, and informative only well above them. Where it is informative it is calibrated: over sixteen (L, η) cells at $L = 6$ –12 the crossing of the trivial bound occurs at a true relative L_1 error between 1.2×10^{-3} and 7.3×10^{-3} , a drift, not scatter, ordered in both arguments and conservative in L , computed without the spectral function it bounds, given the exact probe. A two-constant rule in $1 - w_S$, fitted out of sample, localizes the same crossing more tightly on eleven of the fourteen splits; what distinguishes the certificate from such a rule is that it is an inequality with a proof, whose decision survives replacing w_S by a floor. The Born-ranked, infinite-shot reconstruction of Fig. 3, at 85% of its sector, is non-vacuous in all sixteen channels, at a relative bound of 0.77–1.75 against the trivial 2, 350–900 times its true error. The basis dependence of the boundary term is measured as well. In the determinant basis a device measures in, the coordinate support of the Krylov space $\mathcal{K}(H, \phi)$ is every determinant the reflection about the seed site allows, measured at $L = 6$ to 14, so at full captured weight a subspace holding the whole probe leaks unless it contains all of them; the vacuity at the operating fractions, all below full captured weight, is measured and not deduced from that fact. A momentum-resolved probe measured in plane-wave orbitals has its Krylov space inside one momentum block instead, $1/L$ of the sector, on which the certificate still needs 455–483 of 490 determinants at $L = 8$, a gain of $1/L$ against an exponential (Sec. V E). The method is priced in support, resolution and shots (Sec. VII): the determinant support that reconstructs $A(\omega)$ to relative $L_1 < 0.05$ at $\eta = 0.15t$ grows as $e^{1.05L}$ over $L = 4$ –14, and the shot budget grows from 7.8×10^2 to 1.9×10^7 over $L = 4$ –12. The full-sector references reproduce the opposite finite-size scaling of the charge and spin gaps, the spin gap closing as $\approx 2/L$ with an independent effective-Heisenberg control (Table V).

The certificate does not certify the reconstructions at the operating fractions of the resource scan, and the regime where it would costs more shots than the method saves: moving from the operating fraction to the non-vacuous band multiplies the shots per snapshot needed for the marginal configuration by 7.5 to 10 at $L = 8$ and by 20 to 35 at $L = 12$, and at the acceptance criterion this work uses elsewhere, relative $L_1 < 0.05$, the certificate needs at least 0.98 of the sector at $L = 8$, a fraction at which diagonalizing the whole sector is what one would do instead, and one not reached on the grid at $L = 6, 10$ or 12. The bound refers both spectra to the exact ground-state energy (H1); in a shot-based pipeline E_0 is itself estimated on a sampled subspace, and the bound then holds only after an additive term the certificate does not control; for the shift of $6.7 \times 10^{-4}t$ that 60% of the N sector gives at $L = 8$ that term moves the bound by only +0.20%, while the measured error grows by a factor 16 (Sec. S11). It builds the probe from the exact ground state (H4); from an approximate one it bounds the error of that approximate probe's own spectral function, whose distance to the exact one nothing computed here controls, and no certificate in this work has been evaluated with an approximate probe. All spectra here are at exact-diagonalization sizes, the largest momentum-resolved one at $L = 12$, whereas dynamical DMRG computed the same three channels at 60 to 120 sites in 2007 [3] and correction vectors reach $L = 48$ in the Hubbard chain [4]; one dimension is where tensor networks are near-optimal [102, 103], and the largest minimal bond dimension over the 74 exact ground states deposited here is 42 (Sec. IX).

Whether a certified regime exists at low sector fraction is now a well-posed measurement, and the calibrated crossing is the instrument for making it. Whether it exists at all turns on two open questions: whether the certificate of second order in the boundary coupling that follows from the same identity at full captured weight has a counterpart below it, where the operating fractions lie, that can be evaluated without the answer; and whether some single-particle basis gives $\mathcal{K}(H, \phi)$ a coordinate support that is a small fraction of the sector, the plane-wave block being one such basis for a momentum probe, at about $L(L - 1)$ further two-qubit gates whose effect on the sampled distribution is unmeasured, and with the filling of the block at $L \geq 10$ unmeasured as well. What this work provides toward both is a two-sided bound whose boundary-leakage term is computable from the Ritz pairs and the retained probe, a measured map of where it is informative and at what cost, and the coordinate support that fixes, basis by basis, what zero leakage at full captured weight requires. For a sampled pipeline validated at exact-diagonalization scale, where the exact ground state is in hand, the consequence is that the captured weight is not a convergence diagnostic for the spectral function built on the subspace; what bounds that error, as an exclusion rather than an error estimate, is the weight the subspace

missed together with the boundary coupling, computable there from the same Ritz pairs, the retained probe and one matrix–vector product per Ritz vector.

ACKNOWLEDGMENTS

We acknowledge the use of IBM Quantum services for this work; the two device runs reported in Sec. IX A were executed on `ibm_fez` and `ibm_marrakesh`. The views expressed are those of the author and do not reflect the official policy or position of IBM or the IBM Quantum team.

AI tools were used in the preparation of this manuscript; the author takes full responsibility for its content.

Appendix A: Proof of the leakage certificate and of its closed form

This appendix proves Theorem 1 and the second equality of Eq. (6).

Proof. Write $u = R(z)\phi$ and $u_P = R_P(z)\phi_P$, extended by zero outside $\text{ran } P$.

(1) *The split.* Since $u_P \in \text{ran } P$ and $(z - H_P)u_P = \phi_P$ there, $(z - H)u_P = zu_P - PHu_P - QHu_P = \phi_P - QHu_P$; subtracting this from $(z - H)u = \phi$ gives

$$u - u_P = R(z)[\phi_Q + QH R_P(z)\phi_P]. \quad (\text{A1})$$

This is the Feshbach–Löwdin partition of the resolvent [55, 56], written as an exact identity rather than as an expansion: the first source term is the part of the probe the subspace never held, the second is the part of the Hamiltonian that carries amplitude out of it. Only $P^2 = P$, $P^\dagger = P$, $H = H^\dagger$ and $\eta > 0$ are used; no property of the determinant basis, of the ranking, or of the sampler enters anywhere below.

(2) *Pointwise.* $A - A_P = (\eta/\pi)(\|u\| + \|u_P\|)(\|u\| - \|u_P\|)$ and $|\|u\| - \|u_P\|| \leq \|u - u_P\|$, so $|A - A_P| \leq (\eta/\pi)(\|u\| + \|u_P\|)\|u - u_P\|$ at every ω .

(3) *Integration.* By Cauchy–Schwarz in ω ,

$$\|A - A_P\|_1 \leq \frac{\eta}{\pi} \left[\int (\|u\| + \|u_P\|)^2 d\omega \right]^{1/2} \left[\int \|u - u_P\|^2 d\omega \right]^{1/2}.$$

The first bracket is at most $(\pi/\eta)^{1/2}(\|\phi\| + \|\phi_P\|)$, by Minkowski in $L^2(d\omega)$ and the two sum rules. For the second, apply Minkowski to Eq. (A1): its first term obeys the sum rule with seed ϕ_Q , contributing exactly $(\pi/\eta)^{1/2}\|\phi_Q\|$; its second obeys $\|R(z)v\| \leq \eta^{-1}\|v\|$, since H is self-adjoint and $\text{Im } z = \eta$, and the definition (6) then contributes at most $(\pi/\eta)^{1/2}\Lambda_P(\eta)/\eta$. Multiplying the three factors gives the leakage branch of (7).

(4) *The other two branches.* A and A_P are nonnegative with known integrals, so $\|A - A_P\|_1 \leq \int A + \int A_P = \|\phi\|^2(1 + w_P)$ under no hypothesis at all; and $|\int A - \int A_P| \leq \|A - A_P\|_1$ gives the lower bound $\|\phi\|^2(1 - w_P)$.

(5) *The closed form.* $R_P(z)\phi_P = \sum_m c_m(z - \tilde{E}_m)^{-1}|\tilde{m}\rangle$, so

$$\|QH R_P(z)\phi_P\|^2 = \sum_{m,m'} \frac{c_m \bar{c}_{m'} \langle g_{m'}, g_m \rangle}{(z - \tilde{E}_m)(\bar{z} - \tilde{E}_{m'})}.$$

With $x = \omega + E_0$ the remaining integral is $\int dx [(x - \tilde{E}_m + i\eta)(x - \tilde{E}_{m'} - i\eta)]^{-1}$, which has one pole in each half plane; closing above picks up $2\pi i/[(\tilde{E}_{m'} - \tilde{E}_m) + 2i\eta]$, and multiplying by η/π gives Eq. (6). $\square$

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Supplemental Material

Captured weight and boundary leakage bound the error of sample-based spectral functions

S1. THE PRIMITIVE: PROVENANCE DECLARATIONS IN FULL

Section II states the measurement primitive, fixes the one definition the rest of the paper turns on, and then says — channel by channel, figure by figure — which objects here were reconstructed from a finite number of measurements and which are full-sector references computed classically. That line is routinely drawn nowhere in the text, and a single clause of an abstract is enough to draw it in the wrong place. It is drawn here, explicitly, before any result is quoted.

Two consequences of Rayleigh–Ritz. Two corollaries of Eq. (3) are used later. A determinant acts on A_S through H_S even carrying no seed amplitude: at sector fraction 0.85, 30.2% of the selected determinants at $L = 6$ and 42.0% at $L = 8$ lie outside $\text{supp}(\phi)$, add exactly nothing to the captured weight, and still move the Ritz spectrum. And on the smallest subspace of *unit* captured weight, $S^\star = \text{supp}(\phi)$, of size $(\frac{1}{2} + \frac{1}{L})$ dim at $L = 6$ and 0.571 dim at $L = 8$, the relative L_1 error of Eq. (3) does not vanish either; it is measured there at both resolutions, on both boundary conditions and against its ceiling, in Sec. IIID.

Table I gives, for every dynamical object in this paper, the subspace it was built on and the measurements that built it. Three readings the word *sampled* invites are closed off by it.

The neutral channels are references, not reconstructions. $S(q, \omega)$ and $S^{zz}(q, \omega)$ are computed here by a continued fraction over the *entire* N -particle sector of 853 776 determinants at $L = 12$, at recursion depth $n_\ell = 200$: no subspace is selected and no shot is drawn, and the generating scripts say so in their own headers. The primitive *targets* these two channels — that is what Fig. 1 asserts, and what the seed algebra supports — and targeting a channel is not reconstructing it from measurements; they were not, nor were the $A(k, \omega)$ of Fig. 4 or the N_2 series of Fig. S6. A reference at finite recursion depth is not the exact spectral function either, so its depth is quoted with it, and we quote no error that lies within a factor 20 of the drift of its own reference.

What the Born ranking is, and what it is not. The two objects it would be easiest to call sampled without a budget — Fig. 3(a,b) and the dashed curve of Fig. 5(b) — select S by sorting the accumulated Born weight $\sum_k |\langle d | e^{-iHt_k} \phi \rangle|^2$ of the evolved state over the $K+1$ snapshots and keeping the top fraction. That weight is the distribution the device draws from, not the ground-state amplitudes and not the exact Lehmann residues, so the ranking is *not* a clairvoyant oracle: the repository contains one of those, in `amortized_recovery.py`, and labels it as such there. It is the infinite-shot limit $T \rightarrow \infty$ of the declared protocol, approached from above in error by any finite budget — at matched subspace size finite sampling is worse by 1.50 ± 0.25 at $L = 6$ and 1.96 ± 0.12 at $L = 8$ (Table S6) — and in that limit it selects nothing the device could never return: the only determinants of exactly zero accumulated weight are the reflection-forbidden ones (4, 18, 48, 200 at $L = 6-12$), which rank last and never enter S . What it is not is a finite-shot experiment, and it must not be called one. The replacement is measured and quoted with its budget: at $L = 8$, $T = 2.60 \times 10^6$ shots over $K+1 = 19$ snapshots reach sector fraction 0.835 ± 0.0025 at relative L_1 error $(4.81 \pm 0.27) \times 10^{-3}$ over eight independent shot streams (Fig. 3(c)), and the measured ladder from 2.4×10^4 to 1.9×10^7 shots for $L = 6$ to $L = 12$ is Fig. 5(c) and Table IV. The literal configuration of Fig. 3(a,b)— $\eta = 0.18t$, $K = 16$, all sixteen channels—has also been run at finite shots, at two budgets a factor two apart: 5.2×10^6 per channel suffice for panel (b)’s accuracy, but on a *larger* subspace, and at half that budget, at a fraction just under the announced one (0.839 against 0.850), the penalty is a measured 2.61 and not the 2.05 Table S6 would transfer (numbers in the caption of Fig. 3).

What the a posteriori checks establish. The branch sum rules $\int A_p^\pm d\omega$ are identities of the ground state, not measurements of a reconstruction. Integrated on the grid the figures use, at $L = 6$ and $\eta = 0.15t$, the exact and reconstructed addition branches carry 0.48659 and 0.48647 — agreement to 2.4×10^{-4} relative, both falling 2.7% short of the analytic $1 - \langle \hat{n}_p \rangle = 0.5$ because Lorentzian tails of width η leave the finite frequency window. That is a genuine consistency check on a zeroth moment, and not an error bound. The certificate is the two-sided bound of Theorem 1, and where it is non-vacuous is measured, not asserted, in Sec. V — an essential property given that the classical hardness of a specific instance is empirical rather than a theorem (Sec. IX). What is guaranteed in advance is narrower: a subspace containing the order- K Krylov space reproduces that branch’s exact spectral moments through order $2K+1$ (Theorem 2), and $T \geq \varepsilon^{-1} \log(1/(\varepsilon\delta))$ shots capture every configuration of Born probability at least ε (Prop. 1); the second does not imply the first (Sec. IV).

S1.1. The two budgets

The two halves of this paper are the two terms of one inequality. With Q_S the projector complementary to P_S and $w_S = \|P_S \phi\|^2 / \|\phi\|^2$, the upper branch of Eq. (7) reads

$$\|A - A_S\|_1 \leq (\|\phi\| + \|P_S\phi\|) \left(\underbrace{\|Q_S\phi\|}_{\text{missed weight}} + \underbrace{\Lambda_S(\eta)/\eta}_{\text{boundary leakage}} \right).$$

The captured-weight term $\|Q_S\phi\| = \|\phi\|\sqrt{1-w_S}$ vanishes once the sampler has exhausted the Born support of the probe, and Table I prices that exhaustion in measurements. The leakage term $\Lambda_S(\eta)/\eta$ falls as S grows along the sampler’s ranking (Sec. V), but the captured weight does not control it, and it does not vanish with it. On the subspace that *is* that support — $S^* = \text{supp}(\phi)$, of size $(\frac{1}{2} + \frac{1}{L}) \dim$ at $L = 6$ and $0.571 \dim$ at $L = 8$, fixed by the probe and not by any ranking, so that $w_S = 1$ exactly by construction, with no tie-breaking and no zero-amplitude padding — the weight term is identically zero and the relative error is still 0.43 at $L = 6$ and 0.35 at $L = 8$ at $\eta = 0.18t$, rising to 1.15 and 0.73 at $\eta = 0.05t$. $\Lambda_S(\eta)$ is defined by the resolvent at resolution η and exists only inside the spectral half of this work: the obstruction theorem of Sec. VI takes it as its subject and would have no subject without Sec. III. Conversely, Sec. III leaves open precisely the question Sec. VI answers — whether the a posteriori certificate can be replaced by an a priori predictor computed from the state. Neither half can be removed without leaving the other incomplete in a way we can name.

S2. THE LEAKAGE CERTIFICATE AND THE WEIGHT-ONLY BOUND: HYPOTHESES AND DETAILS

The hypotheses of Theorem 1, and its proof. Four hypotheses carry the theorem, each a property of the pipeline rather than of the inequality: (H1) A and A_P are referred to the same E_0 ; (H2) the L_1 norm is taken over the whole line and normalized by $\|\phi\|^2$; (H3) A is the exact spectral function of the full sector; (H4) the probe is built from the exact ground state. They are stated with the theorem in Sec. III, and each is checked against this work’s own pipeline, with numbers and with the measured consequence of its failure, in Sec. S11. The proof of the theorem and of the closed form, Eq. (6), is Appendix A of the main text; its steps are numbered (1) to (5) there.

What object is certified. If the resolvent inside $\text{ran } P$ is itself evaluated by a Krylov recursion of depth n_ℓ , then Λ computed on that Krylov space bounds the error of the depth- n_ℓ reconstruction *and of nothing else*. The convenient inequality $\Lambda_K \geq \Lambda_S$ that would transfer the statement to the dense A_S is false: 1169 of 6000 random triples violate it, at a minimum ratio of 0.5258. Every table of Sec. V carries the depth of the object it certifies for that reason. Full reorthogonalization is mandatory throughout: without it P_K is not a projector—the orthogonality defect reaches 0.238 at $L = 8$, where the captured weight then reads 1.4868, an impossible value—and Λ is inflated thirtyfold.

Relation to the classical error theory, and the scope of the novelty claim. Nothing in the proof of Appendix A is new as analysis. The split, Eq. (A1), is Löwdin partitioning [55], the same algebra as the Feshbach projection of scattering theory [56]; and sixty years of Rayleigh–Ritz and Krylov error analysis for resolvents and matrix functions [52, 57] contains every rung of steps (2) and (3), along with far sharper residual, Temple and Kato–Temple estimates for individual eigenpairs [58–60], their extension to the noisy, ill-conditioned eigenvalue problem of quantum subspace diagonalization [28], and the loss-of-orthogonality analysis of Paige [61] that makes step (5) conditional on reorthogonalization, and the polynomial-restart convergence theory of Beattie, Embree and Sorensen [62]. Step (3) in particular is the reduced-basis bound on a Galerkin projection onto an arbitrary subspace—the dual norm of the residual over an inf-sup constant [63, 64]—with that constant equal to η .

Nor is “a posteriori” ours to claim, and we name the prior art rather than wait to be shown it. Stopping criteria for rational matrix functions of Hermitian matrices, computed from the residual alone, are Frommer and Simoncini’s [65]; Jia and Lv’s title is this section’s claim word for word—a *posteriori error estimates of Krylov subspace approximations to matrix functions* [66]—and Frommer and Schweitzer carry that line to Stieltjes functions [67]; Lanczos approximations to matrix functions carry a posteriori bounds from the same Cauchy-integral argument as step (5) [68, 69], and stochastic Lanczos quadrature carries them on an approximated spectral measure itself, in Wasserstein and Kolmogorov–Smirnov distance, through Gauss-quadrature brackets that hold for each seeded (weighted) spectral measure and for the global density of states that averages them [70]—bounds on a full-space Lanczos quadrature, not on a projection onto a sampled determinant subspace with boundary leakage as bounded here; and in April 2026, before this work, on *quantum* Krylov, Oliveira, Ziemis and Glaser introduce “two new metrics, the imaginary and unitary filters, that successfully assess the reliability of the obtained solutions without any knowledge of the true eigenspectrum” [71]. That is the framing of this section, established earlier and by others, and the priority is theirs. On the selection side, that a subspace chosen for the energy misses configurations a response needs, and a selection criterion built for the response instead, are Reinholdt, Kjellgren and Kongsted’s, for classical selected configuration interaction [72].

What we did not find in that literature is the conjunction of three things, and it is the whole of the novelty claimed here. First, a closed form for the frequency-integrated residual $\Lambda_P(\eta)$, assembled from the Ritz pairs, on a subspace of *coordinates* carrying no Krylov structure, no invariance and no spectral-gap assumption—so that the estimate attaches

to the set of determinants a sampler returned, which is not the span of the iterates of any recursion, and Eq. (A1) is evaluated on that set directly. Second, that the bound is two-sided on the same object: the lower branch is fixed by the captured weight and the upper branch adds what the weight does not control, and Sec. V uses the distance between them rather than either alone. Third, a *measured* calibration of where that bound crosses the trivial one, over sixteen (L, η) cells whose true errors span five decades (Sec. VD), in place of an asymptotic statement about the crossing. Whether such a bound is *useful* at the fractions sampling reaches is a separate question; it is not settled by the proof, and Sec. V answers it—in the negative—by measurement.

The pooled sweep of Fig. S5, and its limits. The pooled sweep behind Fig. S5 contributes 621 subspaces from five Hamiltonian families and eight construction rules, written by two independent implementations; 15 reproduce their target to double-precision round-off and define no ratio. On the 606 that remain the weight-only bound fails on 469, and restricted to the regime this work operates in, $w_S \geq 0.99$, it fails on 363 of 363, the mildest by a factor 2.00 (Sec. S9). The compliant cases are subspaces so poor that the relative error is already of order unity, where the bound asserts nothing. In the configuration of Fig. 3 itself— $L = 8$, $\eta = 0.18t$, the 3920-dimensional $(N+1)$ sector, momentum seeds $\hat{c}_{k\uparrow}^\dagger |\Psi_0\rangle$, both branches, eight momenta, subspaces grown in order of accumulated Born weight—the median violation factor at sector fractions 0.50, 0.70, 0.85, 0.95 is 18.5, 23.6, 26.5, 29.9, the 64 rows spanning 15.1 to 57.3 overall and 15.1 to 33.7 at the fraction 0.85 of Fig. 3. Those medians *rise* monotonically, by a factor 1.6, while the measured error over the same range falls by three orders of magnitude, from the 10^{-1} level at the lowest fraction to 1.0×10^{-4} at 0.95: the weight-only bound fails worst exactly where the reconstruction is best, which is the opposite of what an error bound is for. That is a defect of the functional form and not a pathology at an edge of parameter space, and it is therefore measurable: Fig. S5 plots it against sector fraction and against η , beside the replacement bound on the same subspaces.

Three properties of that sweep are declared with it, because each of them limits how it may be read. Its ranking uses $K = 16$ slices, one protocol step short of the $K = 18$ of the resource scan of Sec. VII. Its momentum seeds have support on 3848 to 3920 of the 3920 determinants and captured weight below one at every fraction, so the support floor derived below for the local seed does not apply to it; the members of a translation orbit carry equal Born weight, so the cut can fall inside a tied orbit. How far a tie-breaking can move an error was measured on a local-seed subspace at $L = 6$, fraction 0.85, whose cut fell among zero-amplitude determinants: ten tie-breakings moved the measured error by $\pm 20\%$, which is why the violation factors are quoted to three figures at most and why the rise across fractions is read as a direction and not as a rate. (That $\pm 20\%$ is the spread of the *measured error* over tie-breakings of a zero-amplitude cut. It is a different quantity from the 6.6×10^{-4} by which ties move the *bound* in the sweep of Sec. V, where the cut falls below the support floor and only two determinants are tied.) And the error at fraction 0.85 has two independent determinations that differ by a factor 2.3, so only its order of magnitude is quoted. None of this touches the conclusion: the two-level counterexample is analytic, and the counting argument of the next paragraph but one uses no ranking, no protocol and no numerics at all. The sweep is corroboration of a conclusion that does not depend on it.

Why the weight-only bound fails, in full. The reconstruction is Rayleigh–Ritz on $H_S = P_S H P_S$ (Sec. II), not the restriction of A to a subset of its own eigenpairs: the Ritz energies $\tilde{E}_m$ are not eigenvalues of H , so weight the subspace *keeps* is nonetheless moved in ω . The obvious sketch elides exactly this. Having correctly noted $\int A = \|\phi\|^2$ and $\int A_S = \|P_S \phi\|^2$, it bounds the L_1 discrepancy by twice the missing weight—a step legitimate when $[P_S, H] = 0$, where A_S is A with the outside contributions deleted and $\|A - A_S\|_1 = \|\phi\|^2(1 - w_S)$ identically, and controlling nothing otherwise. The two-level example is the extreme: nothing is discarded there, and every unit of retained weight is displaced by g . What the integrals do give, with no commutator hypothesis, is $|\int A - \int A_S| \leq \|A - A_S\|_1$, that is $\|\phi\|^2(1 - w_S) \leq \|A - A_S\|_1$ —a *lower* bound, with 0 violations across the 606 subspaces of the pooled sweep. It survives intact as the left-hand half of Theorem 1: the captured weight does say how wrong the reconstruction must be, it never said how wrong it can be.

The normalization of the witness errors. (Relative here and throughout this subsection means divided by $\|\phi\|^2$, the normalization of Theorem 1; dividing instead by the in-window integral of the exact A , as the repository’s own relative L_1 does, moves them by about 1% at $\eta = 0.05t$ and by 2.6 to 3.3% at $\eta = 0.18t$, measured in Sec. S11, and the two conventions are not interchanged anywhere in this paper without saying so.)

S3. MOMENT EXACTNESS: THE PROOF ROUTE, THE CONJECTURE AND THE SHOT BOUND

That proof is three lines of Rayleigh–Ritz, and it is not the proof this statement is usually given. The usual route argues that an order- $(K+1)$ quadrature “matches the first $2K+1$ moments exactly”, and $2K+1$ is exactly what Table II measures: the *reach* is right, the route is not. What does not hold here is the identification of A_S with a Gaussian-quadrature rule built by a nonorthogonal Lanczos/Krylov construction. No such object is built anywhere in this work: every reconstruction diagonalizes $P_S H P_S$ on a coordinate subspace, so $d\mu_S$ carries $|S|$ nodes, not $K+1$, and the orders that separate the two objects measure the distance between them, not an error in the published reach.

The conclusion is right, the route is not, and the correct route is also the more general one (Sec. S10.1).

Equation (11) invites a corollary: “consequently, at fixed broadening $\eta > 0$, the smoothed reconstruction converges geometrically, $\|A - A_{\mathcal{K}_K}\|_1 \leq C(\eta) \rho(\eta)^{-K}$.” It does not follow, and it is not a corollary here. Moment exactness is a list of $2K + 2$ linear identities, and the L_1 distance between two Lorentzian-broadened measures is not a finite linear functional of them; moment exactness supplies neither of the two ingredients a conversion would need (Sec. S10.3), and neither $C(\eta)$ nor $\rho(\eta)$ is derived or exhibited anywhere in this work or in the deposit. We therefore state the claim as what it is.

One constraint on any answer is already available. The two-level family of Sec. III D has $\mathcal{K}_0 = \text{span}\{\phi\}$, and by direct integration at $\eta = 0.15$ its $\|A - A_{\mathcal{K}_0}\|_1$ is 1.983 at $g/\eta = 2 \times 10^2$ and 1.998 at $g/\eta = 2 \times 10^3$, rising to $2\|\phi\|^2$. Evaluating Conjecture 1 at $K = 0$ therefore forces $C(\eta) \geq 2\|\phi\|^2$: the same family that forces the trivial constant on the weight-only bound forces it here on the prefactor. Whatever content the conjecture has must be carried by $\rho(\eta)^{-K}$ alone, and in particular it cannot be non-vacuous at the small K at which a shot-limited subspace operates.

One figure of this work is read against that conjecture and not against a theorem. A geometric collapse $\text{rel-}L_1 \sim \rho^{-K}$ with $\rho \approx 1.7$, fitted to panel (b) of Fig. 2, is easily reported as “precisely the geometric convergence guaranteed by the moment-exactness theorem”. It is guaranteed by no such thing—that is the corollary ruled out above—and the fit does not carry the claim on its own either. The panel is one of the two places in this work where the sampling is genuine and its budget is declared—multinomial draws at 4000 shots per time step, eight seeds, $\eta = 0.15t$, so $T = 4000(K+1)$ and $T = 5.2 \times 10^4$ at the last point—and the fit uses six of its thirteen measured points, inside a window selected after the fact ($4 \leq K \leq 9$, set as a literal in `spectral_validation_max.py`). Across that window the mean subspace the sampler returns grows from 179 to 236 determinants of the 300-dimensional sector, so decay in the Krylov order is confounded with growth in $|\mathcal{S}|$; and the K of the abscissa is the number of measured time slices, not the K_S of the weight-only bound, which is -1 at all thirteen points by Eq. (9). No rate is fitted to that window here, and the caption claims none. What the panel does show, over all thirteen points, with its eight-seed band and with T attached, is that the error falls by a factor 32 as the measured Krylov order is climbed at fixed η , from 0.59 at $K = 0$ to 0.0185 at $K = 12$, with a single non-monotone step at $K = 3$ inside the discovery phase; that is the statement its caption makes.

There is a sharper reason not to read Eq. (11) as an error bound, and Table II contains it. For a coordinate subspace P_S is diagonal, so $\text{span}(\mathcal{S}) \supseteq \mathcal{K}_K$ at any $K \geq 0$ already forces $\text{supp}(\phi) \subseteq \mathcal{S}$, that is $w_S = 1$ exactly—as the fourth column records for all three rows. Moment exactness here is strictly stronger than unit captured weight, and unit captured weight is precisely the condition under which Sec. III measures the error to be $O(1)$. The two meet in the first row: the $K = 0$ member of the ladder is the witness $\mathcal{S}^* = \text{supp}(\phi)$ of Sec. III, of size $(\frac{1}{2} + \frac{1}{L})D$ at $L = 6$ and fixed by the probe rather than by any ranking. It is moment-exact through order 1, it holds the entire Born weight of the probe, and at $\eta = 0.18t$ its relative L_1 error is 0.28 on the open chain of Table II at $L = 6$, and 0.43 at $L = 6$ and 0.35 at $L = 8$ on the periodic rings on which Sec. III reports it: a measured range of 0.28–0.43 at unit captured weight, whose lower end is the very system on which the method validation of Sec. II is performed. (All three are $\|A - A_S\|_1$ divided by $\|\phi\|^2$, the normalization of Theorem 1; the size of the conversion to the repository’s own in-window relative L_1 is measured in Sec. S11, and the two conventions are never interchanged silently.) Exactness of $2K + 2$ numbers is what the structure certifies; the curve is certified, at fixed η and on the subspace in hand, by Theorem 1, whose own non-vacuity is measured rather than assumed (Sec. V).

What Eq. (13) does *not* say belongs in the same breath, and in the main text rather than in an appendix. The parameter ε thresholds a Born probability; it is not a spectral accuracy, and no step of the argument connects it to $\|A - A_S\|_1$. Arbitrarily many configurations may each carry $p(x) < \varepsilon$ while jointly carrying most of the mixture; and, decisively, even at $w_S = 1$ exactly—nothing left to capture at any ε —the relative error is $O(1)$, as Sec. III measures and the first row of Table II exhibits. Capturing the high-probability configurations is a statement about the sampler, not about the spectrum.

Two compositions of the bound follow easily from it, and neither is made here. “A sampling-capture bound polynomial in $|\mathcal{S}|$ and independent of the Hilbert-space dimension” is a pair of clauses that are literally true and jointly empty, because this same work measures $|\mathcal{S}| \sim e^{1.05L}$ for these lattices, Eq. (20), so a budget polynomial in $|\mathcal{S}|$ is exponential in L , and independence of the Hilbert-space dimension is no saving when the quantity the budget is polynomial in grows exponentially itself. And the fall of the required sector fraction with L is not “the favourable face of the polynomial-shot bound”: it is not a face of that bound at all. Equation (13) relates a shot budget to a Born-probability threshold and contains no statement about $|\mathcal{S}|/D$, about L , or about any trend in either.

What stands in their place is the narrower statement Eq. (13) does support, and it is the stronger one for being narrow: that many shots buy, at confidence $1 - \delta$, every configuration above Born weight ε , and nothing further. What the resulting subspace costs, and what it is worth, are separate measurements, reported as three independent prices—support, resolution and shots—in Sec. VII.

S4. THE NON-VACUITY THRESHOLD: PROTOCOL, TABLES AND LIMITS

Throughout this section we call FR^* the *non-vacuity threshold of the certificate of Theorem 1*: the sector fraction above which the leakage branch of the bound first improves on the trivial bound $\|\phi\|^2(1 + w_S)$. We do not call it a frontier of the method, and it is not one. It is a property of the inequality chain that proves Theorem 1, measured here for the first time, together with one structural fact about the determinant basis. Two measurements separate the two contributions. First, at fixed system and fixed accuracy target the threshold sits far above the fraction the reconstruction actually needs: at $L = 8$, $\eta = 0.18t$, a relative L_1 error of 0.05—the acceptance criterion used throughout this paper—is reached at $\text{FR} \approx 0.52$, while the certificate does not reach 0.05 until $\text{FR} \approx 0.98$, and the intervening shift of ≈ 0.45 in fraction is the looseness of the proof. Second, at full captured weight $\Lambda_S = 0$ requires $\text{span}(S) \supseteq \mathcal{K}(H, \phi)$, whose coordinate support is every determinant the reflection about the seed site allows, at every size measured—a statement that the leakage is nonzero, not about its size, and not the source of the vacuity measured below, which is at $w_S < 1$ —and a Krylov-diagonal oracle ranking gives a *worse* certificate than the Born ranking the device realizes at $L = 6$, $\eta = 0.18t$, at every fraction checked from 0.50 on. Section V E measures both. What is a property of the method, and is new, is the calibration of Sec. V D: the crossing occurs at an essentially fixed level of true error, so the crossing itself is worth computing even where the bound around it is not.

What is evaluated, in full. We evaluate Theorem 1 on the subspaces of Fig. 5 at $L = 6, 8, 10$ and 12—sector dimensions 300, 3920, 52 920, 731 808—with the ranking protocol of Sec. VII ($K = 18$, $\delta t = 0.5$, 16 substeps, 19 snapshots), over fractions 0.20–0.95 and $\eta = 0.10$ –0.25 t . Those subspaces are the $T \rightarrow \infty$ limit of that protocol and not finite-shot draws, and every statement below inherits that limit: the marginal determinant at the $L = 12$ threshold carries Born probability $1\text{--}2 \times 10^{-9}$, so at the budgets of Table IV that subspace barely exists as a sampled object. Of the 204 evaluations, 187 are converged in Krylov depth, 6 marginal and 11 not converged, at a total cost of 2874 s with a peak of 1.64 GiB. *No row flagged not converged is quoted anywhere in this section*; the tables mark the ones that are marginal.

To that grid we add twenty evaluations that are not on it. A grid point near a published fraction is not the published subspace, and the difference is not cosmetic: this sweep reads 0.80, 0.56, 0.40 and 0.18 against published fractions of 0.8200, 0.5561, 0.3625 and 0.1700. The rows of Table III are placed instead at the integer determinant count $|S| = \text{round}(\text{frac} \times \text{dim})$ that Fig. 5 reports—246, 2180, 19 183, 124 407 and 824 504 determinants at $L = 6, 8, 10, 12$ and 14—at the same four resolutions and under the same protocol, at depths $n_\ell = 245$ (exhausting the subspace), 900, 700, 300 and 250, with orthogonality defect at most 4.9×10^{-15} and $|w_K - w_S|$ at most 6.4×10^{-15} . The sixteen up to $L = 12$ pass the depth test defined below; at $L = 14$, where Λ_{out} was not recorded, only its indicator half can be applied, and it passes at all four resolutions. All twenty clear the reference-drift rule of Sec. S11, the weakest— $L = 14$ at $\eta = 0.10t$, where the reference itself drifts by 1.9×10^{-3} on halving its depth—by a factor 51. They cost 129 s of core time up to $L = 12$ and 2238 s of wall time at $L = 14$, and they add the size the sweep never reached.

Three conventions are fixed before any number is read. Full reorthogonalization is used throughout, without which the projector is not a projector (Sec. S11 gives the measured defect). What is certified is the object the code forms— Λ on $\mathcal{K} = \text{Krylov}(H_S, \phi_S, n_\ell)$ bounds the error of A_K at that depth and of nothing else, the convenient inequality $\Lambda_K \geq \Lambda_S$ that would extend it to the dense A_S being false (1169 of 6000 random triples violate it, minimum ratio 0.5258), and A_S is never formed. And depth is certified per point by the internal indicator $\Lambda_{\text{in}}/\Lambda_{\text{out}}$, the truncation part of Λ over its boundary part, rather than assumed from a fixed recursion depth [20] or from a priori Krylov convergence bounds [28]: a point passes the depth test, and is counted converged, when that indicator is below 0.05 and Λ_{out} is constant to 10^{-3} relative over the last three depths of its ladder, and is marginal when the indicator is below 0.30 otherwise (`fsum.py`); Sec. S11 bounds what the depth cap does to FR^* .

Four hypotheses carry the bound, each a statement about the pipeline rather than about the theorem; they are stated in full with the theorem in Sec. III, and all four are checked against this pipeline in Sec. S11, along with the ranking, tie and support declarations of the cut. It is (H3), the exactness of the reference, that enforces the rule that no measured error is quoted within a factor 20 of its own reference drift. One asymmetry runs through every table, and it is narrower than it looks: neither term of Eq. (7) is free of Ψ_0 . The leakage needs the retained probe ϕ_S , the amplitudes of Ψ_0 on the determinants the probe operator maps into S , which the reconstruction itself uses and which is exact here by (H4); the weight term needs in addition the full norm $\|\phi\|^2$, and estimating w_S against an already-truncated Ψ_0 is biased optimistically, while Λ_S evaluated with such a Ψ_0 belongs to another probe (Sec. III). What the tables report is accordingly one object and not a family of them: a bound on $\|A - A_K\|_1/\|\phi\|^2$ for the reorthogonalized Krylov reconstruction at a declared depth, against a reference whose drift is declared and enforced.

The two largest sizes. The largest classically converged case carries the evidence. At $L = 12$ and the published subspace itself— $|S| = 124\,407$ of 731 808 determinants—the captured weight is $w_S = 0.998622$, so Eq. (14) asks for $\hat{\Lambda}_S/\eta < \tau(w_S) = 0.9625$ and the measured value is 4.236 at $\eta = 0.18t$, putting the leakage branch at 8.544 against a trivial branch of 1.9986. That is converged in depth: the depth indicator is $\Lambda_{\text{in}}/\Lambda_{\text{out}} = 2.8 \times 10^{-3}$ and between depths 250 and 300 Λ_{out} moves by 3.7×10^{-7} relative and the bound by 1.4×10^{-5} , while at the neighboring grid

TABLE S1. **Where the certificate stops being vacuous, and what the reconstruction actually needs, measured on the same sweep.** Protocol of Sec. VII, full reorthogonalization, depth certified per point. Column 3 is the fraction Fig. 5 reports; column 4 the control, where this sweep—independent code, the normalization of Theorem 1, the whole line, a deep reference—first reaches relative $L_1 < 0.05$, the parenthesized value being the log-linear interpolation inside the bracket and not a measurement at three digits; column 5 is FR*; columns 6–7 are where the certified value itself first falls below 0.50 and 0.20. Every bracket is delimited by two converged evaluations unless marked: * means the upper end is marginal in depth, *n.c.* that the bracket rests only on evaluations failing the depth test and is therefore not quoted. Columns 5–7 are statements about A_C at the declared depth, not about the dense A_S . Columns 4 and 5 must be read together: FR* alone falls with size and would suggest the certificate improves as the problem grows, whereas what the reconstruction needs falls with it and the gap widens—a gap we do not quote as a ratio, both columns being interpolations inside brackets 0.02–0.10 wide.

L	$\dim(N+1)$	published	rel- $L_1 < 0.05$	FR*	cert < 0.50	cert < 0.20
<i>$\eta = 0.18t$, the resolution of the momentum-resolved channel</i>						
6	300	0.82	[0.70, 0.80]	[0.90, 0.92]	> 0.97	> 0.97
8	3 920	0.56	[0.50, 0.56]	[0.75, 0.80]	[0.90, 0.92]	> 0.95
10	52 920	0.36	[0.30, 0.40]	[0.60, 0.70]	[0.80, 0.85]	[0.85, 0.90]
12	731 808	0.18	[0.14, 0.18]	[0.40, 0.45]	[0.60, 0.70]*	<i>n.c.</i>
<i>$\eta = 0.15t$, the resolution at which Fig. 5 is drawn</i>						
6	300	0.82	[0.80, 0.85] (0.816)	[0.92, 0.95]	> 0.97	> 0.97
8	3 920	0.56	[0.50, 0.56] (0.553)	[0.80, 0.82]	[0.92, 0.95]	> 0.95
10	52 920	0.36	[0.30, 0.40] (0.356)	[0.60, 0.70]	[0.80, 0.85]	[0.85, 0.90]
12	731 808	0.17–0.18	[0.14, 0.18] (0.168)	[0.45, 0.50]	[0.60, 0.70]*	<i>n.c.</i>

point FR = 0.18 the same quantity was checked twice more— Λ_{out} changing by 7.5×10^{-7} between depths 250 and 300, and an independently written re-implementation moving it by 8.6×10^{-5} between depths 100 and 300. In all sixteen deposited depth series Λ at every depth from 100 on lies at or above its deepest value—it is not monotone in depth in general, and rises by up to 2.2% from one depth to the next below depth 100 at $L = 12$ —so the bound at any depth from 100 on, including the depth 150 at which Fig. 5 is drawn, is no smaller, so the vacuity is not an artifact of Lanczos truncation in either direction. Nor is it peculiar to these subspaces: over the 606 live subspaces of the pooled sweep of Sec. S9, Fig. S5, the leakage branch is the smaller of the two on only 61. The reconstructions at the fractions this paper publishes—Fig. 5 and the operating points of Secs. VII and VIII—are therefore uncertified. Figure 3, drawn at 85% of its sector, is the exception: there the certificate of the depth-500 Krylov reconstruction is non-vacuous in all sixteen channels, at a relative bound of 0.77–1.75 against the trivial 2, and loose, 350–900 times the true error of the channel integrated over the whole frequency line (`cert_akw.json`).

One row is quoted with a caveat attached. $L = 14$ is the size at which Fig. 5 reports its headline fraction of 0.08, and the certificate is evaluated there at $|S| = 824\,504$ of 10 306 296 determinants—one determinant above the round($\text{frac} \times \text{dim}$) = 824 503 of the deposited grid—with $w_S = 0.998388$ and a leakage branch of 17.21, 11.18, 9.21 and 6.50 against a trivial branch of 1.9984. Ground state, seed and order there are the repository’s stored checkpoint—residual 2.7×10^{-9} , the order verified to be a permutation of the sector—and the $K = 18$ ranking protocol, re-run at that size, reproduces the stored selection as a set to 1.000000 at $|S| = 824\,504$ and at the fractions 0.04, 0.16 and 0.32, as it does at $L = 10$. The caveat is the depth indicator $\Lambda_{\text{in}}/\Lambda_{\text{out}} = 0.0453$ at $\eta = 0.10t$, the loosest value quoted anywhere in this section, comparable to the 4.7×10^{-2} of the last row of Table S3. It is not repaired, and the row is reported because it makes the verdict of this section worse rather than better.

One column of Table S1 is a positive control and has to be read before the rest. Applying this paper’s own acceptance criterion to the same sweep—independent code, the theorem’s normalization, the whole line rather than a window, a deep reference rather than one recomputed at the subspaces’ own depth—returns 0.816, 0.553, 0.356 and 0.168 at $\eta = 0.15t$ for $L = 6$ –12, against the published 0.82, 0.56, 0.36 and 0.18: agreement to 7% at worst, and to under 2% at every size once each value is compared with the fraction the scan itself reports rather than with the two-figure entry of the replication grid (0.5561, 0.3625, 0.1700 at $L = 8, 10, 12$). None of the three ingredients is shared. Figure 5 is thereby reproduced independently.

The last column of Table III is that control taken one step further, and it is the sharper of the two: instead of re-deriving the fraction it evaluates the relative L_1 error at the published subspace, at the resolution and against the threshold that define it ($\eta = 0.15t$, rel- $L_1 < 0.05$). The five values—0.0477, 0.0478, 0.0484, 0.0485 and 0.0463—all sit just below the threshold and are flat to within 5% of one another across a sector dimension running from 300 to 10 306 296, $L = 14$ included. The criterion that defines the published fractions is therefore reproduced at the subspaces themselves, at every size the figure plots above $L = 4$, again with nothing shared. What fails in this section is not the reconstruction and not that curve; it is the word *certificate* attached to the fractions at which the curve is read.

In shots the distance between those two columns is a budget and not a rhetorical gap (Table S2): reaching the

TABLE S2. **The threshold in the only currency a sampler has.** T_{marg} is the shots per snapshot at which the marginal determinant of the cut appears once in expectation—the reciprocal of its Born probability in the accumulated distribution of the 19 snapshots, from the protocol’s own cut weights; covering all of S costs a further factor $\log|S|$. *It is a different quantity from the measured budgets T of Table IV*, which are totals over the 19 snapshots at which the finite-shot reconstruction reaches relative $L_1 < 0.05$; the two carry different names for that reason, they are not interchangeable, and only ratios within this table are used. Column 2 uses the grid point nearest the published fraction: 0.80 against a published 0.82 at $L = 6$, and 0.40 against 0.36 at $L = 10$, where the published fraction is the worse of the two. Factors are from unrounded budgets and need not reproduce from the two-figure entries. The marginal Born probability at the $L = 12$ threshold is 2.2×10^{-9} at $\text{FR} = 0.40$, 1.2×10^{-9} at 0.45.

L	published fraction $\rightarrow T_{\text{marg}}$	$\text{FR}^* \rightarrow T_{\text{marg}}$	factor
6	$0.80 \rightarrow 2.2 \times 10^4$	$[0.90, 0.92] \rightarrow 3.1\text{--}3.6 \times 10^4$	1.4–1.6
8	$0.56 \rightarrow 2.1 \times 10^5$	$[0.75, 0.80] \rightarrow 1.6\text{--}2.1 \times 10^6$	7.5–10
10	$0.40 \rightarrow 5.2 \times 10^6$	$[0.60, 0.70] \rightarrow 3.4 \times 10^7\text{--}1.1 \times 10^8$	6.7–21
12	$0.18 \rightarrow 2.3 \times 10^7$	$[0.40, 0.45] \rightarrow 4.7\text{--}8.1 \times 10^8$	20–35

TABLE S3. **The strongest certified statements available from converged data.** Each row is the tightest certified value at that size whose bracket is delimited by evaluations passing the depth test; rows that would be stronger but rest on evaluations failing it are not listed. “True error” is the measured relative L_1 at the same point, in the normalization of Theorem 1, given only where it exceeds its own reference drift by more than a factor 20 (Sec. S11, H3); at $L = 10$ it does not, and is withheld rather than rounded. At $L = 6$ the recursion runs to $n_\ell = |S|$, so the evaluation is exact and the depth indicator is not defined. The gap between the certified value and the true error is three to four decades: the object is a certificate and not an error estimate, and Sec. VD states what it is good for.

L	η/t	fraction	certified rel- $L_1 \leq$	true rel- L_1	$\Lambda_{\text{in}}/\Lambda_{\text{out}}$
6	0.18	291/300 = 0.97	0.502	3.94×10^{-4}	exact ($n_\ell = S $)
8	0.18	3 724/3 920 = 0.95	0.219	4.33×10^{-5}	8×10^{-4}
10	0.18	47 628/52 920 = 0.90	0.136	withheld (H3)	1.6×10^{-3}
12	0.18	439 085/731 808 = 0.60	0.717	1.89×10^{-4}	7.0×10^{-3}
12	0.15	0.60	0.878	2.66×10^{-4}	1.4×10^{-2}
12	0.25	512 266/731 808 = 0.70	0.218	1.99×10^{-5}	4.7×10^{-2}

non-vacuous band multiplies the shots per snapshot needed for the marginal determinant by 7.5–10 at $L = 8$ and by 20–35 at $L = 12$, and only by 1.4–1.6 at $L = 6$, where the sector is small. Driving the certified value down to 0.10 costs more again—a further factor 241 at $L = 8$ (fraction 0.95) and 634 at $L = 10$ (fraction 0.90), the corresponding $L = 12$ fraction resting only on evaluations that failed the depth test, so that no budget is quoted for it. This section reports the measured regime of validity: four sizes, four resolutions, 204 evaluations, every bracket delimited by converged rows, and a shot budget attached to each.

The strongest certified statements. Table S3 collects what the sweep does certify, taking at each size the strongest statement whose bracket is converged—at $L = 12$, that 60% of a 731 808-dimensional sector gives relative $L_1 \leq 0.717$, tightening to ≤ 0.218 at 70% and $\eta = 0.25t$. Each row is a statement about A_K at the declared depth, and none of them uses the exact spectrum.

One restriction belongs with that table. At the acceptance criterion this paper uses everywhere else—relative $L_1 < 0.05$ —the certificate requires at least 0.98 of the sector at $L = 8$, measured at integer subspace size rather than interpolated: the certified value is 0.084 at 3802 of 3920 determinants, 0.060 at 3830 and 0.039 at 3855, so the crossing lies in $[0.977, 0.983]$; on our grid it is not reached at all at $L = 6$, $L = 10$ or $L = 12$. At such fractions diagonalizing the whole sector is already what one would do instead, and a certificate informative only there does not certify the regime in which sampling is of interest. We state it without qualification: at the accuracy this paper accepts elsewhere, Theorem 1 does not certify the operating points of Fig. 5, and the shot cost of the regime where it would exceeds the saving that figure reports.

What the table does certify at a fraction a sampler can plausibly reach is weaker, and it is not empty. The trivial bound caps the relative error at $1 + w_S \leq 2$ for *any* subspace, a reconstruction of pure noise included; at $L = 12$ and 60% of a 731 808-dimensional sector the certified 0.717 sits a factor 2.8 below that cap, and at 70% and $\eta = 0.25t$ a factor 9.2 below it. Those are exclusions of gross failure made from the subspace alone, at a dimension where the exact spectrum is expensive and, at the sizes this method is aimed at, unavailable.

The calibration and its limits, in full. The calibration is the reason the sweep is reported. Across all sixteen (L, η) cells the crossing of the trivial bound occurs at a true relative L_1 error of 1.2×10^{-3} to 7.3×10^{-3} , median 2.8×10^{-3} : a spread of 5.9 over a table whose true errors span five decades, and in every cell at least 47 times the drift of our

own reference. $\hat{\Lambda}_S(\eta)/\eta$ is therefore a calibrated a posteriori risk tracker—its crossing of the trivial bound locates a true error of $\approx 3 \times 10^{-3}$ to within a factor of three over $L = 6$ –12 and $\eta = 0.10$ –0.25 t , and it is computed from H , S , the retained probe ϕ_S , the probe norm $\|\phi\|$ and η , without A , given the exact probe (H4). That factor of three is a drift and not scatter, and the deposited grid settles it in a few lines: the sixteen crossing errors are ordered without a single exception, falling with L at every resolution—5.65, 3.50, 2.52 and 1.60×10^{-3} at $\eta = 0.18 t$ for $L = 6$ –12, a factor 2.8–3.7—and rising with η at every size by 1.6–2.1, the product of the two being the observed 5.9. The drift runs the conservative way in the variable one would extrapolate: the larger the system, the smaller the true error at which the certificate first says anything, so reading the crossing as $\approx 3 \times 10^{-3}$ beyond the sizes measured overstates the error rather than understating it. In η , which the user chooses and which lies inside the measured range, it does not. The objection to anticipate is that this is Fig. 5 relabeled. It is not, and the difference is the content: that figure is built knowing the exact spectrum and this quantity is computed without it. That the two track each other to within a factor 5.9 across five decades of error is not a redundancy—it is the calibration, and it is what makes the quantity usable where the second curve cannot be drawn.

Four limits come with it. The crossing is one point per cell, so the tracker locates one error level and does not estimate the error away from it: at a *fixed* subspace the looseness of the bound against the measured error is constant to within $\pm 17\%$ ($L = 6$) and $\pm 25\%$ ($L = 8$) over a decade in η , but across subspaces it varies by a factor 51, and at the threshold itself it is 3×10^2 to 2.4×10^3 —the object is a certificate, not an error estimate, and calling it the latter would be refutable in one line. Neither of its two terms is free of the ground state (Sec. III), and the comparison needs more than the leakage: it is $\hat{\Lambda}_S/\eta$ against $\tau(w_S)$, and τ needs w_S , which a user does not have. That dependence can be discharged conservatively, and doing so is what makes the decision itself a posteriori. Equation (14) is decreasing in $1 - w_S$ for $w_S \geq 0.066$, with $\tau(1) = 1$, and positive for $w_S > 0.296$, so any assumed floor on the captured weight above 0.296 yields a usable threshold: at the $L = 12$ published operating point $w_S = 0.998622$ and $\tau = 0.9625$, and every subspace with $w_S \geq 0.995$ has $\tau \geq 0.9280$. A user willing to assume only that the sampler captured 99.5% of the probe’s Born weight can therefore run the comparison as $\hat{\Lambda}_S(\eta)/\eta < 0.928$ and use no other ground-state information, at the cost of at most 7% in the threshold. Where not even that floor can be assumed, or where w_S is estimated against an already-truncated Ψ_0 —an estimate biased optimistically—the test is not conservative and the bound is not a posteriori. All of this presupposes the exact probe of (H4): with a Ψ_0 estimated on a sampled subspace, $\hat{\Lambda}_S$ itself is evaluated for another probe. And sixteen cells over four sizes and four resolutions is the whole of the evidence: no exponent is fitted to it and no extrapolation to $L = 14$ is made, the local log–log slopes of $\text{FR}^*(L)$ already varying by a factor 4.1 across the four sizes measured.

The fourth limit is the one a reader holding the deposited grid will test first, and it is a real restriction on what the crossing is worth. A sampler already holds the captured Born weight, so the question is whether a stopping rule indexed on $1 - w_S$ alone locates the same error level, in which case the crossing would be that weight relabeled. It is not: over the sixteen cells, $1 - w_S$ at the crossing ranges over a factor 45 against the true error’s 5.9, and still over 9.4 against 3.8 once the extreme cell at each end is dropped. A rule with constants fitted to the grid, however, comes close enough. Take $1 - w_S < c\eta^p$, fit c and p on some of the four sizes with its operating point matched to the certificate’s, evaluate it on the sizes held out, and repeat over all fourteen train/test splits: out of sample it localizes the crossing error about as tightly as the parameter-free certificate does, and more tightly on eleven of the fourteen (`null_ws7.py`). Out of sample the certificate has no edge in sharpness over the captured weight as a tracker of the error; its edge is provability (Sec. VD). The difference lies elsewhere. The fitted rule fires at $1 - w_S \sim 10^{-5}$ – 10^{-4} , one to two orders of magnitude finer than the floor $1 - w_S \leq 5 \times 10^{-3}$ at which the threshold above still works, and it asks that precision of the one quantity this section has already said a user does not have and, estimated against an already-truncated Ψ_0 , has optimistically; strip its second parameter and an η -blind $1 - w_S < c$ is the looser of the two on thirteen of the fourteen splits. Theorem 1, for its part, is an inequality with a proof: it holds at a size no grid has seen, and its decision survives replacing w_S by a floor. The fitted rule is a two-constant regularity with no guarantee outside its fitted range—the only place either would be used.

Within those limits the procedure is operational and identical at every size we measured: assemble $\hat{\Lambda}_S(\eta)/\eta$ from the Ritz vectors already in hand and compare it with $\tau(w_S)$. Above τ the certificate says nothing, which is the regime Fig. 5 operates in; at τ the true relative error is $\approx 3 \times 10^{-3}$ within a factor of three; below τ it is smaller than that, by an amount the bound states loosely and the calibration does not sharpen. That is the whole content of the certificate, and it is a number a user can compute before knowing whether the reconstruction is right.

Where the threshold comes from, in full. Where does the ≈ 0.45 of sector fraction between what the reconstruction needs and what the certificate certifies go? We measured the displacement rung by rung along the chain of inequalities that proves Theorem 1, at $L = 8$, and the answer removes the one repair that looked available: the pointwise step from a scalar distance to a distance in the Hilbert space accounts for 44% of it, the operator-norm step $\|R(z)\| \leq 1/\eta$ applied to a source that depends on ω for 38%, the Feshbach/Minkowski split for 11%, and the Cauchy–Schwarz step in ω for 4.6% (medians over 15 of 18 cells; Sec. S11). The working hypothesis that Cauchy–Schwarz is what places the

threshold is therefore refuted by its own measurement: its total margin for tightening is a factor 1.05 typically and 3.29 at the worst point measured, which moves the threshold by a fraction of one grid interval.

The second measurement concerns the subspace itself, and it is the one that turns part of the result into a statement about the problem. $\Lambda_S = 0$ holds if and only if $\text{span}(S)$ contains $\mathcal{K}(H, \phi_S)$, the Krylov space of the retained part of the probe (Sec. III); for a subspace that holds the whole probe, $w_S = 1$, that space is $\mathcal{K}(H, \phi)$ itself, so the coordinate support of $\mathcal{K}(H, \phi)$ is a floor under every coordinate subspace that holds the whole probe without leaking. Measured, it is 296 of 300 determinants at $L = 6$, where $\dim \mathcal{K} = 65$, and 3902 of 3920 at $L = 8$, where $\dim \mathcal{K} = 596\text{--}663$. (It is the *coordinate support* of $\mathcal{K}$ that is measured, not $\mathcal{K}$ itself, whose dimension is three orders of magnitude smaller; containment, not spanning, is what $\Lambda_S = 0$ requires.) The determinants missing are not an accident of threshold. The reflection $j \rightarrow -j$ about the seed site commutes with H , and ϕ is an eigenvector of it, so every vector of $\mathcal{K}$ vanishes exactly on the reflection-fixed determinants of the opposite parity: 4, 18, 48, 200 and 600 at $L = 6, 8, 10, 12$ and 14. Every other determinant is in the support. At $L = 10$ the same construction—a Lanczos recursion on the whole sector to depth 700 with full reorthogonalization, orthogonality defect 1.6×10^{-15} —leaves no row exactly zero in floating point, although 48 of them vanish in exact arithmetic: round-off is not a support.

Beyond $L = 10$ that basis cannot be stored, and we use a lower bound instead: every $H^k \phi$ lies in $\mathcal{K}$, so the union over k of the coordinate support of $H^k \phi / \|H^k \phi\|$ is contained in the support of $\mathcal{K}$. A count of this kind is meaningless without its threshold, so both are given. Projected onto the reflection sector of ϕ at every power (`z_suppK_sym.py`), so that round-off cannot populate the forbidden sector, the union over $k \leq 36$ is every symmetry-allowed determinant at every size: 296, 3902, 52 872, 731 608 and 10 305 696 at $L = 6$ to 14, at an amplitude threshold of 10^{-12} , and the same at 10^{-10} except at $L = 14$, where 232 genuine amplitudes lie between the two thresholds. Unprojected, the power iteration amplifies round-off in the forbidden sector until it crosses either threshold and reports the entire sector from $L = 8$ on (`z_suppK_power.py`); that count is not a support.

At every size measured, therefore, a coordinate subspace that holds the whole probe leaks unless it contains every symmetry-allowed determinant—the sector minus its forbidden determinants contains $\mathcal{K}$ and does not leak—so at full captured weight no choice of determinants short of all of them makes Λ_S vanish, and that is a property of the basis in which the device measures, not of the bound and not of the ranking. It is a statement that $\Lambda_S \neq 0$, not about its size, and the vacuity at the published fractions is not deduced from it: that is the measurement of Table III. Below full captured weight the space that decides is $\mathcal{K}(H, \phi_S)$, which changes with S , and nothing structural is claimed there. Nor is the ranking the weak link: replacing the Born ordering by a Krylov-diagonal oracle—determinants ordered by the diagonal of the projector onto $\mathcal{K}(H, \phi)$, an ordering no sampler could produce—gives a *worse* certificate at $L = 6$, $\eta = 0.18t$, at every fraction checked from 0.50 on, 12.97 against 4.27 at 0.85 and 5.82 against 0.911 at 0.95; only at 0.30, where both are far above the trivial value, is the oracle marginally better, 21.76 against 24.52.

At fixed size the displacement is the looseness of the proof, and part of the looseness has a structural reason: the leakage branch is of first order in the boundary coupling, the error it bounds of second. For a subspace that holds the whole probe, eliminating Q exactly replaces H_P by $H_P + PHQ(z - QHQ)^{-1}QH P$, and the two sum rules then give $\|A - A_P\|_1 \leq \|\phi\|^2 \|QH P\|^2 / \eta^2$. Sharper, with $\phi_Q = 0$ Eq. (A1) and Cauchy–Schwarz in ω give $\|A - A_P\|_1 \leq \Lambda_P(\eta) \tilde{\Lambda}_P(\eta) / \eta^2$, where $\tilde{\Lambda}_P$ is Eq. (6) with the energy difference in its kernel reversed in sign, equal to Λ_P for a real Hamiltonian and probe. The ratio of the leakage branch, $2\|\phi\|\Lambda_P/\eta$, to the error is therefore at least $2\eta\|\phi\|/\tilde{\Lambda}_P$, and grows without limit as the subspace approaches invariance; at full captured weight the product $\Lambda_P \tilde{\Lambda}_P / \eta^2$ is itself an a posteriori certificate of second order. Every published fraction lies below full captured weight, and whether a certificate of second order exists there is open; we have not constructed one. Section VI closes the picture from the other side: nothing cheaper than the crossing of Sec. VD—no invariant of the state, computable before the subspace exists—can be put in its place.

One route out of the determinant basis has been measured, and it closes this section as an outlook and not as a result, because the same measurement shows where it stops. The qualifier *in the determinant basis* above is load-bearing, and it is measurable. An orbital rotation is a Givens network and the circuit of Sec. II already contains them, so the measurement basis is a choice; for the momentum-resolved probe of Fig. 3 it is a forced one, since H commutes with lattice translation and ϕ_k is an eigenvector of it, so that $\mathcal{K}(H, \phi_k)$ lies inside a single momentum block—rigorously, at every L —and that block is exactly $\dim(N+1)/L$ plane-wave determinants. At $L = 6$ and $L = 8$ the Krylov support fills nearly all of the block, and the certificate becomes non-vacuous only close to the full block: at 48–50 of its 50 determinants at $L = 6$ and at 455–483 of its 490 at $L = 8$, over the momenta of the addition branch (`z_kblock.py`), a factor 5–7 fewer than the site determinant basis requires; on the block itself Λ_S vanishes identically and the bound is zero rather than vacuous. The gain, though, is $1/L$ against an exponential: as a fraction of the fraction Fig. 5 publishes the block is 0.20 at $L = 6$ and then 0.22, 0.28, 0.49 and 0.89 at $L = 14$, the two crossing just past $L = 14$. It is also a laptop Lanczos—60 984 determinants at $L = 12$, 736 164 at $L = 14$ —so the regime it certifies is one that momentum-resolved exact diagonalization has enumerated since the 1970s. Three things are unmeasured here and would have to be measured before more is said: whether the support fills the block at $L \geq 10$, where the threshold then sits, and what the $\approx L(L-1)$ extra two-qubit gates do to the sampled distribution.

S5. ONE-BODY MAGIC, THE GEMINAL WITNESS AND THE DETERMINANT SUPPORT: FULL ANALYSIS

The candidate measures. The candidates have a sharp algebraic origin. Matchgate (free-fermion) circuits are classically simulable [104–107], and what leaves that manifold is fermionic non-Gaussianity: any pure non-Gaussian state injected into a matchgate circuit promotes it to universality [108], the fermionic counterpart of qubit magic [109, 110]. A ladder of measures quantifies it—Gaussian rank and extent [111], non-Gaussian gate counts [112, 113], including the count that makes a state efficiently learnable [114], covariance-matrix measures and monotones [16, 17], and convolution- and entropy-based measures built on the violation of Wick’s theorem [115]; distinct from these is the qubit non-stabilizerness that Gaussian states themselves carry [116].

The collapse, in detail. Equation (17) is exact on any number-conserving state and verified state by state to 10^{-13} across the dissociation of N_2 and the suite of Table S12, that is on the 38 molecular ground states of this work. For spin-restricted natural orbitals ($g_{a\uparrow} = g_{a\downarrow} = n_a/2$) it equals $2N_u$, the quadratic effectively-unpaired-electron index $N_u = \sum_a n_a(2 - n_a)$ of Takatsuka–Fueno–Yamaguchi and of Staroverov–Davidson [74, 75], and $\text{tr}[\gamma(1 - \gamma)]$ is one quarter of the Gigena–Rossignoli one-body quadratic (linear) entropy, $S_2 = 4 \text{tr}[\gamma(1 - \gamma)]$ on a number-conserving state [76], a trace-form one-body entanglement entropy in the sense of Gigena, Di Tullio and Rossignoli [77]. Here $2N_u$ is the quadratic index, distinct from the linear $\sum_a \min(n_a, 2 - n_a)$ of Head-Gordon [117]. A cleanly singly-occupied spin orbital contributes 0 to $\mathcal{F}_1$ but 2 to $2N_u$, so on the five spin-polarized members of Table S12 (O_2, NO, CN, OH, BeH) the two differ and we report $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$ directly. Outside number conservation the collapse fails: a paired Gaussian state hides fractional occupations in anomalous correlators and keeps $\mathcal{F}_1 = 0$.

The algebra is not new. Equation (17) is an immediate number-conserving specialization of the antiflatness of Ref. [16]; N_u has been the standard diagnostic of unpaired density since 1978 [74]; and the relation was noted independently and contemporaneously by Świątek *et al.* [78], who state it for fixed-particle-number states and write its ensemble-averaged form for random superpositions of Gaussian states, their model of interacting-integrable eigenstates. What we add is the exact, state-by-state *dictionary*: three literatures—fermionic magic, effectively-unpaired electrons, one-body entanglement—name one quantity. The verification is the 38 molecular states just quoted; the further equality to $2N_u$ is checked on the closed-shell members, the five spin-polarized ones being excluded for the reason given above. The 36 lattice ground states of this work are number conserving, so Eq. (17) applies to them by the same algebra, but they were not part of that state-by-state check and we do not count them in it. The consequence we carry forward is structural and equally exact: being a function of the occupation spectrum alone, $\mathcal{F}_1$ inherits the orbital-rotation invariance of the eigenvalues of γ . It is a genuine, cheap, exactly computed invariant of the state, and it is that invariance the two following subsections use.

The witness and the fibers. The proof is four short block-diagonal computations (Sec. S12); the step worth flagging here is that at $\theta = \pi/4$ a single maximally-paired geminal is the two-determinant cat, whose Majorana covariance *vanishes identically*, so *every* order of antiflatness saturates its maximum at once. The values of $\mathcal{F}_1$, $\mathcal{F}_2$ and $|\mathcal{S}|$ are reproduced by exact diagonalization for $K = 1, \dots, 4$, and so is $\chi = 2$, the largest Schmidt rank over all cuts of the pairing ordering (Sec. S12).

The orbit is the argument. The witness is not the trivial “a product state is easy” (Fig. S1). Because $\mathcal{F}_1$ and $\mathcal{F}_2$ are invariant under number-conserving orbital rotations [16], the whole orbit $\{U|\Psi_K\rangle\}$ carries the *same* $(\mathcal{F}_1, \mathcal{F}_2) = (4K, 4K)$ while the cost in the fixed computational basis—Born-sampling time, through the bond dimension—is $O(K)$ at the localized representative and, at generic rotated points, not established here. Two conclusions follow, each in the only direction the construction supports: a Gaussian-invariant fermionic magic is not a sufficient certificate of hardness and does not lower-bound the bond dimension, nor the cost beyond linear order in K (there $\chi = 2$ while $\mathcal{F}_k = 4K \rightarrow \infty$); and $\mathcal{F}_k$ determines the cost only if the rotated points are in fact easy, being constant on an orbit that contains an $O(K)$ -cost point and points at which sampling is plausibly hard. In a generic rotated basis the Born sampling of $|\Psi_K\rangle$ is plausibly classically hard—the free-fermionic-linear-optics-with-magic-input regime of the Fermion Sampling advantage scheme [118]—while $\mathcal{F}_k$ is unchanged: any hardness that appears belongs to the sampler’s fixed frame. We are silent on whether magic is *necessary* for hardness, and we anchor the easy end to an external result rather than our own, since for this paired class the additive-error estimation of amplitudes, overlaps and number correlators under free-fermionic dynamics is classically efficient [119].

The same orbit reaches the leakage. The orbit argument was written for $|\mathcal{S}|$; it extends to the second budget of Eq. (7). An invariant upper bound on $\Lambda_S(\eta)/\eta$ *does* exist—it is exhibited inside the proposition below, and we tested it on 1600 hostile cases (GOE/GUE, near-degenerate, banded, $\|H\| \sim 300$) with zero violations and with its first link attained, minimum ratio 1.0000000000. The correct statement is therefore one of non-vacuity, not of non-existence: the same logical form the $|\mathcal{S}|$ no-go always had, where the sector dimension is likewise an invariant upper bound that says nothing.

Table S4 is the exhibit. The two endpoints of each fiber are the *same* rotation angle applied within one geminal block and across two blocks, which is why the captured weight agrees bit for bit. An exhaustive search over all 276

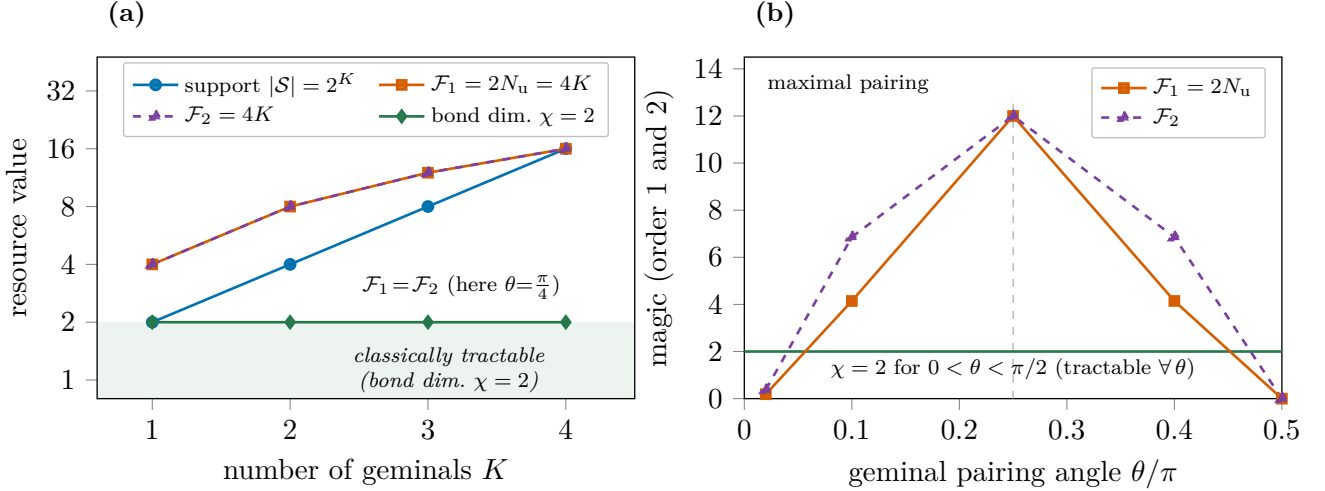


FIG. S1. **The geminal orbit witness (Theorem 3).** Antisymmetrized products of strongly orthogonal geminals in their localized (pairing) basis, where $\chi = 2$ and the state is a classically tractable tensor network. **(a)** As K grows, $\mathcal{F}_1 = 2N_u = 4K$, $\mathcal{F}_2 = 4K$ and $|\mathcal{S}| = 2^K$ all diverge while the bond dimension is 2; the χ markers are the largest Schmidt rank over all cuts of the pairing ordering, computed by the deposited diagonalization (Theorem 3); at the central cut alone it is 2, 1, 2, 1 for $K = 1, \dots, 4$, because that cut falls between two geminals when K is even. **(b)** At $K = 3$, sweeping the pairing angle drives both $\mathcal{F}_1$ and $\mathcal{F}_2$ to their maximum at $\theta = \pi/4$ —where the covariance vanishes and all orders saturate—with $\mathcal{F}_2 > \mathcal{F}_1$ off the peak, and $\chi = 2$ at every interior angle, dropping to 1 at $\theta = 0$ and $\theta = \pi/2$, where the state is a single determinant (the line drawn at $\chi = 2$ is the analytic interior value). Because $\mathcal{F}_k$ is orbital-rotation invariant the same values persist on the whole orbit, which contains the $O(K)$ -cost pairing point and rotated points at which Born sampling is plausibly hard (not established here).

TABLE S4. **Fibers of fixed invariants with moving leakage**, at $s = \pi/16$, $\delta = 0.01$, $\eta = 0.10$, $|\mathcal{S}| = 2^{K-1}$, $w_S = 0.961940$, $\tau(w_S) = 0.795$. Along each row the spectrum, every Lehmann weight, $\|\phi\|^2$, $|\mathcal{S}|$ and all $\mathcal{F}_k$ are identical between the two endpoints (w_S to 2.2×10^{-16}), so the first term of Eq. (7) is fixed and only the second moves. Quantities are relative ($\hat{\Lambda}_S = \Lambda_S/\|\phi\|$) and “certificate” is $\min\{(1 + \sqrt{w_S})(\sqrt{1 - w_S} + \hat{\Lambda}_S/\eta), 1 + w_S\}$, the relative form of Theorem 1; “al.” and “rot.” label the aligned and rotated endpoints of the fiber. Bold marks a non-vacuous certificate, below the trivial bound $1 + w_S$. A continuous angle sweep is in Sec. S12.

K (dim($N+1$))	$\mathcal{F}_1=\mathcal{F}_2=\mathcal{F}_3$	$\hat{\Lambda}/\eta$ al.	$\hat{\Lambda}/\eta$ rot.	cert. al.	cert. rot.
2 (24)	8	0.237	2.326	0.856	1.962 (trivial)
3 (300)	12	0.323	2.336	1.026	1.962 (trivial)
4 (3920)	16	0.391	2.346	1.161	1.962 (trivial)

coordinate subspaces of that size at $K = 2$ settles both ends. At the aligned endpoint the global minimum of the certificate over the 276 is 0.856 and it is attained by the protocol’s own subspace: the subspace the sampler would return is the best available one. At the rotated endpoint the minimum over the same 276 is exactly the trivial bound, attained only by subspaces of vanishing captured weight: *no* determinant subspace of that size certifies anything there.

What this exhibit is not. Four limits are declared with it, and they are why we call it a fiber and not a sweep of the orbit. *(i)* The evidence is a matched pair at $s = \pi/16$ for $K = 2, 3, 4$. At the maximal-pairing angle $s = \pi/4$ the Born distribution of ϕ is exactly flat over 2^K determinants, so “the top $|\mathcal{S}|$ of the Born distribution” does not define a subspace at all (6, 70 and 12870 equally valid outputs at $K = 2, 3, 4$); any contrast quoted at that angle is a tie-breaking artifact and not a property of the fiber, and the sweep of Sec. S12 stops at $s = 0.654$ for that reason. *(ii)* Nothing here grows with the system, and two separate quantities shrink. The ratio of the two leakages falls, $9.80 \rightarrow 7.23 \rightarrow 6.01$ from $K = 2$ to $K = 4$, and it dies as the perturbation is turned up, from 28.7 at $\delta = 0.0025$ to 1.03 at $\delta = 0.3$. More to the point—because it is the quantity the proposition actually rests on—so does the margin of the certificate over the trivial bound: $1.962/0.856 = 2.29$, $1.962/1.026 = 1.91$, $1.962/1.161 = 1.69$, a fall of 26% across the only three sizes available. Where that fall would reach unity we do not say: a linear extrapolation in K , a log-linear one and the last step alone place it between $K = 6$ and $K = 8$, the decrements are themselves decelerating, and three points do not support a number—which is exactly the objection this paper makes to its own fraction headline in Sec. VII. A statement of non-vacuity needs one fiber and not a magnitude, and the fiber at $K = 4$ is still non-vacuous by 69%; but “the effect grows with K ” is false here. *(iii)* The small-leakage endpoint is realized only by the aligned basis, which

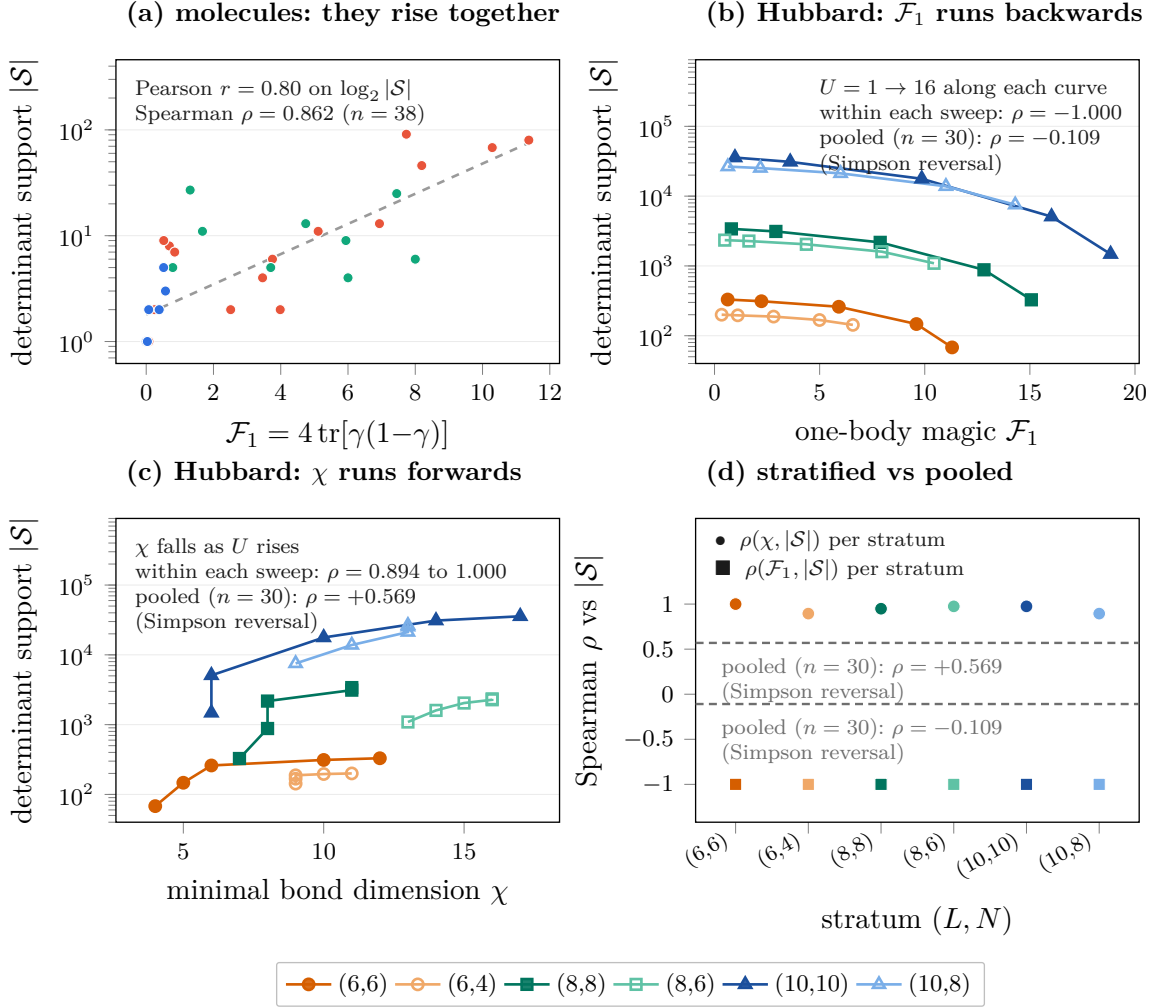


FIG. S2. **Inside every stratum one-body magic $\mathcal{F}_1$ orders the static determinant support $|\mathcal{S}|$ perfectly and with the sign inverted; what fails is the *pooled* coefficient, not $\mathcal{F}_1$.** (a) Nineteen molecular active spaces at equilibrium and at dissociation (38 points): in the static-correlation regime $\mathcal{F}_1$ and $|\mathcal{S}|$ rise together (Pearson $r = 0.80$ on $\log_2 |\mathcal{S}|$; Spearman $\rho = 0.862$). The grey dashed line is the least-squares fit $\log_2 |\mathcal{S}| = 0.474 \mathcal{F}_1 + 0.850$ over those same points, a guide to that correlation and not a model of the cost. Color: covalent bond-breaking (red, 20 points), multireference/open-shell (green, 10) and ionic/weakly correlated (blue, 8), the groups of Table S12. (b) The Hubbard interaction sweep is not one cloud of 30 points but six strata (L, N) of five on-site repulsions $U = 1, 2, 4, 8, 16$; one curve per stratum. Inside every one of the six, raising U raises $\mathcal{F}_1$ and lowers $|\mathcal{S}|$, and does so with a perfect rank reversal: $\rho(\mathcal{F}_1, |\mathcal{S}|) = -1.000$ in all six. Pooling across strata collapses the coefficient to $\rho = -0.109$; that collapse is a Simpson reversal, not an absence of signal. (c) The same six strata against the minimal bond dimension: $\rho(\chi, |\mathcal{S}|)$ runs from $+0.894$ to $+1.000$ within strata, with a mean of 0.948 , and does not change sign between them. Its pooled coefficient, like $\mathcal{F}_1$'s, is a Simpson reversal and is not used: pooling six deterministic five-point sweeps destroys within-stratum structure whichever way the result falls, and it fell in our favour on one of the two. (d) The six within-stratum coefficients against the two pooled ones. Stratified, the $\mathcal{F}_1$ evidence is one-sided and exact: Kendall $\tau = -1.000$ with 0 concordant of 60 within-stratum pairs and no ties, one-sided exact permutation $p = 3.35 \times 10^{-13}$ over the $(5!)^6 = 2.99 \times 10^{12}$ within-stratum relabellings. The pooled reading, $\rho = -0.11$ ($n = 30$, $p \approx 0.56$), is not used. Two limits of this evidence, which the stratified test does not remove. First, the five points of a stratum are a deterministic monotone sweep in U , so the effective number of independent observations is 6, not 30; the exact p counts relabellings, not independent samples. Second, this is a statement about rank direction only: inside a fixed sweep $\mathcal{F}_1$ orders the five costs *perfectly*, so the claim is not that $\mathcal{F}_1$ is uninformative about $|\mathcal{S}|$ but that its sign is not stable—inverted here, positive over the molecular set of panel (a)—which is exactly what an orbital invariant compared against a basis-dependent cost must do.

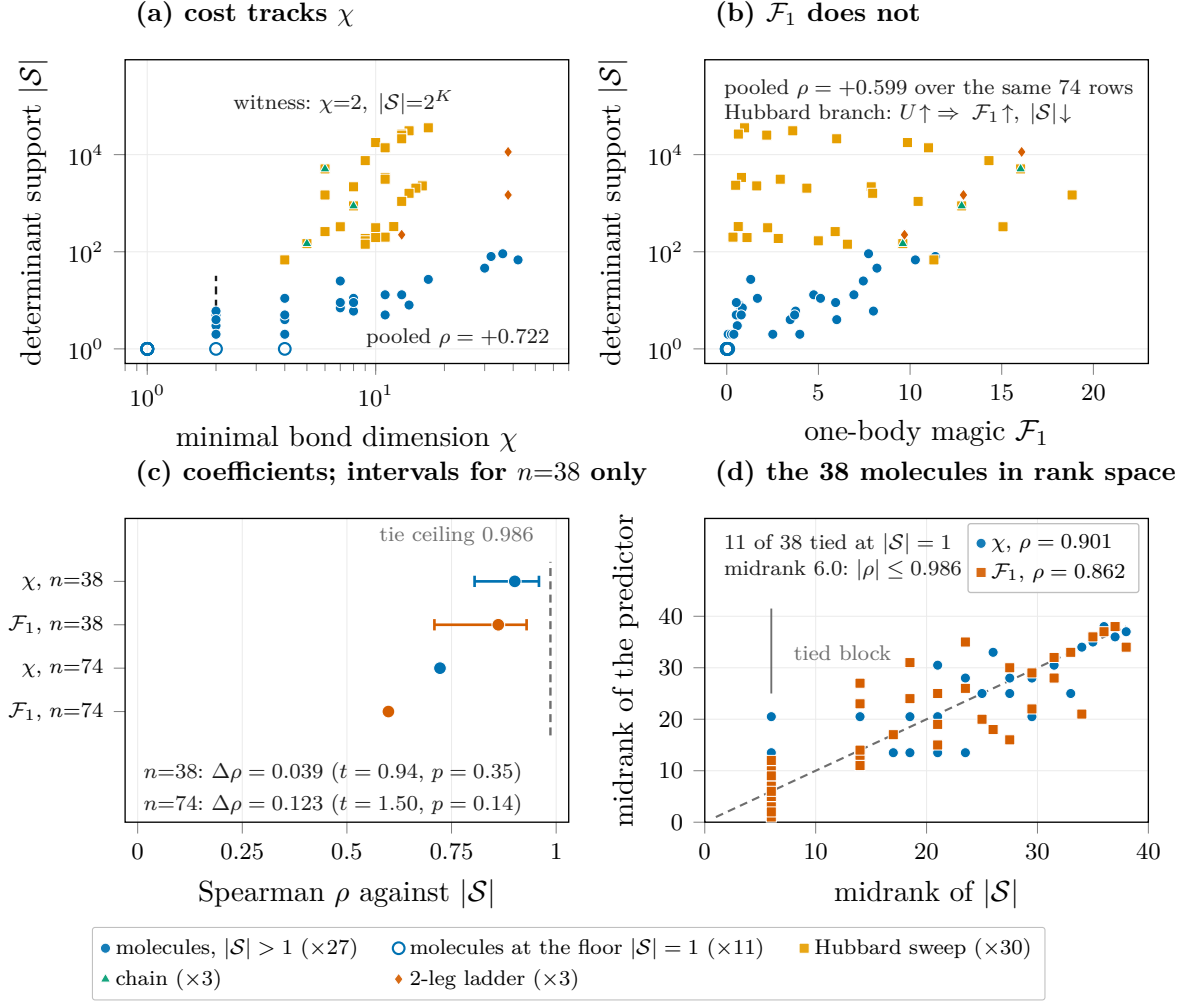


FIG. S3. **The resource thesis over the 74 deposited rows (71 distinct exact ground states).** Nineteen molecules at equilibrium and dissociation, a 30-point Hubbard interaction sweep, and a matched chain/two-leg-ladder pair; all by exact diagonalization. (a) The determinant support tracks the minimal bond dimension in every family (pooled Spearman $\rho = +0.722$ over all 74 rows, $+0.746$ over the 71 distinct states—see *Provenance* below), with the geminal witness $\chi=2$, $|\mathcal{S}|=2^K$ the deliberately loose lower bound. Open circles are the 11 of 38 molecular points pinned at the floor $|\mathcal{S}| = 1$ —29% of the molecular set—where the support cannot resolve anything and from which both molecular coefficients draw much of their leverage. (b) One-body magic does not ($\rho = +0.599$ over the same rows, $+0.592$ over the 71 distinct states): the Hubbard branch runs the wrong way ($U \uparrow$ raises $\mathcal{F}_1$ and lowers $|\mathcal{S}|$) and the chain/ladder pair sits at $\mathcal{F}_1$ values agreeing to within 0.9% while $|\mathcal{S}|$ differs by 1.52, 1.68 and 2.24 $\times$. (c) The two coefficients with the intervals the deposited data actually support. Bars are 95% cluster bootstrap intervals over the 19 molecules ($B = 20\,000$, cluster = molecule): χ gives 0.901 [0.805, 0.959], $\mathcal{F}_1$ gives 0.862 [0.709, 0.929]. *The molecular comparison does not resolve the two resources:* $\Delta\rho = 0.039$, Williams $t = 0.94$ on 35 df, $p = 0.35$ (Steiger $z = 0.94$), bootstrap 95% CI of the difference $[-0.070, +0.189]$; neither does the pooled comparison ($\Delta\rho = 0.123$, $t = 1.50$, $p = 0.14$). The $n = 74$ entries are point estimates—only the difference was resampled. The dashed line is the ceiling $|\rho| \leq 0.986$ imposed by the ties in $|\mathcal{S}|$, against which χ reaches 91% and $\mathcal{F}_1$ 87% of what is attainable. What discriminates between the two candidate resources is therefore not this figure but the stratified Hubbard sweep of Fig. S2(b,d). (d) The same 38 molecular points in the rank space in which a Spearman coefficient is actually computed, against the identity line. The 11 floor points share midrank 6.0—the vertical block at the left—which is the origin of the ceiling quoted in (c). *Provenance and caveats.* The three “chain” rows are the same states as the $U = 8$ half-filled entries of the Hubbard sweep (identical $|\mathcal{S}|$ and χ , identical $\mathcal{F}_1$ to the precision at which they are stored), re-plotted as the ladder’s matched partner; the 74 rows are therefore 71 distinct states and three of them enter the pooled coefficients twice. Recomputed over the 71 distinct states the pooled coefficients are $\rho(\chi, |\mathcal{S}|) = +0.746$ and $\rho(\mathcal{F}_1, |\mathcal{S}|) = +0.592$. The 38 molecular points are 19 molecules $\times$ 2 geometries, i.e. dependent pairs (effective $n \approx 31.3$), which is why every molecular interval here is clustered by molecule.

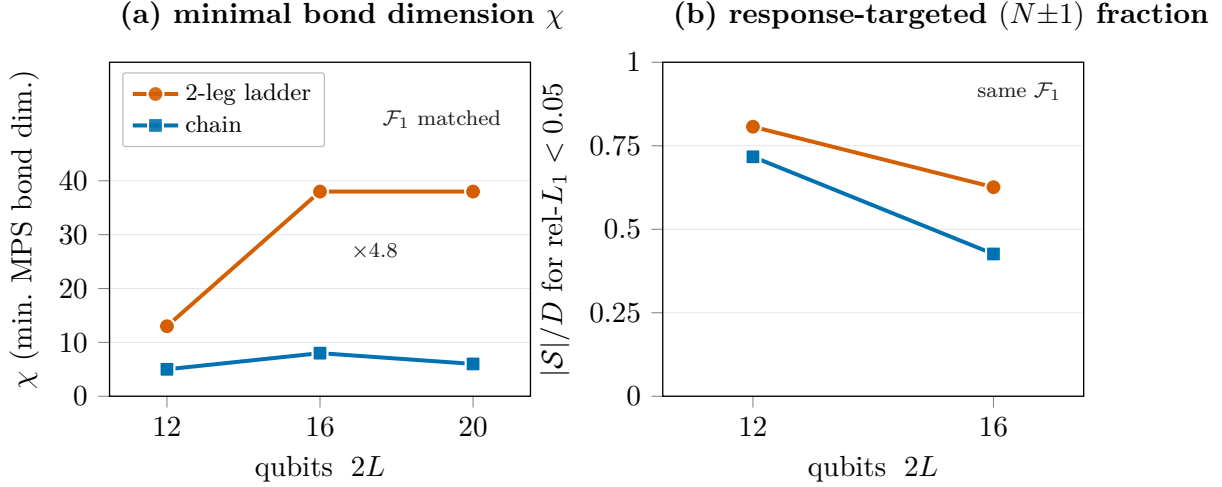


FIG. S4. **The cost is tracked by entanglement, not one-body magic—a controlled chain-versus-ladder contrast.** A two-leg Hubbard ladder versus a chain of the *same* site count (half filling, $U/t = 8$; exact diagonalization). At matched fermionic magic $\mathcal{F}_1$ (open chain vs. ladder: 9.60/9.69, 12.82/12.91, 16.03/16.09 at 12/16/20 qubits— $\mathcal{F}_1$ is a geometry-blind one-body invariant) the ladder’s perpendicular cut has an area-law boundary of two rather than one. **(a)** Its minimal matrix-product-state bond dimension χ is far larger; the chain-to-ladder ratio grows over $12 \rightarrow 20$ qubits ($\times 2.6$, $\times 4.8$, $\times 6.3$), the ladder χ saturating at 38 while the chain χ stays small and does *not* grow (5, 8, 6 at 12/16/20 qubits—non-monotone, with no systematic increase over this range, consistent with the at most logarithmic entanglement growth of the spin sector of the one-dimensional chain)—the reported χ are threshold-truncated Schmidt ranks. **(b)** The response-targeted $(N\pm 1)$ fraction needed for spectral relative- $L_1 < 0.05$ is correspondingly higher for the ladder (these union $(N\pm 1)$ response fractions differ from the single-sector $(N+1)$ $A(\omega)$ fractions of Fig. 5, a different observable). Same one-body magic, higher cost: the determinant support is lower-bounded in the raw count and empirically tracked by the basis-dependent bond dimension χ (Sec. S5.3), not by $\mathcal{F}_1$. A ladder of this width ($\chi \approx 38$) remains classically near-exact for dynamical DMRG; it is a resource contrast and no claim of advantage is attached to it.

is the stabilizer of the state ($|\langle \Psi(0) | \Psi(U_{\text{in}}) \rangle| = 1.000000000000$); at the fiber angle $s = \pi/16$, every generator that actually moves the state gives $\hat{\Lambda}/\eta \in [2.34, 5.61]$. That range is a statement about generators at one angle and not about the angle sweep of Sec. S12, whose rotated endpoint descends to 1.61 as $s \rightarrow 0$ because the rotation itself is being switched off. The rhetorical parallel with the cost witness—where the orbit contains an $O(K)$ -cost point and points of plausibly exponential cost—is therefore lost; Sec. S5.2 states that asymmetry. (iv) The scale is a toy: $2K \leq 8$ orbitals, sector dimension ≤ 3920 , no instance at the $L = 12$ Hubbard scale of Sec. V.

What the exhibit delivers survives all four limits and is measured: on fibers where every invariant is fixed to 10^{-14} and the captured Born weight to 2.2×10^{-16} , the certificate of Theorem 1 separates a non-vacuous value from the trivial bound in every row of Table S4, and a quantity constant on those fibers cannot make that distinction. The a priori predictor asked for at the head of this section does not exist in the invariant class; what stands in its place is the subspace-aware certificate, whose non-vacuous region is mapped, not asserted, in Sec. V.

S5.1. The obstruction is bilateral, and its sign is not stable

The same symmetry closes the cost side, in both directions (Fig. S2). The cost statistic of this section is the *static* determinant support $|\mathcal{S}|_{\varepsilon_w}$: the number of fixed-basis Slater determinants of the *ground state* carrying all but ε_w of its weight. Three tolerances called 0.05 or 0.01 appear in this paper and they are three different objects, so we subscript all of them. ε_w cuts the support: $\varepsilon_w = 10^{-2}$ for the molecular suite (`n19_suite.py`, 99% of the weight) and $\varepsilon_w = 2.5 \times 10^{-3}$ for the lattice families (`cost_vs_entanglement.py`, 99.75%, written there as $1 - \varepsilon^2$ with $\varepsilon = 0.05$). $\varepsilon_\chi = 0.05$ cuts the Schmidt spectrum for χ , by discarded tail $\leq \varepsilon_\chi^2$, in both families. Neither is the third 0.05 of this work, which is a threshold on a relative L_1 spectral error at a stated resolution and defines the *dynamical* support $|\mathcal{S}|_\eta$ of Sec. VII; and neither is the S of Theorem 1, which is whatever the sampler returned. This subsection is about $|\mathcal{S}|_{\varepsilon_w}$ and about nothing else. Where the *raw* support—determinants of nonzero amplitude—is meant we say so, because the two do not obey the same bounds. Unlike $\mathcal{F}_1$, $|\mathcal{S}|$ is basis dependent: a single Slater determinant has $|\mathcal{S}| = 1$ in its own natural orbitals and exponentially many terms in a generic basis. An orbital rotation is a free operation that leaves the occupation spectrum, hence $\mathcal{F}_1$, exactly fixed while changing $|\mathcal{S}|$; a quantity invariant under operations that change $|\mathcal{S}|$ at fixed value cannot determine it.

Both directions are realized and neither is a limiting case. A free-fermion ground state at $U = 0$ —one Slater determinant, $\mathcal{F}_1 = 0$, the minimum—has site-basis support growing exponentially, $|\mathcal{S}|_{\varepsilon_w} = 35, 336, 3496, 37361$ for $L = 4, 6, 8, 10$ on the open chain at the lattice threshold $\varepsilon_w = 2.5 \times 10^{-3}$, the discarded weight being at most ε_w (`free_fermion_support.py`; at $L = 4$ the four smallest determinant weights equal ε_w exactly, and a strict cut gives 36): minimal invariant, divergent cost. Along a Hubbard chain at fixed filling, raising U raises $\mathcal{F}_1$ while lowering $|\mathcal{S}|$ as the ground state localizes: rising invariant, falling cost. Across the nineteen molecules of Table S12 the two rise *together* (Pearson $r = 0.80$ on $\log_2 |\mathcal{S}|$), both merely tracking static correlation. The relationship is two-sided and its sign is not stable—the signature of comparing an orbital invariant to a basis variant.

We state what this is and what it is not. It is not a refutation of anyone’s claim: Ref. [16] proposes $\mathcal{F}_k$ as a measure of fermionic magic (non-Gaussianity) for matchgate circuits—faithful, Gaussian invariant and (sub)additive—which is what it is, and no one proposed it as a predictor of the determinant support. It is our own no-go, about our own cost, and a corollary consistent with—not a competitor to—the two families of Tarabunga *et al.* [17], who in July 2026, before this work, showed that occupation entropies depend only on the covariance matrix, two members of which (orders 1 and 2) are strong monotones that lower-bound the non-Gaussian gate count, while natural-orbital participation entropies depend on the full state and *upper-bound the classical simulation cost* in an orthonormal Gaussian basis. Both of their families are Gaussian invariant; the basis dependence at issue here enters only because $|\mathcal{S}|$ is counted in a fixed determinant basis rather than in the state’s own natural orbitals. That is the quantity on the useful side of the division, and the priority is theirs. On the invariant side the same acknowledgement is owed to Leone and Bittel [18], who in July 2026, before this work, proved that the fermionic entropy $M_f = n(1 - \|\Gamma\|_2^2/2n)$ on n modes—an affine function of the squared Frobenius norm of the correlation matrix—is a strong pure-state Gaussian monotone (its convex roof one for mixed states)—on $n = 2n_o$ modes M_f is $\mathcal{F}_1$ itself and Tarabunga *et al.*’s quadratic occupation entropy, whose strong monotonicity the revised version of Ref. [17] proves independently—and that its fermionic purity $\|\Gamma\|_2^2/2n$ has an unbiased estimator using $O(\varepsilon^{-2})$ two-copy measurements. What our construction adds to that division is the constructive instance it does not contain: a family (Theorem 3) on which every order of the invariant is maximal while the cost is $O(K)$, and fibers (Proposition 2) on which the invariants are frozen to 10^{-14} while the certificate of Theorem 1 still separates.

S5.2. Two blindnesses, and why we do not sell them as one symmetry

The two halves of this section are not one statement wearing two hats. On the cost side the invariance is a genuine physical group: the Gaussian orbital rotations $U(n_o)$ are free operations, they act on the whole problem, and along one orbit the cost is $O(K)$ at one point and, at others, plausibly exponential but not established here, at fixed invariant—a symmetry argument in the ordinary sense. On the precision side it is weaker and of a different kind: $\Lambda_S(\eta)$ reads the off-diagonal block $P_S H Q_S$ in the working basis, and an invariant is blind to that block—informational blindness, not a gauge. The orbit there does not sweep an extensive range; as declared above, the small-leakage endpoint is realized only by the stabilizer of the state, and the contrast shrinks with K rather than growing.

What joins the two is therefore not a symmetry at all, and nothing stronger is claimed for it than what it is: the “+” of Eq. (7), one inequality whose two summands are the two halves of this paper, one controlled by the captured weight and one not, each carrying its own obstruction. What makes that join a measurement rather than an analogy is that both summands have been evaluated on the same object, $\mathcal{S}^* = \text{supp}(\phi)$, where the first summand is identically zero while the second still leaves a relative L_1 error of 0.43 at $L = 6$ and 0.35 at $L = 8$ on the periodic rings at $\eta = 0.18t$, and 0.28 on the open chain (Secs. III and IV).

S5.3. What does track the cost: χ , and the statistics the right way up

The computable lower bounds on $|\mathcal{S}|$ are themselves basis dependent. The orbital-Schmidt rank across a bipartition—the tensor-network bond dimension χ [103]—is a rigorous floor on the *raw* support,¹ and, empirically, a tracker of the measured $|\mathcal{S}|_{\varepsilon_w}$. It is not the winner of a contest: χ is measured rather than predicted, and it has no purchase on the

¹ For any state and any bipartition into (L, R) , $\chi \leq |\mathcal{S}|$ with $|\mathcal{S}|$ the number of determinants of nonzero amplitude: each determinant factorizes across the cut, so the matricization $A_{D_L, D_R} = c_D$ has at most $|\mathcal{S}|$ nonzero entries and rank at most $|\mathcal{S}|$. It is elementary, which is why we record it as an observation, and it is proved for the raw support only: once amplitudes are thresholded the inequality need not survive—in the molecular suite $|\mathcal{S}|$ is cut at 99% of the weight while χ is cut at a discarded tail of $\varepsilon_\chi^2 = 2.5 \times 10^{-3}$, i.e. at 99.75%, so the two are not truncated at the same place—and the measured $|\mathcal{S}|_{\varepsilon_w}$ falls below χ in 7 of the 38 molecular points. It is also one-sided and can be exponentially loose: Theorem 3 has $\chi = 2$ at $|\mathcal{S}|_{\text{pair}} = 2^K$.

leakage at all. On the fibers of Table S4 it is *identical* at the two endpoints—2 and 2 at $K = 3$, 1 and 1 at $K = 4$, at equal raw support—while $\hat{\Lambda}/\eta$ moves by a factor 6–7; in a control at $K = 3$ it jumps from 2 to 16 with $\hat{\Lambda}/\eta = 2.336$ and relative $L_1 = 0.238$ unchanged to three figures; and across a subfamily of 58 points on which χ and the raw support are *exactly* equal, $\hat{\Lambda}/\eta$ still varies by 2.8 and the measured error by 8.3. The only tracker of the leakage is the boundary coupling $\|P_S H Q_S\|$ of Sec. III. What follows is a statement about $|\mathcal{S}|_{\varepsilon_w}$ and about nothing else.

The Hubbard sweep, stratified. The 30 lattice points are not 30 independent observations. They are six (L, N) strata—(6, 6), (6, 4), (8, 8), (8, 6), (10, 10), (10, 8)—each swept over the same five values $U = 1, 2, 4, 8, 16$. Within every stratum the rank correlation of $\mathcal{F}_1$ with $|\mathcal{S}|_{\varepsilon_w}$ is -1.0000 , exactly, in all six; the corresponding $\rho(\chi, |\mathcal{S}|_{\varepsilon_w})$ are 1.000, 0.894, 0.949, 0.975, 0.975, 0.894, mean 0.948. The stratified concordance statistics $(n_C - n_D)/(n_C + n_D)$ are -1.000 for $\mathcal{F}_1$, with 0 concordant pairs out of 60 and no ties, where the statistic is Kendall’s τ , and $+1.000$ for χ , with 50 concordant, 0 discordant and 10 tied, where it is the Goodman–Kruskal γ , which drops the tied pairs; Kendall’s τ_b , which charges them, is below 1. Because each $\mathcal{F}_1$ stratum is a perfect reversal of five points, the exact one-sided stratified permutation p -value is $(1/120)^6 = 3.35 \times 10^{-13}$, against 1.93×10^{-10} for χ , which has ties. Both take a product across strata and therefore assume the six independent. They are not six independent experiments: they are three lengths crossed with two fillings, on one code path and one U grid. The per-stratum coefficients, Fisher’s more conservative combination and the tie structure are in Sec. S14.

Two pooled coefficients are available over these same 30 points: $\rho = 0.57$ for χ , which invites the reading that the bond dimension tracks the cost, and $\rho = -0.11$ for $\mathcal{F}_1$, which invites the reading that it carries no evidence. Neither reading is used here, and for one reason. The 30 points are six deterministic five-point sweeps; the effective n is ≈ 6 , not 30; and pooling strata with different offsets destroys within-stratum structure. Here it destroys it in our favour on one coefficient and against us on the other, which is precisely why neither may be kept. What is used instead is the within-stratum evidence above, which is stronger on both sides and symmetric between them: $\mathcal{F}_1$ anti-orders the support perfectly in all six strata, and χ orders it at a mean ρ of 0.948.

That the evidence is stronger does not make it what we were about to call it. A rank correlation of exactly -1 in all six strata is the strongest evidence a rank statistic can give of a *deterministic monotone* relation between $\mathcal{F}_1$ and the support inside a stratum—with the wrong sign. In this family $\mathcal{F}_1$ is not decoupled from the cost; it is an excellent rank predictor of it whose *sign* reverses from family to family: $+1.000$ in the chain and ladder sweeps below, $+0.86$ across the molecules, -1.000 within every Hubbard stratum. That is the failure mode the no-go of Secs. VIB and S5.1 predicts for an orbital invariant compared against a basis variant, and “unstable in sign”, not “decoupled”, is the phrase the measurement supports. The physical content is one mechanism observed six times—raising U localizes the ground state, so $|\mathcal{S}|_{\varepsilon_w}$ falls and χ falls while $\mathcal{F}_1$ rises—and a p -value of 3.35×10^{-13} is the probability of six perfect reversals under independent random orderings within each stratum—a null that six strata sharing one mechanism do not satisfy—and not evidence that it generalizes beyond the one-dimensional Hubbard family in which it was measured.

The molecular set does not discriminate. The $n = 38$ molecular points are 19 molecules at two geometries—dependent pairs, not 38 draws—and 76% of them lie in some tie in $|\mathcal{S}|_{\varepsilon_w}$, 11 pinned at the floor $|\mathcal{S}| = 1$. The comparison the molecular set is asked to settle is not resolvable by any of three procedures: $\rho = 0.90$ for χ against 0.86 for $\mathcal{F}_1$ differ by 0.039, a molecule-level cluster bootstrap puts a 95% interval of $[-0.070, +0.189]$ on that difference, and Williams’s and Steiger’s tests for dependent correlations give $p = 0.35$. The pooled pair over all 74 states, 0.72 against 0.60, is likewise unresolved. The molecular set therefore does not separate the two candidate resources (Sec. S14), and the whole weight of the separation rests on the stratified Hubbard sweep above.

The two families where $\mathcal{F}_1$ wins, written out. In the matched chain and two-leg ladder families of Fig. S4 ($n = 3$ each), $\rho(\mathcal{F}_1, |\mathcal{S}|) = +1.000$ exactly, against $\rho(\chi, |\mathcal{S}|) = 0.50$ and 0.87. We report it and then dismantle it. The parameter swept there is L , not U : for the open chains $\mathcal{F}_1/L = 1.6008, 1.6020, 1.6031$ at $L = 6, 8, 10$, that is $\mathcal{F}_1 = 1.60 L$ to three figures, so $\rho = +1$ restates “both quantities grow with system size” and nothing else, and conditioning on L leaves zero degrees of freedom. The three chain rows are moreover literal duplicates of three of the 30 Hubbard rows—the $U = 8$, half-filled entries at $L = 6, 8, 10$, identical to the last digit in $\mathcal{F}_1$, $|\mathcal{S}|$ and χ —so they are not independent evidence. And with $n = 3$ the smallest attainable permutation p -value is $1/6$: less evidence than three coin flips. What the chain/ladder pair does establish is the controlled contrast it was built for, which needs no correlation coefficient to read: at $\mathcal{F}_1$ equal to within 1%, the ladder’s support exceeds the chain’s by up to $\sim 2.2\times$, while χ moves from 5 to 13 at $L = 6$ and from 8 to 38 at $L = 8$.

What stands, and where the tracker fails too. Figure S3 consolidates the picture over all 74 exact ground states: the determinant support is bounded below by the bond dimension in the raw count, and the bond dimension is the better empirical tracker of the thresholded one. It is not a monotone tracker, and the exception is in the same file as the rule. Within every Hubbard stratum χ orders the support at $\rho = 0.89$ –1.00. Along the chain family, where the swept parameter is L and not U , it does not: $\chi = 5, 8, 6$ at $L = 6, 8, 10$ while $|\mathcal{S}|_{\varepsilon_w} = 147, 880, 5084$, so χ *falls* across the last step while the cost grows by 5.8, and $\rho(\chi, |\mathcal{S}|_{\varepsilon_w})$ there is 0.50; in the ladder family χ is flat at 38 over that same step while the cost grows by 7.7. That non-monotonicity of the chain χ is stated here explicitly, and it is a limit on χ itself. What $\mathcal{F}_1$ does instead is fail in two *additional* and visibly distinct ways—the Hubbard

branch runs the wrong way at a stratified $\tau = -1.000$, and the chain/ladder points sit at identical $\mathcal{F}_1$ with $|\mathcal{S}|_{\varepsilon_w}$ differing by up to $\sim 2.2\times$. Neither statement supplies the predictor this section set out to find, χ being itself basis dependent, measured rather than predicted, and blind to the leakage. What $\mathcal{F}_1$ *is* remains intact and useful: a faithful, efficiently measurable detector of static (multireference) correlation, generalizing to fermionic non-Gaussianity, and to the nineteen molecules of Table S12, the qubit non-stabilizerness peak of molecular bonding [120] and its reported proportionality to Hartree–Fock weight, itself reported to break down beyond the Coulson–Fischer point [121]; that a single-particle invariant can be maximal while bond entanglement stays $O(1)$ is the fermionic counterpart of the separation between magic and entanglement reported for qubit matrix-product states [122]. The cost is a property of the sampler’s frame, read after the fact by χ and priced in Sec. VII; the error is a property of the subspace’s boundary, read after the fact by $\Lambda_S(\eta)$ and mapped in Sec. V. Per-species $|\mathcal{S}|_{\varepsilon_w}$, D and χ are in Sec. S15; the full stratified analysis—the six per-stratum coefficients, the exact one-sided permutation values, the tie structure, the cluster bootstrap and the dependent-correlation tests—is in Sec. S14.

S6. THE THREE PRICES: THE RESOLUTION GRID AND FINITE SHOTS

The support exponent and its fit range. Restricting the fit of Eq. (20) to $L = 4\text{--}12$, the five sizes Table S5 covers, gives 1.082 at $R^2 = 0.998$; we quote the fit range with the exponent throughout, because it matters at the second decimal. The same caution applies to the object fitted: 1.082 is the fit to the published points, while the 1.090 in the bottom block of Table S5 is the fit to that table’s own coarser replication grid at the same resolution. They are two fits to two datasets and the difference is theirs, not a rounding.

The two exponents are one exponent. It would be convenient to report that the support exponent, unlike the fraction exponent, is resolution-independent. It is not, and the arithmetic is immediate: the fraction *is* the support divided by the sector dimension—exactly, to the last digit, at all six sizes—and the sector dimension does not depend on η . The two exponents therefore differ by a constant, $b_{|\mathcal{S}|}(\eta) - b_{\text{frac}}(\eta) = b_{\text{dim}} = 1.291$ on $L = 4\text{--}12$, and they move together: across the decade $\eta = 0.05 \rightarrow 0.50 t$ both shift by 0.148 in absolute value (Table S5). There is one resolution-dependent exponent in this analysis, and quoting the fraction and the support separately reports it twice.

What differs is the base it is measured against, and that is not cosmetic. On the fraction exponent 0.148 is three quarters of the quantity itself: $e^{-0.13 L}$ becomes $e^{-0.27 L}$, and the headline changes qualitatively with a choice the model does not make. On the support exponent the same 0.148 carries it from 1.165 to 1.017—a seventh of its own value, 13% measured against the larger end and 15% against the smaller—and “exponential in L with an exponent near one” survives at every resolution in the decade. We therefore report the support, and we report it for two reasons, neither of which is stability: it is the absolute resource a device has to populate, and it is not divided by a sector dimension that grows for a reason—the combinatorics of the Hilbert space—that has nothing to do with the method. The fraction is reported as the derived, resolution-dependent quantity that it is, with its full grid attached.

The headline these numbers offer—that the fraction needed for a fixed spectral accuracy falls from 0.92 at $L = 4$ to 0.08 at $L = 14$, as $e^{-0.25 L}$ —is tunable, and no headline of that form is claimed here. Over the decade of η the fraction exponent moves by a factor 2.2, from -0.1266 at $\eta = 0.05 t$ to -0.2743 at $\eta = 0.50 t$. At $\eta = 0.50 t$ the fraction at $L = 12$ is already 0.10, a quarter above the $L = 14$ value at the resolution used throughout this work and reached two sizes earlier: any headline of this form is reachable by choosing the broadening.

Nor is the fraction exponential even at fixed η . The local logarithmic slopes between consecutive published points are $-0.056, -0.194, -0.214, -0.379, -0.377$ —a factor 6.8 across five intervals, steepening monotonically over the first four. A single-exponential fit over $L = 4\text{--}14$, with $R^2 = 0.94$, is a straight line drawn through a curve of increasing slope, and the fitted exponent depends on where the fit stops: $L = 4\text{--}10$ gives -0.159 , $L = 4\text{--}12$ gives -0.209 , and the -0.2477 of the fit over $L = 4\text{--}14$ requires reaching $L = 14$.

The mechanism that makes the fraction move with η is available without derivation, and Sec. IV states it as Conjecture 1 rather than as a theorem: the polynomial-approximation route to a geometric rate in the Krylov order places the kernel poles at $\omega \pm i\eta$, so any such rate approaches 1 as $\eta \rightarrow 0^+$ and any prefactor diverges (Sec. S10.3). At finer resolution more Krylov order—hence more of the sector—is needed for the same L_1 error. We use that as a mechanism and not as a bound: no $C(\eta)$ or $\rho(\eta)$ is exhibited anywhere in this work, and none is needed to read Table S5, which is a measurement.

One further constraint has to be declared rather than inherited. The reference against which the threshold is applied is itself a finite-depth Lanczos computation, recomputed at the same depth as the subspaces so that the comparison is internally consistent ($n_\ell = 150$; `scaling_lanczos_mf.py`). Two consequences follow, and both belong in the open. First, the published relative L_1 is a distance between two depth-150 objects, not between the reconstruction and the exact A . Second, that reference drifts with depth, and the drift is resolution-dependent: between $n_\ell = 100$ and $n_\ell = 200$ at $L = 12$ the relative L_1 change of the reference *alone* is 11.1% at $\eta = 0.05 t$, 3.0% at $\eta = 0.10 t$, 0.93% at $\eta = 0.15 t$ and 0.05% at $\eta = 0.25 t$. At the depth used here the protocol is self-consistent only for $\eta \gtrsim 0.15 t$ —which is

TABLE S5. **Sector fraction required for relative $L_1 < 0.05$, versus system size and resolution.** Same ground state, same time-evolution ranking (protocol $K = 18$, $dt = 0.5$, 16 substeps, 19 snapshots; `scaling_lanczos_mf.py`), same threshold, same depth $n_\ell = 150$; only η changes. The scan over subspace size in this replication is coarser than the bisection that produced the published points, and returns the smallest scanned fraction meeting the threshold, so the $\eta = 0.15t$ column is an upper bound on them: against the published 0.9167, 0.8200, 0.5561, 0.3625, 0.1700 it reads 0.917, 0.820, 0.560, 0.380, 0.180, agreeing at the two smallest sizes and standing 0.7%, 4.8% and 5.9% high at the three largest. The same granularity is why the fraction exponent of this column, -0.2012 , is not identical to the -0.2093 obtained by fitting the published points themselves over $L = 4$ –12. Bottom block: least-squares exponents. *Both* fit windows are given for the support, because the choice moves the third digit and this work criticizes exactly that elsewhere in this section. The fraction exponent moves by a factor 2.2 over the decade, from -0.1266 to -0.2743 , while the support exponent moves from 1.165 to 1.017; in absolute value both move by 0.148, which is the whole content of the comparison.

L	η/t							
	0.05	0.075	0.10	0.15	0.18	0.25	0.35	0.50
4	0.917	0.917	0.917	0.917	0.917	0.917	0.917	0.875
6	0.900	0.890	0.870	0.820	0.800	0.730	0.680	0.630
8	0.740	0.680	0.640	0.560	0.540	0.480	0.420	0.380
10	0.520	0.480	0.440	0.380	0.340	0.260	0.220	0.200
12	0.340	0.280	0.240	0.180	0.160	0.140	0.120	0.100
fraction exponent, $L=4$ –12	-0.1266	—	-0.1681	-0.2012	—	-0.2395	—	-0.2743
R^2	0.904	—	0.918	0.929	—	0.970	—	0.982
support exponent, $L=4$ –12	1.165	—	1.123	1.090	—	1.052	—	1.017
R^2	0.999	—	0.999	0.998	—	0.999	—	0.999
support exponent, $L=6$ –12	1.136	—	1.088	1.053	—	1.022	—	0.992

exactly where the published column sits, at the margin. Fig. 8, a different channel drawn at $n_\ell = 200$, sits *below* that margin, at $\eta = 0.10t$, which is why its caption quotes its lineshape with its depth and not as exact.

The fraction is therefore reported as a declared window rather than as a number: only for $\eta \geq 0.15t$ at $n_\ell = 150$, with the full $\text{frac}(L, \eta)$ grid of Table S5 published beside it.

The shot budget, in full. Populating a subspace by measurement is a coupon-collector problem on the Born distribution of the time-evolved seed, so the budget exceeds the support it buys. Table IV reports that budget *measured*, by multinomial sampling of the declared protocol, at the five sizes where the measurement is affordable; only $L = 14$ carries no measurement. Two independent runs contribute, and we name them as two rather than merging them: run A draws eight seeds at $L = 4, 6$ and 8, run B draws four seeds at $L = 4, 6, 8$ and 12 and six at $L = 10$, with different seeds and a different budget grid. Where they overlap they agree—2.43 against 2.44×10^4 at $L = 6$ and 2.70 against 2.61×10^5 at $L = 8$, that is to 0.2% and 3.2%—which is why the two larger rows are reported as measurements and not as extrapolations of the smaller ones. Every budget is for *one* orbital seed and *one* branch; the momentum-resolved observable consumes the full set of orbital seeds in both $(N \pm 1)$ sectors, sixteen channels at $L = 8$.

The published curve and finite shots, in full. Precision about the label matters more here than anything else. The ranking that produced the published fractions is `argsort` applied to the Born distribution of the time-evolved state accumulated over the 19 snapshots of the protocol (`scaling_lanczos_mf.py`, `stage_rank`)—that is, to the distribution the device would itself measure, in the limit of infinitely many shots. It is *not* a clairvoyant ordering by exact eigenvector amplitudes; the repository contains such an oracle elsewhere (`amortized_recovery.py`) and labels it as one. But it is also not a finite-shot reconstruction, and to describe it as *sampled* without a shot count is to mislabel it. The published curve is the $T \rightarrow \infty$ limit of the declared protocol, and it is labelled that way from here on.

The published curve is therefore a strict lower bound on the finite-shot cost. At $L = 4$ the two agree because the sampler exhausts the sector; at $L = 6$ and $L = 8$ the finite-shot fraction exceeds the published one by 0.037 and 0.071. That is a direction across three sizes, one of which is degenerate, and we state it as a direction and do not fit an exponent to it—which is the same objection this section raises against the fraction headline, applied to our own finite-shot curve. At the published subspace itself, $L = 8$ and $|\mathcal{S}|_\eta = 2180$, the infinite-shot ordering gives relative $L_1 = 0.0490$, just inside the threshold, against 0.1006 ± 0.0036 for finite shots at $T = 1.47 \times 10^5$: a factor 2.05 ± 0.07 , and the reason the finite-shot fraction has to rise to 0.63.

The finding that runs the other way is the more useful one, and it is what this subsection closes on: the penalty is budgetary, not structural, and it extinguishes with a budget that is affordable at this size. On the same pair as the last row of Table S6— $L = 8$, $|\mathcal{S}|_\eta = 2331$ —raising the budget from $T = 2.04 \times 10^5$ to $T = 2.60 \times 10^6$, a factor 12.7, takes the ratio of the finite-shot error to the infinite-shot-ordering error from 1.96 ± 0.12 to 1.06 ± 0.01 , through 1.51, 1.18 and 1.09 at the intermediate budgets. No sampling penalty in this family can be quoted without its budget, because the penalty is a statement about T and not about the method. We therefore report finite-shot fractions only

TABLE S6. **Finite shots, with the budget attached.** Multinomial sampling of the declared protocol, eight independent seeds, $\eta = 0.15t$; $\pm$ is the sample standard deviation across seeds, not the standard error of the mean. Top block: the sector fraction the finite-shot reconstruction needs to reach relative $L_1 < 0.05$, and the budget at which it reaches it. Bottom block: at *matched* subspace size, the error of the infinite-shot ordering against the error of finite shots at the stated budget, and the ratio between them.

	$L = 4$	$L = 6$	$L = 8$
fraction, $T \rightarrow \infty$ (published)	0.9167	0.8200	0.5561
fraction, finite shots	0.9167 ± 0.0000	0.8571 ± 0.0135	0.6275 ± 0.0031
budget T at which it is reached	$(7.8 \pm 2.0) \times 10^2$	$(2.43 \pm 0.34) \times 10^4$	$(2.70 \pm 0.13) \times 10^5$
matched $ \mathcal{S} $ (size the sampler returned)	22	247	2331
budget T at that size	7.8×10^2	1.74×10^4	2.04×10^5
relative L_1 , $T \rightarrow \infty$ ordering	1.4×10^{-14}	0.0469	0.0362
relative L_1 , finite shots	1.4×10^{-14}	0.0702 ± 0.0115	0.0709 ± 0.0042
ratio	1.00	1.50 ± 0.25	1.96 ± 0.12

with T and the seed count attached, and we report the infinite-shot ordering for what it is—the floor that any budget approaches from above, at the price Table IV now states.

S7. PHYSICAL VALIDATION: SPIN SCALING, POLE CENSUS, SUM RULE AND SIZE

The charge scale, in full. $\Delta(L)$ decreases monotonically and stays above the Lieb–Wu thermodynamic value $4.67952t$ at every size, the residual falling from $0.679t$ at $L = 6$ to $0.289t$ at $L = 12$ —that is, to 6.2% of the thermodynamic value (Table V). $\Delta(L = 12) = 4.96876t$ reproduces the $4.97t$ this work has quoted throughout, and is the same number that appears as the Hubbard-band separation of Fig. 4 and as the low-frequency scale of Fig. 7. We fit no extrapolation exponent: a straight line in $1/L$ through the four points extrapolates to $4.561 \pm 0.044t$, which undershoots Lieb–Wu by $0.12t$ and is therefore evidence that the functional form is not $1/L$, not evidence about the limit. The statement supported is that the charge scale converges to a finite thermodynamic value and is already within 6.2% of it at the largest size reached here.

Spin. The lowest pole of $S^{zz}(q = \pi, \omega)$ gives $L\Delta_s = 2.091, 1.917, 1.996, 1.996$: a $1/L$ closing with a coefficient of about 2 (mean 2.000, range 1.917–2.091). Two fits make that precise and bound it. Fitting $\Delta_s = c_0 + c_1/L$ over these four sizes gives an intercept $c_0 = -0.020 \pm 0.020$ —consistent with zero at one standard error—against $c_0 = 4.561 \pm 0.044$ for the charge gap: one intercept is consistent with zero and the other is not, and that separation, not either intercept on its own, is the result. It has to be read with the control that limits it, which we state here rather than leave to the figure: the pure Heisenberg ring, whose spin gap is known to vanish in the thermodynamic limit, gives a linear-in- $1/L$ intercept of $0.0142 \pm 0.0014t$ —above zero, not consistent with it—so at these sizes a vanishing intercept is not by itself evidence that a gap closes. The statement that survives without a fitted model is the flatness of $L\Delta_s$ below. Fitting instead $L\Delta_s = d_0 + d_1L$ gives $d_0 = 2.09 \pm 0.17$ with $d_1 = -0.010 \pm 0.018$, a slope consistent with zero: over $L = 6$ – 12 the product $L\Delta_s$ is flat to the resolution of four points. What is *not* supported is a third significant figure on the coefficient. The four values span 8% and are not monotone, and we quote $L\Delta_s \approx 2$ with the spread attached, not 2.00; the intercept $d_0 = 2.09$ is the fit’s value at $L = 0$, not the level over $L = 6$ – 12 .

The independent control is the effective spin model. The pure Heisenberg ring at $J = 4t^2/U = 0.5t$ —no charge degree of freedom, no double occupancy, a different Hamiltonian on a different Hilbert space, solved by dense diagonalization on a different code path—gives $\Delta_{\text{ST}}(N = 12) = 0.35585J = 0.17792t$ and reproduces the same $1/L$ law across the same four sizes, with $L\Delta_{\text{ST}} = 2.054, 2.091, 2.116, 2.135$. Two things have to be said about that agreement. First, the two numbers at $L = 12$ differ by 6.5% of the Heisenberg value, equivalently 7.0% of the Hubbard one; we name the denominator because the two differ in the third figure. Second, the residual is not monotone and changes sign—the Hubbard gap is 1.8% *above* the Heisenberg one at $L = 6$ and 8.3%, 5.7% and 6.5% *below* it at $L = 8, 10, 12$ —so it cannot be attributed to a smooth $O(t^4/U^3)$ correction to the effective exchange, which would not change sign. A correction of that order is the expected *scale* of the difference at $U/t = 8$, and it is the right scale; its size dependence is not resolved here and we claim nothing about it. The control confirms the $1/L$ scaling and the magnitude of the coefficient to within 10%; it does not pin the coefficient, and its own $L\Delta_{\text{ST}}$ drifts upward by 0.0134 ± 0.0014 per site over the same range.

What establishes the physics instead is measurement rather than illustration: the four-size scaling of Table V, plotted in Fig. 6, in which the spin scale is a converged pole position at every size and closes as $\approx 2/L$ with an independent effective-Heisenberg control, and the pole census of Table S7, which states what is actually inside the window at each

TABLE S7. **Pole census, across three sizes.** Poles carrying more than 1% of the channel maximum, counted per momentum; each entry is the range over the momenta plotted (7 values of k for $A(k, \omega)$, 6 values of q for the collective channels). The $L = 6$ and $L = 8$ columns are dense diagonalizations, where the count needs no recursion, on the subset of momenta available at those sizes; the $L = 12$ column is the reorthogonalized recursion at $n_\ell = 120$, which reproduces the dense counts pole by pole at the two smaller sizes. “Leading weight” is the fraction carried by the single largest pole at $L = 12$. The addition branch of $A(k, \omega)$ and the charge channel are at $\eta = 0.18t$, the spin channel at $\eta = 0.10t$. The last column is the smallest separation between retained poles in units of the full width at half maximum, which is why the figures show bands: at the tightest momenta the lines are genuinely merged, and the census says how many of them there are.

channel	poles above 1%			dim ratio	leading	$\delta\omega_{\min}/$
	$L=6$	$L=8$	$L=12$		weight	FWHM
$A(k, \omega)$, addition	2–3	2–4	3–7	2440	0.32–0.69	0.12–2.04
$S(q, \omega)$, charge	3–4	3–6	6–9	2440	0.38–0.68	0.24–2.20
$S^{zz}(q, \omega)$, spin	1–2	1–3	1–3	2134	0.80–0.99	1.08–2.37

size instead of naming it.

S7.1. A pole census

At $L = 12$ and the broadenings used here, none of the three momentum-resolved channels is a dense spectrum inside the plotted window; each momentum carries a small number of discrete poles that the broadening merges into a band. Counting poles that carry more than 1% of the channel maximum—Lanczos with full reorthogonalization at $n_\ell = 120$, checked for depth stability at $n_\ell = 60, 120$ and 240 at $q = \pi$ in every channel, and validated pole by pole against dense diagonalization at $L = 6$ and $L = 8$ —gives Table S7; the recursion is the Lanczos one of Refs. [20, 52].

The census is the deposited `pole_structure.json` and is taken on the momentum grid the figures of this section plot, seven values of k and six of q ; a count over a coarser grid is a count of different momenta and is not comparable entry by entry. The census covers the three channels the figures of this section plot; the k -integrated $A(\omega)$, which is a sum over the same branch, and the removal branch are not counted, and no statement is made about them here.

The count does grow with size, and the size of the growth is the point. Between $L = 6$ and $L = 12$ the $(N+1)$ sector grows by a factor 2440, while the largest per-momentum pole count grows from 3 to 7 in the addition branch, from 4 to 9 in the charge channel and from 2 to 3 in the spin channel. A line count that roughly doubles while the Hilbert space grows by three orders of magnitude is a discrete spectrum whose resolvable lines are being broadened together, not a dense one forming. The spin channel is the extreme case: at every momentum a single pole carries between 80% and 99% of the weight, at separations of at least 1.08 full widths, and at $q = \pi$ that single pole is the $0.16635t$ scale of Table V. What Fig. 8 shows at $L = 12$ is one dominant antiferromagnetic mode plus at most two satellites.

This has a consequence for how the reference itself must be described: $A(k, \omega)$ is not computed exactly, as the Lehmann sum would be. With $n_\ell = 150$ and no reorthogonalization in a sector of dimension 731 808, what is computed is a 150-node Gauss quadrature of the spectral measure [20, 52]. It is accurate enough here—and it is accurate *precisely because* the measure it integrates is a handful of poles; a dense spectrum at this dimension would not be reproduced by 150 nodes. It is not, however, converged: the same depth moves the reference by 0.93% at $\eta = 0.15t$ and by 3.0% at the $\eta = 0.10t$ of Fig. 8 when it is taken from $n_\ell = 100$ to $n_\ell = 200$ (Sec. VII). The reference is therefore labelled for what it is: a finite-depth continued fraction, checked against dense diagonalization at $L \leq 8$ and quoted with its depth. The pole positions of Table V are a separate and stronger object—they are computed at full reorthogonalization and are stable across $n_\ell = 60, 120, 240$ —which is why the physical claim of Sec. VIII A rests on them and not on a lineshape. The census counts poles, and the count does not depend on the broadening either; only the picture does.

S7.2. The sum rule, as measured

The analytic identity $\int A^+ d\omega = 1 - \langle \hat{n}_p \rangle = 0.5$ is exact and grid-independent, but it is an identity of the *ground state*: evaluating it costs one ground-state expectation value, $\langle \Psi_0 | \hat{c}_p \hat{c}_p^\dagger | \Psi_0 \rangle$, and never touches the reconstruction. It is therefore not an a posteriori check of A_S , and the caption of Fig. 2 does not present it as one.

Integrating the plotted curves instead, over the 600-point window at $\eta = 0.15t$:

$$\int_{\text{window}} A_{\text{exact}} d\omega = 0.486588, \quad \int_{\text{window}} A_S d\omega = 0.486471. \quad (\text{S1})$$

Both fall 2.7% below the analytic 0.5 because Lorentzians of width η leak outside any finite window: the analytic in-window mass is 0.486589, the tail outside it is 0.013411, and the residue of the trapezoidal quadrature itself is 3×10^{-6} relative—so the deficit is entirely the window and not the reconstruction. The a posteriori statement the reconstruction does support is the comparison of the two integrated curves, which differ by 1.16×10^{-4} absolute, that is 2.4×10^{-4} relative. That is the number we report, and it is consistent with the $\sim 2\%$ window effect that Fig. 4 states.

S7.3. Size: the convention, and what it does not license

The system size is not hidden and is not the weak point. The exact-diagonalization convention for this observable is $L = 12$ –16 at $\eta \approx 0.1$ – $0.2t$: Xu, Lu, Tohyama and Shao [100] compute this spectral function by Lanczos at $L = 12$ and $\eta = 0.2t$ at half filling, and at $L = 16$ at quarter filling. Their model is the extended Peierls–Hubbard chain, and their $L = 16$ run is a dimerized lattice away from half filling; it is their half-filled $L = 12$, and the plain-Hubbard $L = 12, 8, 6$ of their Appendix A, that sit alongside ours. Our $L = 12$ at $\eta = 0.10$ – $0.18t$ is that convention, not an outlier within it.

What the convention does not license is the vocabulary of the thermodynamic limit, and the criterion that says so is in the dynamical-DMRG literature this paper already cites. Jeckelmann’s rule is explicit [80]: “ $\eta_0(N)$ must be larger than the distance $\delta\omega$ between two consecutive excited states.” At $L = 12$ the largest spacing between consecutive excited-state energies in the window is $\approx 0.48t$, a factor 2.7 above the $\eta = 0.18t$ used for $A(k, \omega)$ and 4.8 above the $\eta = 0.10t$ used for the spin channel. The rule is not satisfied here. Satisfying it would demand $\eta > 0.48t$. That is only a tenth of the Mott scale, so the Hubbard bands would survive; but it is $0.96J$ and 2.9 times the $L = 12$ spin gap itself, so the entire low-frequency structure of Fig. 8 would merge into one feature. The number the rule demands is set by the spacing, $0.48t$; we take it from the spacing and not from a rounded-up interval. The correct response is not a larger η : it is to stop using words that require the rule to hold. The dynamical-DMRG standard fixes the product ηL and lets η fall with the size: Benthien and Jeckelmann use $\eta = 5t/L$ in one place, the $L = 100$ spin calculation, and declare $\eta L = 12t$ for the finite-size scaling that carries the zone-boundary charge peak to $L = 120$ [3]; Jeckelmann’s own prescription is $\eta(N) = c/N$ with $c = 6.4$ and $12.8t$, on chains of up to $N = 256$ [80]. This work is at $L = 12$ with $\eta L = 1.2$ – $2.2t$, a factor of two to ten below every constant in that range, and it is the words, not the size, that we do not use. Section IX C records the same restriction: the spin-charge statement rests on pole positions at four sizes.

S8. HARDWARE AND SCOPE: FULL ACCOUNT

Three claims one could make for this method are larger than the quantity that fixes them: the hardware claim is fixed by a subspace size, the chemistry claim by a set of sector dimensions, and the advantage claim by the same bond dimension this paper puts forward as its own cost tracker. This section states each with its fixing quantity attached, and derives the region in which this method has no advantage over a classical one.

The two hardware runs. Both device runs are reported in full and they are the entirety of the quantum hardware used in this work. The first was executed on the IBM Heron processor `ibm_fez` (job `d9s16avpemt73ct6g8g`): the $L = 6$ periodic Hubbard ring at $U/t = 4$, $\eta = 0.15t$, 12 qubits, 3.5×10^5 computational-basis shots— 5×10^4 on each of seven circuits $k = 0, \dots, 6$, of which $k = 0$ is the $t = 0$ reference—pooled into one subspace, under measurement twirling, Pauli twirling and dynamical decoupling, with no readout-error mitigation applied to the counts and no zero-noise extrapolation (Fig. 9(a) is the result; the circuits prepared the computational-basis $(N+1)$ determinant with orbitals $0\text{--}3\uparrow$ and $0\text{--}2\downarrow$ occupied rather than $\hat{c}_p^\dagger|\Psi_0\rangle$, and their hopping layer applies $R_{xx}R_{yy}$ rotations without Jordan–Wigner strings, so they are not the primitive of Fig. 1; because the retained subspace is the whole sector, neither affects the reported A_{hw}). No configuration recovery was applied. The notebook post-selected on particle number and S_z after reading each bitstring in reversed qubit order (`int(bs[::-1], 2)`), which maps qubit q to $11 - q$: site i to $5 - i$ and spin $\uparrow$ to $\downarrow$. The circuits conserve both quantities and start from four $\uparrow$ and three $\downarrow$ electrons, so under that reading their own output falls in the $S_z = -\frac{1}{2}$ sector and is discarded: in a noiseless statevector simulation of the same seven circuits, 0 of 3.5×10^5 shots are retained, against all 3.5×10^5 and the full sector of 300 under the correct reading (`src/hw_bitorder_check.py`, `data/hw_bitorder_check.json`). The $|\mathcal{S}| = 300$ determinants retained from the device therefore all came from device errors that changed the spin balance; the device counts needed to redo the post-selection are not part of the deposit (Sec. IX D). The recorded relative L_1 is 0.0: the reconstructed A_{hw} agrees with the exact A to 7.9×10^{-15} at every frequency, not byte for byte, because the retained subspace spans the whole sector.

That number is exact *by coverage*, and the distinction is not rhetorical: at $|\mathcal{S}| = \dim$, Rayleigh–Ritz on $\text{span}(\mathcal{S})$ is diagonalization of the full sector, $P_{\mathcal{S}} H P_{\mathcal{S}} = H$, and a vanishing error is an identity rather than a measurement of device fidelity. The run establishes only that the steps of the pipeline execute end to end on a processor—transpiled circuits, device counts, post-selection to a particle-number and S_z sector, and the classical Lehmann assembly of Sec. II.

The post-selection itself was wrong: it read every bitstring in reversed bit order, so every determinant it kept came from a device error, and the run is not evidence that the device sampled the intended distribution. It establishes nothing about accuracy, neither at the complete coverage it reached, where agreement is an identity, nor at incomplete coverage, the regime the rest of this paper is about. The runs are therefore described as *executed on* the named processors, not as demonstrating the method on them, and the coverage qualification applies wherever they are mentioned.

The second run, and a comparison the record cannot decide. Stretched N_2 ($R = 2.0$ Å, CAS(10e,12o)/cc-pVDZ, 24 qubits) was executed on `ibm_marrakesh` (job `da125f2ein7c73bcsqs0`, 10^4 shots, CCSD-seeded local unitary cluster-Jastrow (LUCJ) circuit). Averaged over 8 recovery seeds on the cached device counts it reaches 0.59 ± 0.14 mHa of the exact active-space value (best seed 0.40 mHa), while the noiseless statevector simulation of the same circuit stops at 29.5 mHa (Fig. 9(b)). That contrast invites the reading that device noise *helps*, and the record cannot support it, because the mechanism named for it is a statement about a quantity the record does not contain: recovery is said to turn corrupted bitstrings into valid configurations and thereby *enlarge* the variational subspace—a valid-shot economy, not a fidelity gain—so the comparison is decided by the two recovered subspace sizes, and neither is stored. The same gap runs through the controlled sweeps, whose scripts save at most the size of the naively post-selected subspace and not of the recovered one (the spectral script of Fig. S7 saves neither), so the places where an error *below* the noiseless value is reported beyond the seed bands— NH_3 from 2.53 to 1.24 mHa at $\varepsilon = 10\%$, CO from 1.73 to 0.63 mHa at 20% and C_2 from 0.184 to 0.026 mHa at 30% in Fig. S8(a), and the spectral relative L_1 of Fig. S7, from 0.105 at $\varepsilon = 0$ to 0.045 at 13%—rest on the same unrecorded variable, at the price of one line in each script, both of which already build the subspace.

The mechanism itself is reported elsewhere. On the dissociation of N_2 in a large active space, with the same LUCJ ansatz used for the run above, Vaquero-Sabater, Carreras, Broers, Shirakawa, Yunoki and Casanova report that “sampling noise can enhance Hilbert-space exploration by generating additional configurations beyond those supported by the ideal ansatz,” and that “when combined with configuration recovery, this leads to systematically improved energies” [86]: the same molecule, the same ansatz and the same mechanism, and there it is carried by a controlled comparison under a simple noise model that the work introduces. The same work also reports that “recovery alone (starting from randomly generated configurations) can efficiently construct accurate CI spaces” [86]. Our record contains no control of that kind (uniformly random bitstrings passed through the same recovery with the same eight seeds), so it cannot show that the device counts contributed anything to the 0.59 mHa beyond what recovery supplies on its own. What this work cannot claim is its own measurement of the effect, not the effect: our two arms differ in the one quantity neither of them recorded, so they cannot be evidence either way, whatever the literature independently establishes.

What *is* controlled is the comparison in which both arms consume the same corrupted bitstrings at the same rate with the same seeds and differ only in the recovery step, and there the gain from recovery is measured and grows with the noise: at $\varepsilon = 2\%$ eighteen of nineteen molecules reach chemical accuracy with recovery against seventeen without it, at $\varepsilon = 30\%$ recovery roughly halves the error on N_2 ($31.9 \rightarrow 13.2$ mHa, Fig. S8), and on the periodic Hubbard ring it keeps the spectral relative L_1 near 0.05 for $4\% \leq \varepsilon \leq 16\%$ while naive post-selection on the same bitstrings degrades steadily (Fig. S7)—seed-averaged, with ± 1 s.d. bands, at matched noise. That is the noise result of this work.

Scale. These runs are 12 and 24 qubits, and the same class of observable has been executed elsewhere at larger scale—retarded spin Green’s functions of the one-dimensional Fermi–Hubbard model on IBM Heron r2 at up to 30 qubits [11], an ARPES-type $A(k, \omega)$ of a free-fermion 27-site chain at 54 qubits [12]—and, at a comparable 20 sites on trapped ions, dynamical structure factors against scattering data [15].

In those works the device data determine the reported spectrum, as they do in the four-qubit trapped-ion Hubbard-dimer reconstruction of Ref. [32]; in our spectral run they do not: the retained subspace is the whole 300 -dimensional $(N+1)$ sector and every retained determinant came from a device error, so the spectral function it returns is the exact-diagonalization one, the run is not an entry in that class, and it is not offered as a scale result. The run is also at a size where the coverage question is degenerate, and Sec. VII says how degenerate in shots: the 3.5×10^5 spent—seven circuits at 5×10^4 shots each, pooled into one subspace—are 14.4 times the $(2.43 \pm 0.34) \times 10^4$ that Table IV measures for one orbital seed and one branch at $L = 6$, so complete coverage of a 300-dimensional sector is what the budget predicts for a correctly read run, as the noiseless simulation above confirms; the coverage this run recorded came from device errors and does not test that prediction.² The first size at which a device run would test the reconstruction rather than only the execution of the pipeline is $L = 8$: sector 3920, at $(2.70 \pm 0.13) \times 10^5$ shots per orbital seed and branch—a factor 1.3 *below* what was already spent here, so what confined this run to $L = 6$ was not the shot budget. The momentum-resolved observable at that size is a different matter: sixteen channels, 8.3×10^7 shots as measured for Fig. 3 to reach its relative L_1 of 2.16×10^{-3} , over two orders of magnitude above this job. That is not its cost at the 0.05 threshold used above, which was not measured: half that budget already gives 6.16×10^{-3} .

² Table IV replicates the repository protocol, $K = 18$ with 19 snapshots, at $U/t = 8$ rather than the $U/t = 4$ of the device run; this is a consistency statement between two budgets, not an identity.

The region of no advantage, in full. The resource statement of Sec. VI has a consequence for the standing of this method. What tracks the determinant support is χ , the minimal bond dimension across a bipartition in the working basis (Sec. S5.3). But χ is, by definition, the cost parameter of a matrix-product representation of the same state. A cost claim built on a basis-dependent boundary quantity therefore cannot select for the quantum side of the comparison: wherever our own diagnostic says the determinant support is small it says a matrix-product state is small too, and both methods are charged the same number. The thesis of Sec. VI implies its own region of no advantage, and implies it more sharply than an external objection would.

The instance is in the literature and it is not marginal. Ouyang, Chi and Chan reproduce the 7 260 trajectories of the Fermi–Hubbard dynamics experiment of Hartnett *et al.*—120 qubits, up to 90 Trotter steps [87]—by tensor contraction at bond dimension 32, faster and to higher accuracy than the device run [88]: the same pattern in which tensor-network and Pauli-propagation methods have overtaken earlier hardware demonstrations, the first “quantum utility” claim [89] among them [90, 91], with Pauli propagation since extended to real-time two-point correlators [92], and in which unitary cluster-Jastrow circuits are themselves now simulated classically: the energy of any single-layer circuit in polynomial time, reproducing the largest (77-qubit) experiment of Ref. [35] on a laptop [93], and, to additive approximation, the Born-rule probabilities of hardware-sampled bitstrings by an extended matchgate simulator, at a cost exponential only in the magnitudes of the controlled-phase angles [94]—against which the standing worst-case hardness result for that circuit family, its embedding of instantaneous-quantum-polynomial sampling [123], hard in the worst case [124] and, under average-case conjectures, to additive error [125], is not a contradiction but a statement about a different regime. Two architecture-independent results point the same way: polynomial-time classical estimation of expectation values in noisy circuits [126], and classical estimation in noiseless chaotic circuits across geometries [127]. On the ground-state energy the strong classical selectors [128–130] already win on cost [95, 96], even optimally ordered SQD subspaces need a number of configurations that grows exponentially with the size of Heisenberg and Hubbard lattices [97], and the case for a generic exponential advantage in ground-state chemistry was found wanting before that [131].

One framing does not survive. It is tempting to argue that the energy is where classical methods have caught up and to move the target to dynamics; in one dimension that inverts, because dynamical DMRG computed all three of the momentum-resolved channels reported here nineteen years ago [3] (Sec. I, Sec. VIII). Dynamics in one dimension is where classical methods arrived first, and no framing of this work says otherwise. Within the region this work measures—one-dimensional Hubbard rings at $L \leq 14$, the nineteen active spaces, and a largest minimal bond dimension of 42 over the 74 deposited exact ground states, attained on a molecular active space, the largest on the lattice being 38 in the two-leg ladder—there is no computation performed here that a classical method cannot perform, and we make no advantage claim at any size reached. Deriving one’s own no-advantage region is a stronger position than conceding it; the credit does not change the fact, and the fact is the one just stated.

The same derivation says where the boomerang stops, and that is the measurement this subsection closes on. The controlled contrast of Sec. S5.3 is a search for a target on which the two costs come apart: at one-body magic matched to within 1%, the two-leg ladder carries $\chi = 13$ against the chain’s 5 at $L = 6$ and $\chi = 38$ against 8 at $L = 8$, while its determinant support exceeds the chain’s by 1.52, 1.68 and 2.24 $\times$. The quantity to look for is a target whose minimal bond dimension is large in *every* ordering while the sampled support stays compact—a property of geometry and orbital connectivity rather than of dimensionality, so two-dimensional lattices and molecular orbital graphs are where it would be measured, and nothing here measures it. Section V supplies the second coordinate of that search: in the determinant basis the Krylov space of the probe already has coordinate support on every symmetry-allowed determinant at $L = 8$ (3902 of 3920), so zero leakage at full captured weight requires a different working basis; whether a non-vacuous certificate at small sector fraction does is open (Sec. VE).

S9. COUNTEREXAMPLES TO THE WEIGHT-ONLY BOUND, AND THE POOLED SWEEP

The three-level family. Section IIID exhibits a two-level instance forcing the constant of the weight-only bound (8) to its trivial value at $w_S = 1$. With $w_S < 1$ strictly, the weight term alone fails without bound. Take $H = \text{diag}(\begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}, \delta)$, $\phi = e_0 + \varepsilon e_2$ and $S = \{0\}$, so that $1 - w_S = \varepsilon^2/(1 + \varepsilon^2)$ may be made as small as wished while the weight displaced to $\pm g$ stays $O(1)$. The ratio of the measured error to the weight term then diverges as ε^{-2} at fixed g/η : the error stays of order $\|\phi\|^2$ while the weight term is $2\|\phi\|^2(1 - w_S) = 2\varepsilon^2$, and δ , the energy of the discarded component, does not enter. Any $C > 0$ in the second term caps the ratio.

The pooled sweep behind Fig. S5. Five machine-readable scans, written by two independent implementations, contribute 621 subspaces: 243 from the original sweep, and 162 + 112 + 104 from a re-implementation deposited as `cert_stress.json`, `cert_akw.json` and `cert_teqsci.json`. Fifteen reproduce the target to double-precision round-off (relative $L_1 \leq 10^{-10}$) and define no ratio, leaving 606 live subspaces over five Hamiltonian families — Hubbard chains, XXZ, t - V , Anderson and sparse random matrices — and eight construction rules, of which 156 are finite-shot

realizations at 4000 shots per snapshot and the remainder deterministic weight-ordered or structured selections. On those 606: the weight-only bound fails on 469; among the 363 with $w_S \geq 0.99$ — the regime a sampling experiment aims for — it fails on all 363, the mildest by a factor 2.00, with 50 of them carrying $w_S = 1$ exactly, so that the bound is identically zero and the ratio infinite; below $w_S = 0.99$ it survives on 137 of 243, in every case where the relative error is already of order unity and the bound excludes nothing. Theorem 1 is violated by none of the 606; the largest ratio of error to bound is 0.9999999997, attained on four subspaces with $w_S = 0$ exactly, where $A_S \equiv 0$ and the trivial branch is an identity rather than an inequality. The leakage branch is the smaller of the two on only 61 of the 606: the sweep shows that the bound *holds*, not that it is tight.

Independent re-verification. The 378 rows of the three deposited scans were re-audited by a script sharing no code with the one that produced them. It confirms 378 of 378 for each side of Eq. (7); a largest error-to-bound ratio of 0.9999999997 over the 376 rows that define one; the exclusion lemma (9) on 378 of 378, with 357 rows at $K_S = -1$ and 21 at $w_S = 1$ exactly; and violations of the weight-only bound on 219 of the 219 finite-ratio rows with $w_S \geq 0.99$, the mildest by a factor 2.00, on the same subspace (XXZ, $\Delta = 2$, $\eta = 0.9$, fraction 0.75) that is the mildest violation of the pooled sweep: every row with $w_S \geq 0.99$ lies at least a factor two from compliance.

S10. MOMENT EXACTNESS: STATEMENT, PROOF, SHARPNESS, AND THE SHOT COUNT

We make precise the two things a Krylov-adequate subspace does deliver. Fix the seed $|\phi\rangle = \hat{c}_p^\dagger|0\rangle$ (the electron-addition branch; removal is identical), with weight $\|\phi\|^2 = \langle\phi|\phi\rangle$, and let $A(\omega) = \sum_n |\langle n|\phi\rangle|^2 \mathcal{L}_\eta(\omega - (E_n - E_0))$ be the exact spectral function, where $\{|n\rangle, E_n\}$ diagonalize H in the $(N+1)$ -particle sector and $\mathcal{L}_\eta(x) = \frac{\eta/\pi}{x^2 + \eta^2}$. For a configuration set $\mathcal{S}$ let $P = P_S$ be the orthogonal projector onto $\text{span}(\mathcal{S})$, let $H_S = PHP$ with eigenpairs $\{|\tilde{m}\rangle, \tilde{E}_m\}$, and let $A_S(\omega) = \sum_m |\langle \tilde{m}|\phi\rangle|^2 \mathcal{L}_\eta(\omega - (\tilde{E}_m - E_0))$ be the reconstruction. Write $\mathcal{K}_K = \text{span}\{\phi, H\phi, \dots, H^K\phi\}$ and $w_S = \|P\phi\|^2/\|\phi\|^2$. Moments are taken on the point measures $d\mu = \sum_n |\langle n|\phi\rangle|^2 \delta(\omega - (E_n - E_0))$ and $d\mu_S = \sum_m |\langle \tilde{m}|\phi\rangle|^2 \delta(\omega - (\tilde{E}_m - E_0))$, not on the broadened A , whose moments of order ≥ 2 diverge.

S10.1. What the proof uses

Theorem 2 and its proof are stated in Sec. IV A. The proof uses $P = P^\dagger = P^2$, $H = H^\dagger$ and $\mathcal{K}_K \subseteq \text{span}(\mathcal{S})$, and nothing else. In particular it does not use that $\mathcal{S}$ is a coordinate subspace, that it was obtained by drawing bitstrings, or that any Lanczos basis is well conditioned [28]; it holds verbatim for the orthogonal projector onto a Krylov space, which is the form evaluated in Sec. V. The single asymmetry in Eq. (12)— a runs to K while b runs to $K+1$ —is what produces the odd reach $2K+1$ rather than $2K$. Two structural remarks follow at no cost.

(a) *The node count.* $d\mu_S$ has $|\mathcal{S}|$ atoms, one per Ritz pair, not $K+1$. Equation (11) is therefore a weak quadrature statement for a rule of that size: a Gauss rule of n nodes for $d\mu$ is exact through order $2n-1$ [52], so the $K=1$ row of Table II, with 280 nodes, would reach order 559 if its nodes were the Gauss nodes of $d\mu$. They are not, and it reaches order 3; those 555 orders measure the distance between the two objects. This is the measurable content of the distinction: identifying A_S with an order- $(K+1)$ Gauss rule misdescribes the object and the reach such an object would have; the reach $2K+1$ of Theorem 2 itself is unaffected.

(b) *Unit captured weight is implied, not assumed.* For a coordinate subspace P is diagonal, so $P\phi = \phi \iff \text{supp}(\phi) \subseteq \mathcal{S} \iff w_S = 1$. Since $\mathcal{K}_0 = \text{span}\{\phi\}$, the hypothesis of Theorem 2 at any $K \geq 0$ already implies $w_S = 1$. Every subspace admissible for moment exactness therefore sits at the extreme of the weight budget of Sec. III, where the shot term vanishes identically and the whole residual error is leakage—which is why the two statements are complementary rather than competing.

S10.2. Sharpness, control, and the limits of the validation system

Table II is computed by exact diagonalization of the $(N+1, S_z = +\frac{1}{2})$ sector of the open $L=6$, $U/t=8$ Hubbard chain ($D=300$, $\|\phi\|^2 = 1/2$), with $\mathcal{S} = \bigcup_{a \leq K} \text{supp}(H^a\phi)$. Four measurements support it.

Reach. Through order $2K+1$ the relative discrepancy is $\leq 3.2 \times 10^{-15}$ for $K=0, 1$ and $\leq 7.5 \times 10^{-15}$ for $K=2$. At order $2K+2$ it is 9.5×10^{-4} ($K=0$) and 6.0×10^{-8} ($K=1$), eleven and seven decades above the arithmetic floor. The absolute discrepancies at that order, 3.104583×10^{-2} and 1.340167×10^{-4} , agree to six significant figures across all six implementations listed below and are properties of the construction, not of the arithmetic. The $K=2$ row carries no entry there, for a structural reason and not a numerical one: the ladder has filled the sector, $A_S = A$ identically, and their relative L_1 distance is 0 to machine arithmetic. One run of one of the six routes does report a nonzero

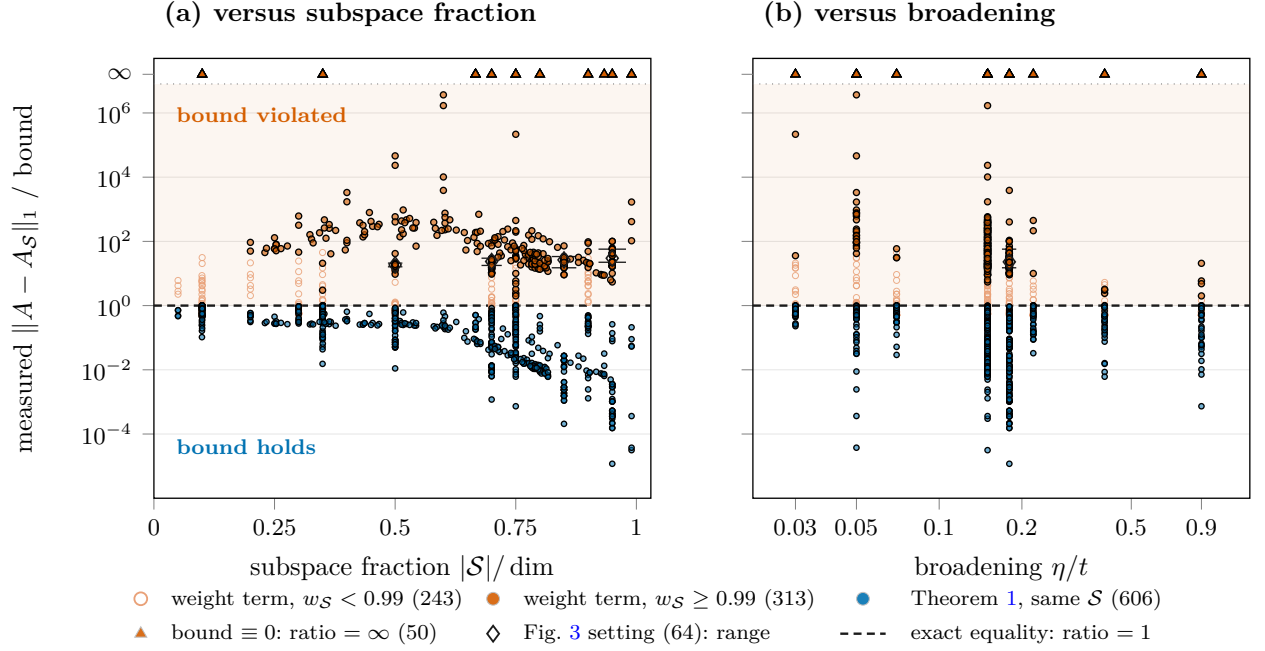


FIG. S5. **A bound indexed on the captured weight alone fails on exactly the subspaces this work relies on; Theorem 1 does not.** For each subspace $\mathcal{S}$ we plot the measured error $\|A - A_S\|_1$ divided by the bound, so every point above the dashed line at 1 is a violation. Orange: the weight term of Eq. (8), $2\|\phi\|^2(1 - w_S)$, split by captured weight (filled, $w_S \geq 0.99$; open, $w_S < 0.99$). Blue: Theorem 1 as stated, $\min\{\|\phi\|^2(1 + w_S), (\|\phi\| + \|\phi_S\|)(\|\phi_Q\| + \Lambda_S(\eta)/\eta)\}$, on the same subspaces. We pooled 621 subspaces from five machine-readable sweeps written by two independent implementations (239 + 4 from the original sweep; 162 + 112 + 104 from an independent re-implementation); 15 of them reproduce the target to within double-precision round-off (relative $L_1 \leq 10^{-10}$) and define no ratio, leaving 606 live subspaces, of which 156 are finite-shot realizations at 4000 shots per snapshot and the remainder are deterministic weight-ordered or structured selections. *The weight term alone is violated on 469 of the 606.* On 50 of them it is identically zero: those subspaces carry the whole weight to double precision ($w_S = 1$ exactly, and $K_S \in \{0, 1, 2\}$, so the remaining term is the undefined constant $C\rho^{-K_S}$); their ratio is infinite and they are drawn on the row marked ∞ , above the dotted rule. Among the 313 finite-ratio points with $w_S \geq 0.99$ —the regime the paper invokes—*every one* is a violation, the mildest by a factor 2.00; with the 50 infinite ones that is 363 of 363, and the finite factors reach 3.66×10^6 . Below $w_S = 0.99$ the bound survives on 137 of 243 points, but only where the relative error is already of order unity, so it excludes nothing there. Diamonds mark the configuration of Fig. 3 ($L = 8$, $U/t = 8$, $\eta = 0.18t$, momentum seeds $c_{k\uparrow}^\dagger|\Psi_0\rangle$, subspace fractions 0.50/0.70/0.85/0.95, both branches, eight momenta): that reconstruction reproduces the deposited `sampld_akw_L8.json` to 3.1×10^{-8} per momentum, and its violation factor is 15.1–57.3, of which 15.1–33.7 at the fraction 0.85 of Fig. 3. Those 64 rows sit at only four abscissae, so they are drawn as a min–median–max whisker at each one rather than as 64 overlapping markers (medians 18.5, 23.6, 26.5, 29.9 at fractions 0.50, 0.70, 0.85, 0.95); each of them is also plotted individually inside the filled orange series, which is where their scatter can be read. Theorem 1 is violated on none of the 606; its ratios span 1.18×10^{-5} to 0.9999999997. That maximum is reached on four subspaces with $w_S = 0$ exactly, where $A_S \equiv 0$ and $\|A - A_S\|_1 = \|\phi\|^2$ equals the trivial cap by algebra—the branch that carries no information—and the leakage branch $\Lambda_S(\eta)/\eta$ is the smaller of the two on only 61 of the 606 rows: the figure shows that Theorem 1 *holds*, not that it is tight. In (b) the η axis is not a sweep within one model: the Hubbard-chain rows exist only at $\eta/t = 0.05, 0.15, 0.18$ and the stress families (XXZ, t -V, Anderson, sparse random) only at 0.03, 0.07, 0.22, 0.40, 0.90, so no trend across the full axis may be read from it.

residue in that row; the same run reports a cyclic-subspace dimension of 301 inside a sector of dimension 300 and emits floating-point overflow while forming the high powers. It is discarded, and the corrected run is the one quoted.

Reproducibility of the residues. The relative discrepancies are quoted, and the absolute ones below order $2K+2$ are not, for a reason worth recording. The exact moments $\mu_j = \langle \phi | (H - E_0)^j | \phi \rangle$ run from 5.0×10^{-1} at $j = 0$ to 1.6×10^5 at $j = 6$, so an absolute residue at the top of the ladder is a roundoff quantity whose size depends on the route taken to the same identity. Across six equivalent routes—Rayleigh–Ritz on the shifted or on the unshifted matrix, crossed with exact moments obtained from the Lehmann sum, from the shifted matrix–vector recursion, or by binomial recombination of unshifted powers—the relative residue below order $2K+2$ stays inside $[1.1, 1.8] \times 10^{-15}$ at $K = 0$ and $[1.0, 3.2] \times 10^{-15}$ at $K = 1$, a spread under a factor 3.3, while the absolute residue at the same orders is not stable to two significant figures. We therefore quote relative residues below $2K+2$, and absolute ones only at $2K+2$, where the discrepancy is physical and route independent.

Negative control. Over 300 uniformly drawn coordinate subspaces of the same size $|\mathcal{S}| = 200$, the zeroth moment is wrong by between 1.86×10^{-2} and 4.18×10^{-1} (median 1.67×10^{-1}), i.e. by 3.7% to 84% of $\|\phi\|^2$ (median 33%), with every one of the 300 draws failing; the Krylov-support subspace of identical size is exact to 4.4×10^{-16} . At $|\mathcal{S}| = 280$ the random draws fail by 4.8×10^{-5} to 2.6×10^{-1} (median 1.12×10^{-2}), again without exception, against 5.0×10^{-16} . The discriminating variable is the identity of the determinants, not their number.

Scale. The Krylov ladder at $L = 6$ occupies 0.667, 0.933 and 1.000 of the sector at $K = 0, 1, 2$, so it exhausts the sector at $K = 2$ and the table has exactly three rows. This system can verify Eq. (11) and locate its exact reach; it cannot exhibit the behavior of $\|A - A_{\mathcal{K}_K}\|_1$ at large K , which is the regime Conjecture 1 concerns, and we make no claim there.

Why the open chain. The ladder terminates when the coordinate subspace contains the cyclic subspace generated by ϕ —not when it fills the sector—and at that point the reconstruction is exact, because $\text{span}(\mathcal{S})$ then carries every eigenvector of nonzero Lehmann weight. On the periodic rings of Sec. VII that happens early: at $L = 6$ the ladder runs $200 \rightarrow 296$ and stalls at 296 of 300 determinants, where $A_{\mathcal{S}} = A$ to 1.0×10^{-14} in relative L_1 and the moments are exact past order $2K+1$ at every $K \geq 1$. The ring therefore offers a single row in which the reach of Eq. (11) is visible at all ($K = 0$: relative discrepancy 4.4×10^{-16} through order 1, 4.6×10^{-3} at order 2); the open chain offers two, and is the system on which the method validation of Sec. II is performed. Within the reach they cover, both agree: the identity is exact, sharp at the next order, and decidable from the subspace and the probe alone.

S10.3. The geometric-convergence claim, and why it is left open

Conjecture 1 is easily taken for a consequence of Eq. (11). It is not one: moment exactness gives $2K+2$ linear identities, and $\|A - A_{\mathcal{K}_K}\|_1$ is not a finite linear functional of them. A proof would require (i) a polynomial approximation theorem for $x \mapsto \mathcal{L}_\eta(\omega - x)$, uniform in ω , on a real set containing $\text{spec}(H)$ and $\text{spec}(H_{\mathcal{K}_K})$, and (ii) a tail estimate controlling the complement. For (i) the natural route is a Bernstein-ellipse bound, in which the resolution enters through the poles of the kernel at $x = \omega \pm i\eta$: the largest admissible ellipse has semi-minor axis below η , so its parameter—and with it any geometric rate—approaches 1 as $\eta \rightarrow 0^+$. That is a mechanism and not a derivation: it exhibits neither $C(\eta)$ nor $\rho(\eta)$, and neither appears anywhere in this work or in the deposit.

The prefactor is moreover pinned from below by the two-level family of Sec. IIID. With $H = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}$ and $\phi = e_0$ one has $\langle \phi | H | \phi \rangle = 0$, so $\mathcal{K}_0 = \text{span}\{\phi\}$ carries a single pole at the origin while A carries two at $\pm g$; by adaptive integration over the whole line at $\eta = 0.15$ (`src/moments_table.py`), $\|A - A_{\mathcal{K}_0}\|_1 = 1.5197, 1.8352, 1.9504, 1.9835, 1.9950, 1.9983, 1.9995$ for $g = 1, 3, 10, 30, 100, 300, 1000$ at $\|\phi\|^2 = 1$, approaching the trivial ceiling $2\|\phi\|^2 = 2$ monotonically. Any $C(\eta)$ valid uniformly on this family therefore satisfies $C(\eta) \geq 2\|\phi\|^2$. A geometric bound with a trivial prefactor is not worthless—it would still decay in K —but it cannot be non-vacuous at small K , and small K is where a shot-limited subspace lives.

Conjecture 1 therefore remains open, and we name its two missing ingredients explicitly so that it can be attacked rather than left unremarked. What this work offers in its place is not a rate in K at all but a bound on $\|A - A_{\mathcal{S}}\|_1$ itself, at the η of the figure and on the subspace actually in hand (Theorem 1), measured subspace by subspace, together with a measurement of where that bound is non-vacuous (Sec. V). That is a weaker statement than a rate and a stronger one than a conjecture, and it is the one the deposit can reproduce.

S10.4. The sample-complexity bound

Proposition 1 is stated in Sec. IV C; this is its proof.

Proof of Proposition 1. Let $\mathcal{N}_\varepsilon = \{x : p(x) \geq \varepsilon\}$. Since $\sum_x p(x) = 1$ and each member contributes at least ε , $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$. For fixed $x \in \mathcal{N}_\varepsilon$, $\Pr[x \notin \mathcal{S}] = (1 - p(x))^T \leq (1 - \varepsilon)^T \leq e^{-\varepsilon T}$ for independent draws from p ; for $T/(K+1)$ draws from each p_k ,

$$\Pr[x \notin \mathcal{S}] = \prod_{k=0}^K (1 - p_k(x))^{T/(K+1)} \leq \exp\left(-\frac{T}{K+1} \sum_{k=0}^K p_k(x)\right) = e^{-Tp(x)} \leq e^{-\varepsilon T}$$

as well. A union bound gives $\Pr[\mathcal{N}_\varepsilon \not\subseteq \mathcal{S}] \leq \varepsilon^{-1} e^{-\varepsilon T}$, which is at most δ for the stated T . $\square$

The cardinality bound $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$ is deterministic, and it is what removes the circularity of the form $T = O(\varepsilon^{-1} \log(|\mathcal{S}|/\delta))$, in which the random size of the sampler's own output appears inside a bound on the number of draws that produce it. Where $\varepsilon^{-1} \leq |\mathcal{S}|$, the new logarithm is also the smaller one, so nothing is lost.

The scope of ε needs the same care. It thresholds a Born probability of the time-averaged mixture, and Proposition 1 makes no statement about $\|A - A_S\|_1$. Two independent obstructions stand between the two quantities. First, the weight left uncaptured is bounded only by $|\{x : p(x) < \varepsilon\}| \cdot \varepsilon$, which for a delocalized state is $O(1)$: many small probabilities can sum to most of the mixture. Second, and decisively, a subspace holding the *entire* Born support of the probe— $w_S = 1$ exactly, nothing left to capture at any ε —still has relative L_1 error 0.28 at $L = 6$ on the open chain of Table II, and 0.43 at $L = 6$ and 0.35 at $L = 8$ on the periodic rings, all at $\eta = 0.18t$ and all relative to $\|\phi\|^2$ (Sec. III); it is the $K = 0$ row of Table II. Proposition 1 is a statement about the sampler and about nothing else. The spectral error at fixed η is the subject of Theorem 1, and what the composition of the two actually costs is reported as three separate prices in Sec. VII.

S11. THE LEAKAGE CERTIFICATE: HYPOTHESES, PROTOCOL, AND THE ANATOMY OF THE THRESHOLD

Theorem 1 holds under four hypotheses. Each is a statement about the pipeline that produced the reconstruction rather than about the theorem, so each has to be checked against the pipeline. We check all four here, with numbers, for the sweep of Sec. V, and add the two protocol declarations and the decomposition that section quotes.

(H1) *Common origin.* A and A_P must be referred to the same E_0 . A shift common to both cancels exactly—verified to 1.000000 for shifts of 10^{-3} , 10^{-2} and $10^{-1}t$ —which is how the hypothesis is satisfied here, a single E_0 being used for both objects. If A_P is instead referred to $E_0 + \delta$, then

$$\|A - A_P^{(E_0+\delta)}\|_1 \leq \|A - A_P\|_1 + \|\phi_P\|^2 (4/\pi) \arctan(|\delta|/2\eta),$$

an additive term verified to 2×10^{-5} in all 48 cases tested and *which the certificate does not control*: the bound is guaranteed only once that term is added. A ground state obtained by Rayleigh–Ritz on 60% of the N -electron sector at $L = 8$ gives $\delta = 6.7 \times 10^{-4}t$, which adds 0.0024 to a bound of 1.2036—+0.20%—while multiplying the *measured* error by 16; and the dependence is not even monotone, a shift of $10^{-2}t$ lowering the measured error at $L = 6$, fraction 0.85, $\eta = 0.18t$ from 2.7×10^{-2} to 1.2×10^{-2} . The measured errors quoted beside a bound in Sec. V are therefore those of a pipeline handed the exact E_0 , and they bound the error of a pipeline that estimates it from neither side.

No cheap a posteriori control on δ is available: residual (Weinstein) and Temple bounds [58, 59] computed on ground-state subspaces of 20–80% of the N sector at $L = 6$ give arctan terms of 1.76, 1.64, 1.40 and 0.92, all vacuous against the ceiling of 2. In a shot-based pipeline E_0 is itself estimated on whatever subspace the shots populated, and the certificate has never been tested that way.

(H2) *Normalization and support of the L_1 norm.* The bound is on $\int_{\mathbb{R}} |A - A_P| d\omega$ normalized by $\|\phi\|^2 = \int_{\mathbb{R}} A d\omega$. A relative error normalized instead by $\int_W A$ over a finite window W and evaluated by finite quadrature—the convention of the repository, in which Sec. VII reports—is a different quantity, differing by the product of $\int_W |A - A_P| / \int_{\mathbb{R}} |A - A_P|$ and $\|\phi\|^2 / \int_W A$. Both factors are measured here: the second is 1.0100, 1.0309 and 1.0373 at $\eta = 0.05, 0.15$ and $0.18t$ at $L = 6$ and 1.0110, 1.0278 and 1.0335 at $L = 8$, while the first lies between 1.0040 and 1.0115 across the 204 configurations of this study. Every number in Sec. V is in the normalization of Theorem 1, and the two conventions are nowhere interchanged.

(H3) *Exact reference.* A denotes the exact spectral function of the full sector of the probe. If the reference is itself a Lanczos reconstruction of finite depth n_ℓ , the theorem does *not* bound $\|A^{(n_\ell)} - A_P^{(n_\ell)}\|_1$, and a measured error below the reference’s own drift measures the reference rather than the reconstruction. That drift runs from 2.6×10^{-5} ($L = 12$, $\eta = 0.10t$) to 2.8×10^{-7} ($\eta = 0.18t$), and at $L = 10$ the reorthogonalization gap, 1.71×10^{-5} , is of the same order as the depth drift. Every measured error in this work is therefore labeled with the depth of its reference and with that reference’s own drift, and we quote no measured error anywhere in Sec. V within a factor of 20 of its own reference drift: twelve rows at $L \geq 10$ fall under that rule and are left blank rather than rounded, the weakest surviving cell— $L = 12$, $\eta = 0.10t$ —standing at a factor 47. One gap is declared and not bounded. The reconstructions this paper publishes are produced by bare Lanczos at depth 150, while the certificate is evaluated on the reorthogonalized Krylov space; the difference between the two has been measured at a single point at $L = 8$ —relative L_1 of 1.2454×10^{-3} against 1.2464×10^{-3} —and never bounded, and at $L = 12$ it has never been measured at all.

(H4) *Exact probe.* A and A_P must be built from one probe, and it must be the one whose spectral function is wanted: $\phi = \hat{c}_p^\dagger |\Psi_0\rangle$, or its removal, density or spin counterpart, with the exact N -electron ground state Ψ_0 . That holds by construction in every certificate evaluated in this work: A and A_P are formed from the same ϕ , and on the lattices that ϕ is built from a ground state computed on the full N -electron sector—at $L = 14$ the stored checkpoint of Sec. S4, residual 2.7×10^{-9} —while the Rayleigh–Ritz ground state of the (H1) test above supplies E_0 alone and leaves ϕ exact. A pipeline whose Ψ_0 is itself estimated on a sampled subspace forms another vector ϕ ; Theorem 1 then

TABLE S8. **Angle sweep of the fiber exhibit**, $K = 3$, $\delta = 0.01$, $\eta = 0.10$, $|\mathcal{S}| = 4$. Certificate columns as in Table S4; w_S is identical at the two endpoints, and bold marks a certificate below the trivial bound $1 + w_S$.

s	w_S (both endpoints)	$\hat{\Lambda}/\eta$ al.	$\hat{\Lambda}/\eta$ rot.	cert. al.	cert. rot.
0.131	0.982963	0.328	1.610	0.913	1.983 (trivial)
0.262	0.933013	0.317	3.010	1.131	1.933 (trivial)
0.393	0.853553	0.299	4.124	1.312	1.854 (trivial)
0.524	0.750000	0.277	4.849	1.450	1.750 (trivial)
0.654	0.629410	0.252	5.139	1.543	1.629 (trivial)

bounds $\|A[\tilde{\phi}] - A_P[\tilde{\phi}]\|_1$, and nothing the pipeline computes bounds the distance from $A[\tilde{\phi}]$ to $A[\phi]$. The certificate has never been tested with an approximate probe.

Ranking, ties and support. Two properties of the cut are declared with the protocol. Only the reflection-forbidden determinants (4, 18, 48, 200 at $L = 6$ –12) carry exactly zero accumulated Born weight, and they rank last, so the cut is never filled with configurations the device would never sample in the infinite-shot limit; ties at the cut involve two determinants (four at $L = 6$, at fractions 0.80 and 0.85) and move the bound by at most 6.6×10^{-4} relative. Above the support floor $|\text{supp}(\phi)|/D$ of Sec. IIID, on the other hand, the ranking selects among determinants outside $\text{supp}(\phi)$: at fraction 0.85 that is 30.2% of S at $L = 6$, 42.0% at $L = 8$ and $\approx 39.6\%$ at $L = 10$. Those contribute nothing to w_S , and they do change H_S . The depth uncertainty quoted in Sec. V is the full recomputation of FR^* with $L = 12$ capped at $n_\ell = 300$: it moves by 0.0024, 0.0021, 0.0012 and 0.0000 at $\eta = 0.10, 0.15, 0.18$ and $0.25t$.

Reorthogonalization and the certified object. Full reorthogonalization is used at every point of the sweep, without which the projector is not a projector: the orthogonality defect is at most 4.7×10^{-15} , and $P_K \phi = \phi_S$ to 1.1×10^{-14} . At $L = 12$ the certificate is evaluated at $n_\ell = 300$ –500 on up to 658627 determinants, the dense A_S never being formed—which is why what the bound controls is A_K at the declared depth and not A_S (Sec. VA).

Anatomy of the displacement. The percentages quoted in Sec. VE are medians over 15 of the 18 (η , target) cells at $L = 8$; three cells are censored from above, the certificate never reaching the target inside the grid, and are excluded rather than imputed. Pooling $L = 6$ with $L = 8$ shifts them to 52%, 31%, 11% and 4.1%, which is why the $L = 8$ figures are the ones quoted. One caveat travels with them, and it is why they are reported as a diagnosis and not as a quantity: of the rungs of the inequality chain only the last is computable by a user from (H, S, ϕ, η) , so the decomposition is of an object the user never sees.

S12. THE GEMINAL WITNESS: PROOF AND ANGLE SWEEP

Proof of Theorem 3. Each geminal occupies its own four spin-orbitals, so (18) is a product across the K blocks and γ is block diagonal with spatial natural occupations $(n_{b_i}, n_{a_i}) = (2 \cos^2 \theta, 2 \sin^2 \theta)$; hence $N_u = 8K \cos^2 \theta \sin^2 \theta = 2K$ at $\theta = \pi/4$, and $\mathcal{F}_1 = 4K$ by Eq. (17). The Majorana covariance is likewise block diagonal, and at $\theta = \pi/4$ a single maximally-paired geminal is the two-determinant cat, whose covariance vanishes identically ($\Gamma = 0$): every order saturates its maximum, $\mathcal{F}_k = 2n_o = 4$ per geminal for all k , giving $\mathcal{F}_2 = 4K$. This equality of orders is a knife-edge of maximal pairing; away from it the orders separate ($\theta = \pi/6$ gives $\mathcal{F}_1 = 3$, $\mathcal{F}_2 = 3.75$ per pair, both still large on the same family). In the pairing basis each geminal is a two-configuration superposition, so the product carries 2^K determinants. Finally, as a product of two-orbital factors the state is a matrix-product state whose every bond crosses at most one geminal, so $\chi = 2$ in that ordering, and contraction and Born sampling then cost $O(K)$. The values of $\mathcal{F}_1$, $\mathcal{F}_2$ and $|\mathcal{S}|_{\text{pair}}$ are reproduced by exact diagonalization for $K = 1, \dots, 4$. So is the value $\chi = 2$ derived here, the largest Schmidt rank over all cuts of the pairing ordering, for $K = 1, \dots, 4$ and at every interior angle of the sweep ($\chi = 1$ at $\theta = \pi/2$); the Schmidt rank at the central spin-orbital cut alone is 2, 1, 2, 1 for $K = 1, \dots, 4$, because that cut falls between two geminals when K is even. $\square$

Table S8 continues the fiber exhibit of Table S4 into a continuous sweep of the rotation angle at $K = 3$. It stops at $s = 0.654$: at the maximal-pairing angle $s = \pi/4$ the Born distribution of ϕ is exactly flat over 2^K determinants, so the top- $|\mathcal{S}|$ rule does not select a subspace and no contrast measured there is well defined. Across every row the captured weight is identical at the two endpoints to 2.2×10^{-16} , so the entire difference sits in the leakage term, and the certificate at the rotated endpoint is the trivial bound $1 + w_S$ in every row.

S13. ONE-BODY MAGIC CANNOT BOUND THE DETERMINANT SUPPORT

Proposition 3 (Coefficient-insensitivity obstruction). *Let μ be any functional of a fixed finite set of reduced density matrices, continuous in the state. Then μ cannot usefully upper-bound the raw determinant support (the unthresholded Slater rank): for every $\epsilon > 0$ there are states with $|\mu - \mu(|D_0\rangle)| < \epsilon$ (for $\mathcal{F}_k$, $\mu < \epsilon$) whose raw support is every determinant of the particle-number sector, so any bound $|\mathcal{S}|_{\text{raw}} \leq f(\mu)$ valid on the sector takes the trivial value, the sector dimension, at values of μ arbitrarily close to $\mu(|D_0\rangle)$. The count is lower- but not upper-semicontinuous; the CP tensor rank fails upper semicontinuity in the same way and, unlike the count, is not even lower-semicontinuous: its sets of rank at most r are not closed [79] (that reference names the two semicontinuities the other way round, calling matrix rank upper-semicontinuous). Nor can the invariant $\mathcal{F}_1$ usefully lower-bound it: for N electrons on $2n_o$ modes it implies $|\mathcal{S}|_{\text{raw}} \geq N/(N - \mathcal{F}_1/4)$, which never exceeds $2n_o/N$, and at half filling ($N = n_o$), where that is 2, any increasing f with $|\mathcal{S}|_{\text{raw}} \geq f(\mathcal{F}_1)$ satisfies $f \leq 2$.*

Proof. Rank is a coefficient-insensitive count. Take $|\Psi_\epsilon\rangle = |D_0\rangle + \epsilon \sum_{k=1}^K |D_k\rangle$ with the D_k distinct determinants of the sector of $|D_0\rangle$. As $\epsilon \rightarrow 0$ every reduced density matrix converges to that of $|D_0\rangle$, so any continuous $\mu \rightarrow \mu(|D_0\rangle)$ (for $\mathcal{F}_k$, to 0), while the raw support stays $K + 1$ for every K up to the sector dimension minus one. For the second part, every natural orbital lies in the span of the at most $N|\mathcal{S}|_{\text{raw}}$ orbitals occupied in the support, so $\text{rank } \gamma \leq N|\mathcal{S}|_{\text{raw}}$; with $\text{tr } \gamma = N$ and $\text{tr } \gamma^2 = N - \mathcal{F}_1/4$, the Cauchy–Schwarz inequality $(\text{tr } \gamma)^2 \leq \text{rank } \gamma \text{tr } \gamma^2$ gives the bound, and $\text{rank } \gamma \leq 2n_o$ caps it at $2n_o/N$. At half filling with $n_o \geq 2$ the two-determinant cat $(|1 \cdots 10 \cdots 0\rangle + |0 \cdots 01 \cdots 1\rangle)/\sqrt{2}$, whose determinants differ in $N \geq 2$ orbitals, has $\gamma = \frac{1}{2}\mathbb{1}$, hence $\mathcal{F}_1 = 2n_o$ (maximal) and Slater rank 2, so $f(2n_o) \leq 2$; for $n_o = 1$ every state is a single particle on two modes, of raw support at most 2. $\square$

The obstruction is about the unthresholded rank; the operative, coefficient-sensitive support $|\mathcal{S}|_\epsilon$ of Sec. VI collapses to 1 along the same $\epsilon \rightarrow 0$ path, so continuity alone does not obstruct it. What obstructs $|\mathcal{S}|_\epsilon$ itself is the orbital-rotation symmetry—no invariant functional survives it—together with the two equal-weight witnesses below, whose amplitudes are $\Theta(1)$ and therefore threshold-robust. Two explicit witnesses rule out the natural repairs. The two-determinant cat of Prop. 3 is maximal in $\mathcal{F}_1$ yet of Slater rank 2, so at half filling no $\mathcal{F}_1$ -monotone bounds $|\mathcal{S}|$ from below beyond 2; a paired (BCS) Gaussian state has Gaussian rank 1 yet fractional occupations, so no occupation functional bounds it from above. The computable lower bounds on $|\mathcal{S}|$ that are not capped in this way are the basis-dependent orbital-Schmidt (matrix-flattening) ranks [103]—the tensor-network bond dimension—to which any Gaussian-invariant magic measure is blind. That lower bound is exact:

Proposition 4 (The bond dimension lower-bounds the support). *For any state and any bipartition of the spin-orbitals into (L, R), the Schmidt rank χ across the cut (the minimal matrix-product-state bond dimension there) satisfies*

$$\chi \leq |\mathcal{S}|, \quad (\text{S2})$$

where $|\mathcal{S}|$ is the number of fixed-basis Slater determinants of nonzero amplitude. Hence the minimal bond dimension is a rigorous, basis-dependent lower bound on the determinant support. The bound is one-sided: the geminal witness of Thm. 3 has $\chi = 2$ yet $|\mathcal{S}|_{\text{pair}} = 2^K$, so $|\mathcal{S}|$ can exceed χ exponentially already in the pairing basis, which for that family is a natural-orbital basis, while the orbital-invariant one-body magic $\mathcal{F}_1$ cannot usefully upper-bound $|\mathcal{S}|$ and at half filling cannot lower-bound it beyond 2 (Prop. 3).

Proof. Write $|\Psi\rangle = \sum_{D \in \mathcal{S}} c_D |D\rangle$ over its support. Each determinant factorizes across the cut, $|D\rangle = |D_L\rangle \otimes |D_R\rangle$, so the matricization $A_{D_L, D_R} = c_D$ has at most $|\mathcal{S}|$ nonzero entries; its rank—the Schmidt rank χ —is therefore at most $|\mathcal{S}|$. $\square$

Since a matrix-product-state simulation costs $\text{poly}(\chi)$, Prop. 4 says that *in the sampler’s own basis a tensor network is never less compact than the sampled subspace*: this is why we make no advantage claim in one dimension, where χ is small, and why any advantage must live in the higher-entanglement regime where χ itself—and with it the classical tensor-network cost—grows (the chain-to-ladder jump of Fig. S4). Empirically χ is also the operative correlate of the measured cost: across the 74 exact ground states of Fig. S3 the thresholded support $|\mathcal{S}|_\epsilon$ tracks χ (pooled Spearman $\rho = 0.72$; $\rho = 0.90$ within the $n = 38$ molecular suite, where however one-body magic scores a comparable $\rho = 0.86$ —the two resources are separated not there but by the Hubbard sweep, where $\mathcal{F}_1$ anti-orders the support in all six strata while χ orders it at a mean ρ of 0.948; the pooled coefficients for that sweep, $\rho = 0.57$ and -0.11 , are Simpson reversals of the six strata and are not the evidence here, Sec. S5.3). Two caveats belong with that number. A rank correlation is not a statement of tightness, and per the geminal witness the bound can be exponentially loose. And the proposition is proved for the raw support: once amplitudes are thresholded the inequality need not survive—here $|\mathcal{S}|$ is cut at 99% of the weight while χ is cut at $\epsilon = 0.05$, a tighter tolerance—and $|\mathcal{S}|_\epsilon$ falls below χ in 7 of the 38 molecular points. What stands is that χ is a rigorous floor on the raw support and an empirical correlate of the measured one, whereas $\mathcal{F}_1$ orders the support with a sign that is not stable from family to family (Sec. S5.3).

TABLE S9. **The six Hubbard strata.** Each is a five-point sweep in U at fixed (L, N) . The one-sided exact permutation probability of a perfect reversal within one stratum of five distinct values is $1/120 = 0.0083$; ties in χ raise it. “distinct χ ” counts how many of the five χ values differ, which is what caps $\rho(\chi, |\mathcal{S}|)$ below unity in five of the six.

L	N	$\rho(\mathcal{F}_1, \mathcal{S})$	$\rho(\chi, \mathcal{S})$	χ over $U = 1, 2, 4, 8, 16$	distinct χ
6	6	−1.000	+1.000	12, 10, 6, 5, 4	5
6	4	−1.000	+0.894	11, 10, 9, 9, 9	3
8	8	−1.000	+0.949	11, 11, 8, 8, 7	3
8	6	−1.000	+0.975	16, 16, 15, 14, 13	4
10	10	−1.000	+0.975	17, 14, 10, 6, 6	4
10	8	−1.000	+0.894	13, 13, 13, 11, 9	3

S14. THE RESOURCE STATISTICS, STRATIFIED

Section S5.3 reports the correlation evidence within the design that produced it. This appendix gives the design, the per-stratum coefficients and the three tests, so that the decision not to use the two pooled coefficients can be checked rather than taken.

The lattice design. The 30 Hubbard points are six (L, N) strata swept over the same five on-site repulsions $U = 1, 2, 4, 8, 16$; they are not 30 independent observations, and the effective number of independent sweeps is six. Table S9 gives each stratum. The rank correlation of $\mathcal{F}_1$ with the static support $|\mathcal{S}|_{\varepsilon_w}$ is -1 in all six, exactly; that of χ runs from $+0.894$ to $+1.000$ with mean $+0.948$, its departures from unity being ties in χ and not inversions.

Three tests, and what each assumes. The stratified concordance statistics $(n_C - n_D)/(n_C + n_D)$ are -1.000 for $\mathcal{F}_1$, with 0 concordant and 60 discordant within-stratum pairs and no ties, so that the statistic is Kendall’s τ , and $+1.000$ for χ , with 50 concordant, 0 discordant and 10 tied, where it is the Goodman–Kruskal γ , which drops the tied pairs; Kendall’s τ_b , which charges them, is below 1. Because each $\mathcal{F}_1$ stratum is a perfect reversal of five distinct points, the exact one-sided stratified permutation probability is the product of the six per-stratum chances, $(1/120)^6 = 3.35 \times 10^{-13}$, over a relabelling space of 2.99×10^{12} ; the same test applied to χ , which has ties, gives 1.93×10^{-10} . Fisher’s combination of the six per-stratum probabilities is conservative here and returns 6.56×10^{-8} and 1.14×10^{-5} ($\chi^2 = 57.4$ and 44.7 on 12 degrees of freedom); we quote the permutation values and record Fisher’s beside them. All are *one-sided*—a perfect reversal is one of the 120 orderings, not two—and all take a product across strata, so all assume the six strata independent. They are not six independent experiments: they are three lengths crossed with two fillings, on one code path and one U grid. What the numbers establish is that the direction is reproduced six times out of six, not that it generalizes beyond this family.

Why neither pooled coefficient is used. Pooled over the same 30 points the coefficients are $\rho(\chi, |\mathcal{S}|) = +0.569$ and $\rho(\mathcal{F}_1, |\mathcal{S}|) = -0.109$, the second with a nominal two-sided $p = 0.57$ and the first with $p = 0.0010$. Pooling six deterministic five-point sweeps with different offsets destroys the within-stratum structure that carries the signal; it is a Simpson reversal in both cases. It happens to destroy the signal in our favour on χ and against us on $\mathcal{F}_1$, and that is exactly why neither may be kept: a rule that is applied to one coefficient and not to the other is not a rule.

The molecular set. The $n = 38$ molecular points are 19 molecules at two geometries, i.e. dependent pairs. The intraclass correlation of $\log_2 |\mathcal{S}|$ within a molecule is 0.214, a design effect of 1.214 and a design-corrected effective n of 31.3. Ties are pervasive: 29 of the 38 (76%) lie in some tied group of $|\mathcal{S}|_{\varepsilon_w}$ and 11 are pinned at the floor $|\mathcal{S}| = 1$, which caps the attainable Spearman coefficient at $\rho_{\max} = 0.9863$; against that ceiling χ reaches 91.4% and $\mathcal{F}_1$ 87.4%. Under a molecule-level cluster permutation (10^6 resamples) both coefficients are beyond the resolution floor of the test, $p \leq 1.0 \times 10^{-6}$. The comparison the molecular set is asked to settle is not resolvable, and three procedures agree that it is not: $\rho = 0.9011$ for χ against 0.8618 for $\mathcal{F}_1$ differ by 0.0393; a molecule-level cluster bootstrap (2×10^4 resamples) puts a 95% interval on the difference of $[-0.070, +0.189]$, which contains zero; Williams’s test for dependent correlations gives $t = 0.94$ on 35 degrees of freedom, $p = 0.35$, and Steiger’s $z = 0.94$, $p = 0.35$. Both parametric tests assume bivariate normality and untied ranks while 76% of these points sit in a tie, which is why the bootstrap is quoted first; at the cluster-level effective n of 19, Williams gives $p = 0.53$. Pooled over all 74 rows the pair is 0.722 against 0.599 and is likewise unresolved. Recomputed over the 71 *distinct* states—three chain rows duplicate three Hubbard rows—it is 0.746 against 0.592.

One artefact of the deposit, recorded because it changes a digit. The pooled $\rho(\mathcal{F}_1, |\mathcal{S}|)$ stored in the deposit is 0.5991984 and recomputing it from the per-species values gives 0.5992028. The difference is not a bug in either: it is a tie in $\mathcal{F}_1$ created when two rows that are the same state are stored at different precisions. The quantity is quoted to three figures throughout for that reason.

TABLE S10. **Reproducibility data for the nineteen-molecule suite** (cc-pVDZ, valence bond-breaking active spaces). $R_{\text{eq}}, R_{\text{diss}}$ in Å; for the polyatomics ($\text{BeH}_2, \text{H}_2\text{O}, \text{NH}_3, \text{H}_4, \text{H}_6$) the representative varied bond length is shown and full Cartesians are in `src/n19_suite.py` of the deposit. D : dimension of the N -electron (neutral, ground-state) sector, in which the tabulated $|\mathcal{S}|$ lives; $|\mathcal{S}|$: ground-state determinant support ($\geq 99\%$ FCI weight); χ : minimal MPS bond dimension across the central spin-orbital cut. The $|\mathcal{S}|/D$ of Table S12 is a distinct ratio, taken in the $(N\pm 1)$ spectral sector.

Molecule	CAS(n_e, n_o)	R_{eq}	R_{diss}	D	$ \mathcal{S} ^{\text{eq}}$	$ \mathcal{S} ^{\text{diss}}$	χ^{eq}	χ^{diss}
F_2	(1+1,2)	1.412	3.00	4	1	2	2	2
N_2	(3+3,6)	1.098	2.40	400	7	80	7	32
H_2O	(2+2,4)	0.958	2.30	36	1	13	1	13
HF	(1+1,2)	0.917	2.30	4	1	4	1	4
NH_3	(3+3,6)	1.012	2.33	400	1	68	4	42
CO	(3+3,6)	1.128	2.40	400	8	46	14	30
H_6	(3+3,6)	0.900	2.00	400	9	91	8	36
H_4	(2+2,4)	0.900	2.00	36	2	11	4	8
H_2	(1+1,2)	0.741	2.00	4	1	2	1	2
BeH_2	(2+2,4)	1.334	2.90	36	1	6	1	8
C_2	(4+4,6)	1.243	2.60	225	13	6	11	2
NO	(4+3,6)	1.151	2.50	300	5	25	11	7
O_2	(5+3,6)	1.208	2.60	120	11	4	4	2
CN	(4+3,6)	1.172	2.50	300	27	9	17	7
OH	(3+2,4)	0.970	2.40	24	1	5	1	4
BH	(2+2,4)	1.232	2.60	36	5	3	2	2
BeH	(2+1,3)	1.343	2.70	9	1	2	1	2
LiH	(1+1,2)	1.596	3.20	4	1	2	1	4
LiF	(1+1,2)	1.564	3.20	4	1	1	1	1

S15. REPRODUCIBILITY DATA FOR THE MOLECULAR SUITE

Table S10 lists, for every molecule of Table S12, the equilibrium and dissociation bond lengths, the active space, the N -electron (neutral, ground-state) sector dimension D , the ground-state determinant support $|\mathcal{S}|$ (the number of determinants carrying $\geq 99\%$ of the FCI weight), and the minimal matrix-product-state bond dimension χ at both geometries. These are the quantities that enter the $\mathcal{F}_1-|\mathcal{S}|$ and $\chi-|\mathcal{S}|$ correlations of Sec. VI, pooled over the $n = 38$ equilibrium-and-dissociation points (Spearman $\chi-|\mathcal{S}|$ $\rho = 0.90$, $\mathcal{F}_1-|\mathcal{S}|$ $\rho = 0.86$; Pearson $\mathcal{F}_1-\log_2 |\mathcal{S}|$ $r = 0.80$). The $|\mathcal{S}|/D$ reported in Table S12 is a *distinct* quantity—the fraction of the $(N+1)$ sector used in the spectral *reconstruction*, not the ground-state support tabulated here. Every active-space FCI energy is converged to better than 10^{-12} Ha (the largest residual across all 38 points is 1.1×10^{-13} Ha); the full Cartesian geometries and the active-space definitions (frozen core, n_e, n_o) are given in `src/n19_suite.py` of the deposit, and the absolute ground-state energies are not deposited, only these FCI residuals (`data/n19_suite.json`).

S15.1. The molecular suite, and the size of its sectors

The nineteen-molecule suite reconstructs the orbital-projected addition spectral function $A(\omega)$ at dissociation in a valence bond-breaking active space, against a subspace grown by multinomial sampling of the time-evolved probe, seeded with the probe’s largest-amplitude determinant, which is inserted rather than drawn: 8000 draws at each of 27 evolution times, 2.16×10^5 per molecule, at $\eta = 0.10$ Ha, from a single fixed random seed and with no spread reported across seeds. What decides the reach of that test is not the shot budget but the sector it is poured into, and Table S11 gives it.

Twelve of the nineteen reconstructions are exact by construction rather than by approximation, and the dimensions say why: seven sectors have dimension six or less, and in five more the six to twelve sampled determinants already contain $\mathcal{K}(H, \phi)$. The suite therefore cannot show that the reconstruction is efficient across chemistry: on twelve of the nineteen there is nothing for efficiency to mean. Its energy check is an active-space CASCI energy against itself, and the residual the appendix records is 1.1×10^{-13} Ha across all 38 points—a convergence check on a deterministic diagonalization, worth keeping in the reproducibility appendix at its true magnitude, but not a validation against an

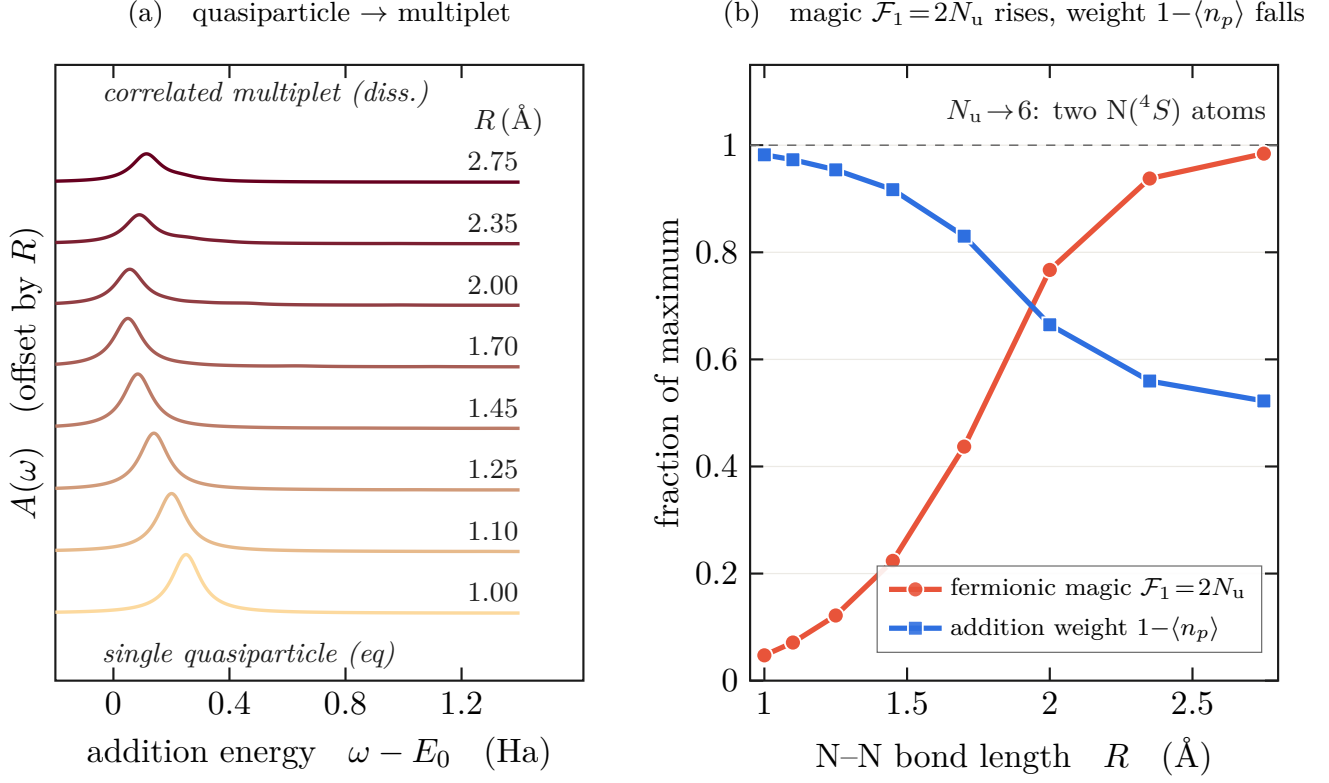


FIG. S6. **Magic and multireference character in N_2 dissociation.** (a) The exact addition spectral function $A(\omega)$ (orbital-projected, CAS(6,6), cc-pVDZ) along the triple-bond stretch, offset vertically, one row per R : the sharp coherent quasiparticle peak near equilibrium ($R = 1.0$ Å, slightly compressed from the $R_{\text{eq}} = 1.098$ Å of Table S10) broadens into a correlated multiplet on dissociation as its spectral weight collapses. (b) Two physically independent quantities along the same coordinate. The fermionic magic obeys the *exact* identity $\mathcal{F}_1 = 2N_u$ [Eq. (17), verified to $< 10^{-13}$], so a single curve carries both: it rises with the orbital-rotation-invariant multireference count—the effective number of unpaired electrons $N_u = \sum_i n_i(2 - n_i)$ from the natural-orbital occupations n_i , approaching its atomic limit of 6 (the two triplet $N(^4S)$ atoms) as the bond fully breaks. The integrated addition weight $\int A^+ d\omega = 1 - \langle \hat{n}_p \rangle$ (from panel a) falls from 0.98 to 0.52 over the same stretch; the two cross as the bond breaks. The active-space CASCI energies are converged to 1.1×10^{-13} Ha, which is a convergence check and not a validation against an independent method (Sec. S15.1).

TABLE S11. **What the nineteen-molecule reconstruction tests.** The suite grouped by whether the reported error is an approximation or an identity. D is the dimension of the $(N+1)$ sector the reconstruction runs in—all nineteen species take the addition branch—and $|\mathcal{S}|$ the absolute subspace size at the end of the protocol. Top row: five sectors of dimension 2, one of 3 and one of 6, where the reported error is identically zero and where in three cases the sampled subspace is the entire two-dimensional sector. Middle row: five further cases whose subspace contains the probe’s Krylov space $\mathcal{K}(H, \phi)$, so what is reported is floating-point round-off (containing the probe’s support alone would not suffice, Sec. IIID). Bottom row: the seven cases in which a truncation costs something. The largest sector in the suite is 300, the dimension of the $L = 6$ Hubbard sector of Sec. IX A, against 731 808 at $L = 12$ on the lattice side of this work.

	count	D	$ \mathcal{S} $	rel- L_1
sector of dimension ≤ 6	7	2–6	1–2	0 exactly
subspace contains $\mathcal{K}(H, \phi)$	5	24–90	6–12	$\leq 1.4 \times 10^{-14}$
genuine truncation	7	20–300	5–149	2.2×10^{-6} – 1.0×10^{-4}

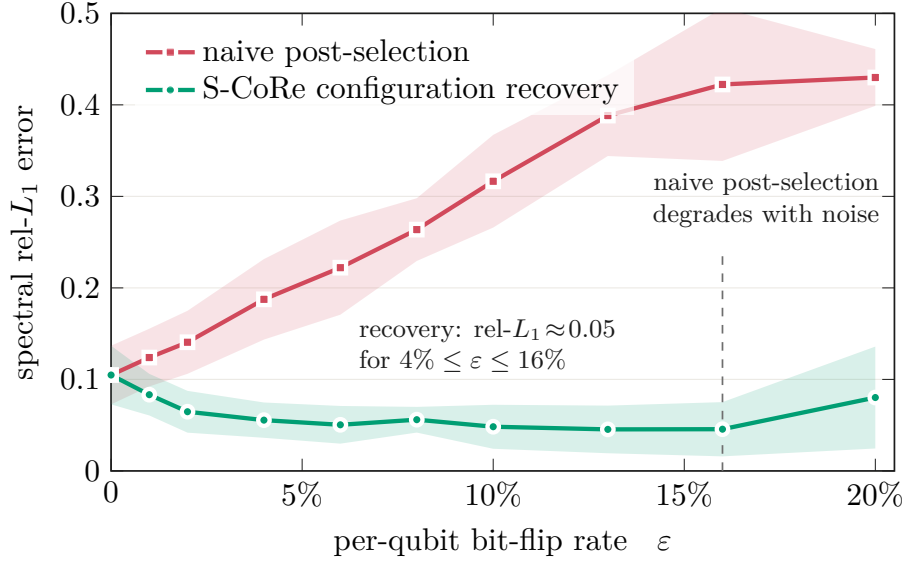


FIG. S7. **Noise robustness (synthetic per-qubit bit-flip channel, $L = 6$).** Spectral relative- L_1 error vs the per-qubit bit-flip rate ε , applied to bitstrings Born-sampled from the exact distribution (this is a controlled classical simulation, not a device run). Naive post-selection degrades steadily, whereas S-CoRe self-consistent configuration recovery, applied to the same corrupted bitstrings, keeps the error near 0.05 for $4\% \leq \varepsilon \leq 16\%$ (periodic Hubbard ring, $U/t = 4$; seed-averaged over 6 seeds at 6000 shots each; bands = ± 1 s.d. across seeds). At $\varepsilon = 0$ the two arms coincide at 0.105, about twice the acceptance threshold 0.05 of the resource scan, and the recovered error at finite ε falls below that noiseless value; that second comparison is not used here (Sec. IX A): it is decided by the size of the recovered subspace, which this sweep does not record.

independent method.

What the suite does test is the bottom row of Table S11, and it is a result. On seven Hamiltonians—O₂, CN, NO, CO, N₂, NH₃ and H₆, in sectors of dimension 20 to 300—the same primitive reproduces the orbital-projected $A(\omega)$ to a relative L_1 between 2.2×10^{-6} and 1.0×10^{-4} from 21% to 50% of the sector. Those seven are independent chemical Hamiltonians rather than one molecule repeated, and their agreement is a consistency check of the primitive outside the Hubbard family. It is not a scaling statement and is not on a common axis with Sec. VII, since the Hamiltonians, the broadening and the error level all differ. Per-species dimensions, subspace sizes and bond dimensions are in Sec. S15, whose column D is the N -electron dimension; the $(N+1)$ dimension of each species is $\binom{n_o}{n_\uparrow+1} \binom{n_o}{n_\downarrow}$ (300 against 400 for N₂).

S16. THE DEQUANTIZATION TEST

The one place where a learned model could remove the quantum sampling from this pipeline is the generation of the dominant determinants, and we tested it on stretched N₂ (cc-pVDZ, $R = 2.5$ Å, CAS(10e,12o), 792 strings per spin, orbitals in the $D_{\infty h}$ symmetry-adapted gauge), where the exact energy is available to score against. A device head is emulated by drawing 1 000 configurations from the exact FCI marginal and corrupting them with a Heron-class noise model—5% depolarizing replacement and asymmetric readout at the median rates of the FakeTorino calibration, 0.0229 for $1 \rightarrow 0$ and 0.0200 for $0 \rightarrow 1$ —after which 24% of the shots carry the wrong particle number. From that one histogram a subspace of $D = 120$ strings per spin is built four ways and scored on the exact energy, over ten seeds paired across the four. A single-pass analogue of the occupation-based configuration recovery of Ref. [35] gives 38.22 ± 2.62 mHa. A cheap classical Epstein–Nesbet perturbative prior [98], which completes the subspace from one application of H to the Hartree–Fock determinant—no additional shots, no additional quantum processing—gives 24.38 ± 4.02 mHa. A generative flow network [99] trained on the recovered histogram fused with that prior gives 22.57 ± 1.60 mHa, and trained on the histogram alone 25.55 ± 1.84 mHa (means and standard deviations over the ten seeds).

The classical step is resolved beyond any reading of the seeds: it lowers the error by 13.85 mHa, a 36% reduction, in all ten seeds, with paired $t_9 = 12.37$; at 500 shots it is larger still, 54% with $t_9 = 36.2$. The generative step is not. On top of the classical prior the fused generator lowers the error by a further 1.80 ± 4.78 mHa, in six of the

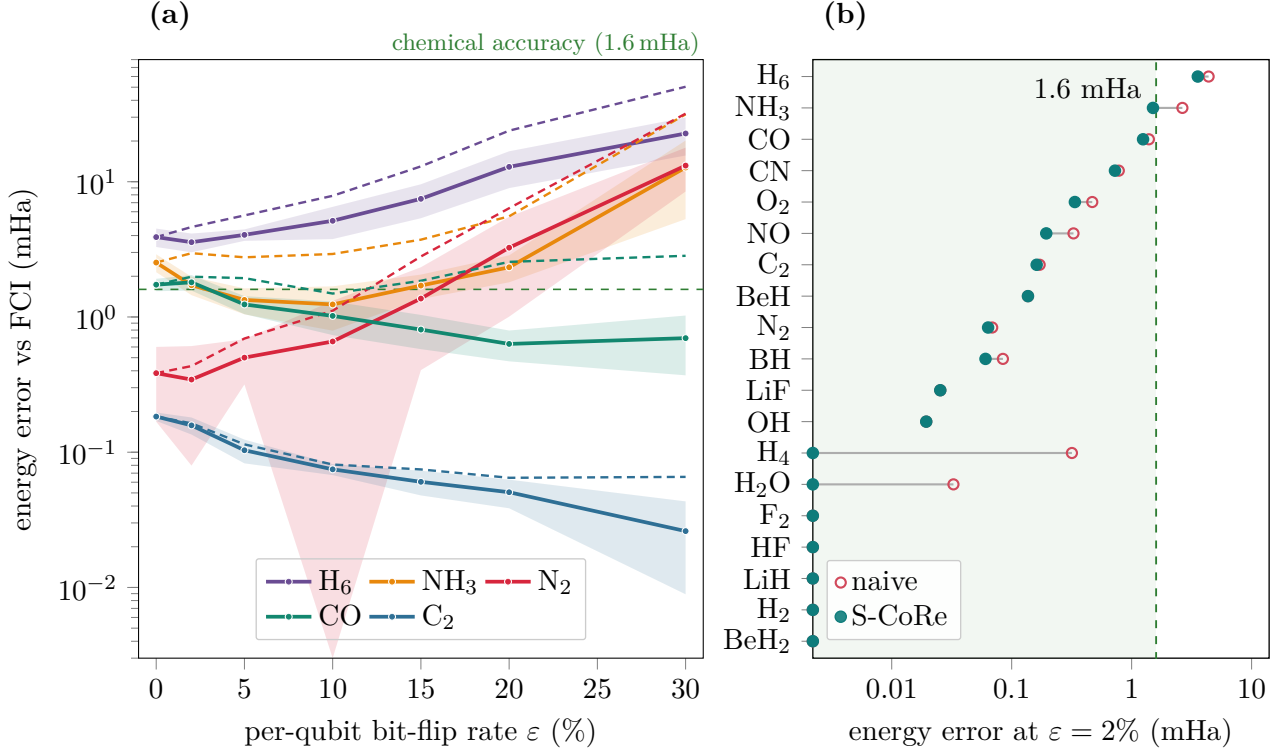


FIG. S8. **SQD ground-state energy under a controlled bit-flip model across the molecular suite.** (a) Energy error vs the per-qubit bit-flip rate ϵ for five representative molecules (H_6 , NH_3 , N_2 , CO , C_2 ; seed-averaged over 8 seeds at 4000 shots each, bands = ± 1 s.d. across seeds, running to the bottom of the panel where the mean minus one s.d. falls below it, as for N_2 at $\epsilon = 10\%$; log scale). Solid = S-CoRe recovery, dashed = naive post-selection. The advantage of recovery over naive post-selection, both arms consuming the same corrupted bitstrings at the same rate with the same seeds, grows with noise. For NH_3 , CO and C_2 the error at finite ϵ also falls below its own $\epsilon = 0$ value, beyond the seed bands (NH_3 $2.53 \rightarrow 1.24$ mHa at $\epsilon = 10\%$); that second comparison is not used here (Sec. IX A): it is decided by the size of the recovered subspace, which this sweep does not record. (b) The full nineteen-molecule suite at a realistic $\epsilon = 2\%$ (log scale, sorted; errors that are exact to round-off, in both arms for F_2 , HF , LiH , H_2 and BeH_2 , where the open circle is hidden behind the filled one, and after recovery for H_4 and H_2O , are drawn at the left edge of the axis): for each molecule the naive post-selection error (open circle) is linked to the S-CoRe value (filled), so the leftward pull is the recovery gain; the shaded band is chemical accuracy (< 1.6 mHa). Eighteen of nineteen land inside it; H_6 , the most multireference case, is the residual challenge. Panels (a) and (b) are independent seed ensembles (panel (a): 8 seeds; panel (b): 5 seeds, 4000 shots), so a given molecule's $\epsilon = 2\%$ value can differ by up to a factor of a few between them (for N_2 , 0.34 vs 0.06 mHa; both far below chemical accuracy).

ten seeds, with paired $t_9 = 1.19$ ($p = 0.26$), and without the prior it is not distinguishable from the classical control (1.17 mHa, $|t_9| = 0.92$). At 500 shots both generative arms are *worse* than the classical control: the fused one by 5.31 mHa ($|t_9| = 3.27$, $p = 0.010$, worse in nine of the ten seeds), the one without the prior by 15.33 mHa ($|t_9| = 10.3$).

The two halves of the ten seeds are the two five-seed runs of the companion study [132], and they show why five seeds do not decide this comparison. Over seeds 0–4 the fused generator is ahead of the classical control by 5.23 ± 3.19 mHa in every seed, with $t_4 = 3.67$, beyond the two-sided critical value 2.776; over seeds 5–9 it is behind, -1.62 ± 3.46 mHa with $t_4 = -1.05$. The companion reports $+5.23 \pm 3.19$ and -1.83 mHa for the same two runs; the first is reproduced here to the fourth decimal and the second to 0.21 mHa. A five-seed t test can therefore clear its critical value and fail to replicate, and a distribution-free test cannot help, since with five seeds the exact two-sided sign test and the signed-rank test both floor at $p = 2/2^5 = 0.0625$; with ten the floor is 0.002, which the classical step reaches. `src/gflow_dequant.py` writes every seed to `data/gflow_runs/`, and `data/gflow.json` holds the paired statistics quoted here.

What the test establishes is the statement we keep. On this instance the dominant determinants are supplied by the sampled histogram (here an emulated one, drawn from the exact ground state and then noise-corrupted, so the test does not measure whether a device state would supply them) and completed by a classical prior, not manufactured by a learned generator: the largest improvement available to an under-sampled head costs no quantum resources at all and should be the default post-processing step, since a demonstration that omits it is measured against a baseline that

TABLE S12. **The nineteen-molecule fermionic-magic suite**, computed in a cc-pVDZ active space; the active-space CASCI energies are converged to 1.1×10^{-13} Ha, which is a convergence check and not a validation against an independent method (Sec. S15.1). The fermionic AntiFlatness $\mathcal{F}_1$ ($k=1$ Majorana antiflatness; $\mathcal{F}_1=0$ iff Gaussian/mean-field, maximum $\mathcal{F}_1^{\max}=2n_o$) is listed near equilibrium and at dissociation, in a valence bond-breaking active space CAS(n_e, n_o). Fermionic magic switches on for covalent bond-breaking, is already large at equilibrium for inherently multireference species (C_2 , O_2), and stays near-Gaussian for ionic and weakly-correlated bonds—so $\mathcal{F}_1$ marks the multireference, strongly-correlated regime rather than bond length *per se*. The last two columns validate the *method*: at the dissociation geometry the reconstruction of the (orbital-projected) spectral function $A(\omega)$ from a subspace drawn in 2.16×10^5 shots matches the exact Lehmann result to relative- L_1 error, on a subspace that is a fraction $|S|/D$ of the full $(N+1)$ sector. In twelve of the nineteen the subspace contains the probe’s Krylov space and the agreement is exact by construction; see Table S11. The listed $\mathcal{F}_1 = 4 \text{tr}[\gamma(1-\gamma)]$ are directly computed spin-orbital values (for the five open-shell species they are *not* $2N_u$; see Sec. VI). Per-molecule bond lengths, active spaces, FCI residuals, and the absolute $|S|$, sector dimension D and minimal bond dimension χ are listed in Sec. S15; the absolute FCI energies are not deposited. [†]: near-degenerate ground state.

Molecule	Bonding	CAS(n_e, n_o)	$\mathcal{F}_1^{\text{eq}}$	$\mathcal{F}_1^{\text{diss}}$	$\mathcal{F}_1^{\text{diss}}/\mathcal{F}_1^{\text{max}}$	rel- L_1	$ S /D$
<i>Covalent bond-breaking</i>							
F ₂	covalent single	(1+1,2)	0.068	3.985	1.00	$<10^{-13}$	0.50
N ₂	covalent triple	(3+3,6)	0.847	11.377	0.95	2×10^{-5}	0.21
H ₂ O	multicentre	(2+2,4)	0.007	6.936	0.87	$<10^{-13}$	0.50
HF	polar single	(1+1,2)	0.001	3.460	0.87	$<10^{-13}$	1.00
NH ₃	multicentre	(3+3,6)	0.080	10.289	0.86	1×10^{-5}	0.47
CO	polar triple	(3+3,6)	0.688	8.192	0.68	1×10^{-4}	0.41
H ₆	H-chain	(3+3,6)	0.521	7.735	0.64	2×10^{-5}	0.50
H ₄	H-chain	(2+2,4)	0.252	5.113	0.64	$<10^{-13}$	0.50
H ₂	covalent single	(1+1,2)	0.038	2.511	0.63	$<10^{-13}$	0.50
BeH ₂	multicentre	(2+2,4)	0.039	3.754	0.47	$<10^{-13}$	0.25
<i>Multireference / open-shell</i>							
C ₂ [†]	covalent double	(4+4,6)	4.743	8.001	0.67	$<10^{-13}$	0.10
NO	polar (radical)	(4+3,6)	0.790	7.449	0.62	2×10^{-6}	0.25
O ₂	covalent double (triplet)	(5+3,6)	1.673	6.000	0.50	1×10^{-5}	0.25
CN	polar (radical)	(4+3,6)	1.307	5.938	0.49	1×10^{-5}	0.24
OH [†]	polar (radical)	(3+2,4)	0.013	3.703	0.46	$<10^{-13}$	0.33
<i>Ionic / weakly correlated</i>							
BH	metal hydride	(2+2,4)	0.518	0.573	0.07	$<10^{-13}$	0.25
BeH	metal hydride (radical)	(2+1,3)	0.035	0.386	0.06	$<10^{-13}$	0.67
LiH	metal hydride	(1+1,2)	0.002	0.075	0.02	$<10^{-13}$	1.00
LiF	ionic	(1+1,2)	0.000	0.001	0.00	$<10^{-13}$	1.00

is too weak, and a learned generator adds nothing on top of it that ten seeds resolve.

The class of learned constructions that select or generate the determinants themselves is by now several—neural quantum states built on selected configurations, autoregressive and dual-channel samplers for selected configuration interaction, and neural-network classifiers [133–136]—and the defensible role for a learned model is the one that consumes the device histogram, as physics-informed generative recovery of hardware samples does [137].

A learned model can also schedule device work, as generative models for measurement grouping do [138, 139], alongside the non-learned error-mitigation and validation protocols that already sit at that interface [140, 141] and learners that process device output coherently in quantum memory [142]; this is the boundary the companion study maps from the ground-state side [132].